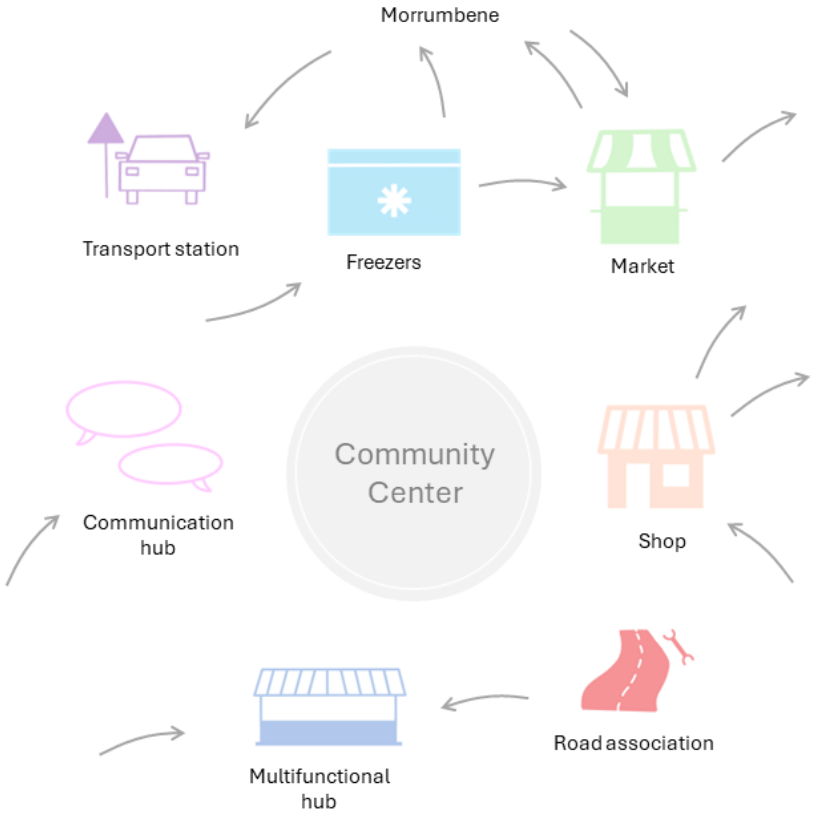
Degree Project in Industrial Design Engineering
Second cycle, 30 credits

Empowering Rural Communities in Moçambique

Development of a community center to improve living standards in a rural Moçambican village

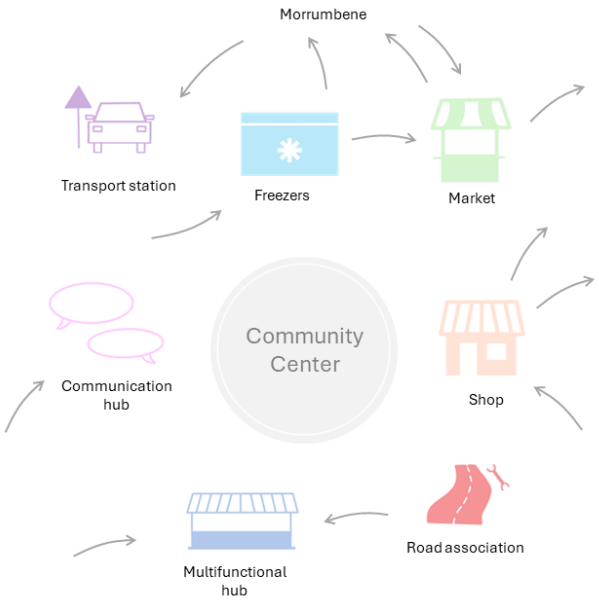
ANDREAS FALK



Empowering Rural Communities in Moçambique

Development of a community center to improve living standards in a rural Moçambican village

Andreas Falk



Master of Science Thesis TRITA-ITM-EX 2024:217
KTH Industrial Engineering and Management
Machine Design
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KTH Industrial Engineering and Management

Examensarbete TRITA-ITM-EX 2024:217

Utveckling av livskraftiga landsbygdssamhällen i Moçambique

Utveckling av ett community center för att förbättra levnadsstandarden i en Moçambikansk by på landsbygden

Andreas Falk

Godkänt	Examinator	Handledare
2024-06-26	Claes Tisell	Magnus Eneberg
	Uppdragsgivare	Kontaktperson
	Project Vita	Michel Olofsson

Sammanfattning

Denna rapport presenterar hur ett community center kan hjälpa befolkningen i en by på landsbygden i Moçambique. Projektet är en masteruppsats i Teknisk Design på Kungliga Tekniska Högskolan. Uppsatsen är i ett samarbete med välgörenhetsorganisationen Project Vita som arbetar i Moçambique.

Projektets syfte var att bidra med ytterligare kunskap i hur community centers ska utvecklas och skapa hållbarare kustbyar på Mozambikanska landsbygden. Projektet kombinerade tjänstedesign och systemdesign tillsammans med ett användarcentrerat och bottom-up tillvägagångsätt, där olika metoder användes så som intervjuer, observationer, och en co-creation workshop. En 8 veckor lång fältstudie genomfördes i den Mozambikanska byn Linga Linga, där dess läge på landsbygden innebär stora utmaningar för befolkningen.

Genom att definiera användarnas behov och designa tjänstelösningar för att passa det lokala sammanhanget, utvecklades ett community centerkoncept. De sju tjänster som föreslås i centret löser på ett holistiskt sätt många problem samtidigt, vilket skapar en kedjereaktion av positiv samhällspåverkan. Det har potential att stärka samhället genom ökad demokrati, effektiviserad informationsspridning, och förbättrad tillgång till resurser och utbildningsmöjligheter, där lokala traditioner och kulturer samtidigt värderas.

Även om samhällen på landsbygden står inför många utmaningar, är en av huvudinsikterna att det finns möjligheter till samhällsförändring. Projektets övergripande slutsats är att community centers kan förbättra livsstandarden för befolkningen på landsbygden genom att aktivt stärka samhällsengagemang snarare än investera i storskaliga lösningar.

KTH Industrial Engineering and Management

Master of Science Thesis TRITA-ITM-EX 2024:217

Empowering Rural Communities in Moçambique

Development of a community center to improve living standards in a rural Moçambiqan village

Andreas Falk

Approved	Examiner	Supervisor
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Abstract

This report presents how a community center can help citizens in a rural village in Moçambique. The project is a Master's Thesis in Industrial Design Engineering at KTH Royal Institute of Technology. The thesis is in collaboration with Project Vita, a non-profit organization in Moçambique.

The purpose of the thesis was to expand the knowledge of community center development and to develop sustainable rural coastal communities in Moçambique. The project combined service- and systemic design with a user-centered and bottom-up approach, employing methods such as various interviews, observations, and a co-creation workshop. An 8-week field study was conducted in a village called Linga Linga, where its rural location poses great challenges for the population.

By defining the stakeholder's needs and designing service solutions to suit the local context, a community center concept was developed. The seven services proposed in the center holistically solve multiple needs simultaneously, creating a chain reaction of positive community impact. It potentially strengthens the community through enhanced democracy, information spreading, resource accessibility, and education opportunities, while valuing local traditions and culture.

Even though rural communities face many great challenges, one of the main project findings is that there are opportunities for change. The overall conclusion of the project is that a community center can improve the quality of life for rural citizens, by building on enhanced community engagement rather than large-scale investments.

Acknowledgment

First and foremost I would like to express my deepest appreciation to my contact person in Moçambique Michel Olofsson, who has spent hours helping me through field trips, by finding interview participants and interpreting the interviews. Without your support, the project would not be possible.

I would like to forward my greatest gratitude to my University tutor Magnus Eneberg. Your guidance, inspiration, and dedication to the project have been of the utmost importance to reaching my goals.

Additionally, I would like to thank the Swedish organizations ICLD, Kvinna till Kvinna Foundation, and SIDA in Moçambique, for providing me with their valuable time and expertise to support my project. Moreover, I want to send a thank you back to Moçambique, to the residents of Linga Linga and the Morrumbene district for happily participating in my interviews and workshop.

Another organization I want to thank is the Karl Engvers Foundation for making my field study in Moçambique possible by providing me with a scholarship.

Lastly, I thank my family and friends for your interest, support, and inspiration for conducting this type of project and fulfilling my goal.

Andreas Falk

Stockholm, June 2024

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1. Introduction

The first chapter provides the background of the study and problem. The purpose, deliverables, and the project's connection to the UN Sustainable Development Goals are also presented. The chapter ends with delimitations and disposition of the report.

1.1 Background and Problem Description

There is rapid urbanization in the world. The UN reports the world's urban population have increased from 30% in 1980 to 55% in 2018 which is projected to be 60% in 2030 (United Nations, 2018). The driving force behind this transformation is economic growth (Li, Huang and Song, 2022). The transformation is occurring both in developed countries like Sweden, as well as in developing countries where the fastest-growing cities are located in Asia and Africa (United Nations, 2018). In poorer countries, urbanization often leads to expanded slums in the cities (Liu and Li, 2017). This is because poor and less educated rural citizens migrate to urban areas in search of job opportunities. Urbanization has highly affected rural communities as well because the rural-urban migration has caused a decline in the quality of life in rural areas and a weakening of rural economies and communities. Consequently, there is a growing motivation for youth to leave their home villages to live in urban areas.

In Moçambique over 62% of the population lives in rural areas with poor infrastructure connected to the cities (UN-Habitat, 2023). The low-income country has a yearly rural-urban migration increase of 0.8%, mainly accelerated by economic reasons, conflicts, and natural disasters. Moçambique is on the UN list of the world's Least Developed Countries (LDC) and has been for over 35 years (UN, no date a). The country confronts large structural impediments to sustainable development and has limited human development resources. The UN and the Government of Moçambique have jointly developed a framework for sustainable development by 2026 in the country (UN Moçambique, 2021). By this year, more people should benefit from more inclusive and sustainable economic growth.

Moçambique's high rural population and rapid urban movement, call for sustainable rural development solutions to prevent unsustainable urbanization and increase resilience of rural communities. Scholars and industry professionals mean community centers can contribute with services to face socio-economic challenges and empower communities and individuals (Holcim foundation, 2008; Colistra, Bixler and Schmalz, 2019; Lee and Na, 2019; Tony *et al.*, 2019). This study explores the capacity of community centers to address rural challenges and foster sustainable rural development.

The project is in collaboration with the non-profit organization Project Vita, who work towards developing communities out of poverty in Moçambique. The project involved a case study of the rural and coastal village called Linga Linga, located in the South-East of Moçambique. Project Vita has seen a desire to meet local needs and gather solutions from previous projects in one place, where a community center will fill this function. Tough living conditions, poor

connection to the larger cities, increasing effects of climate change, and lack of tangible value visualization for previous project solutions, call for a holistic solution to improve the life situation for the population in Linga Linga.

1.2 Purpose

The purpose of the thesis is to expand the knowledge of community center development and build more sustainable rural coastal communities in Moçambique. The two primary research questions are:

RQ.1: What needs do the stakeholders of Linga Linga have that can be met by a community center?
RQ.2: What services could the community center offer to meet the identified needs?

1.3 Deliverables

The project's deliverables are:

- Report on the identified stakeholder's needs.
- A service design concept of the community center and the individual service's importance level.
- A descriptive film about the service design concept, its opportunities, the stakeholder's benefits, and the societal impact.

1.4 Sustainable Development Goals

The United Nations adopted the 2030 Agenda for Sustainable Development in 2015, where 17 Sustainable Development Goals (SDG) were set up to provide a shared framework for peace and prosperity for the planet today and in the future (UN, no date e). The project has certain linking indicators to SDGs 5, 8, and 9. However, it is mainly linked to SDG 11, but also targets SDG 1 and 4:



SDG 1. *No poverty.* The project specifically targets 1.4: "By 2030, ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, as well as access to basic services…" (UN, no date b), by understanding the rural needs and develop solutions to fulfill them.



SDG 4. *Quality education*. The thesis particularly targets 4.a: "Build and upgrade education facilities that are child, disability and gender sensitive and provide safe, non-violent, inclusive and effective learning environments for all*"* (UN, no date c), by considering suitable educational spaces and opportunities for kids and adults.



SDG 11. *Sustainable cities and communities*. The thesis is specifically connected to the target 11.a: "Support positive economic, social and environmental links between urban, peri-urban and rural areas by strengthening national and regional development planning" (UN, no date d), by concentrating on the local's needs and how to satisfy these in a sustainable local- and larger perspective.

1.5 Delimitations

Firstly, the field study had a time constraint of 8 weeks. Another limitation was to only consider fulfilling the stakeholder's needs through services and not including the architecture perspective. Lastly, the geographical limitation of the community center was its location in Linga Linga, where the needs and stakeholders only connected to this area were considered. Therefore, the project's outcome may differ if applied to other locations.

1.6 Disposition

The report starts with an introduction to the project. Thereafter, a theoretical framework (Chapter 2) presents relevant theories to the study, followed by the methodology (Chapter 3) which outlines the study's approach. Chapter 4 involves the pre-study, which is a preparatory phase to the field study. The chapter is divided into three sub-chapters, where the methods are presented, the results, and an analysis. The chapter involves desktop research and interviews with relevant actors. The next chapter (Chapter 5) considers the field study phase, including the methods: in-depth interviews, pop-up interviews, and observations. In this chapter, the stakeholder's needs are identified, and it follows the same three-sided report structure as Chapter 4. The sixth chapter concerns the concept development phase. Here, the insights from the two first phases are integrated, where a co-creation workshop and ideation session is presented, which generate ideas for satisfying the identified needs. Thereafter, in Chapter 7, the final concept is shown, followed by a discussion regarding the final concept, research questions, and methodology. The last chapter includes the conclusion, where the project's practical and scientific contribution is presented, as well as, potential further studies.

2. Theoretical framework

This chapter presents previous research that is relevant to the study, regarding Project Vita, community centers, social innovation, rural development, and intercultural awareness.

2.1 Project Vita

The thesis's partner organization, Project Vita, is located in Moçambique. They have since the start been working on projects to develop solutions for the locals in Linga Linga. This rural and coastal village is seen as a pilot location where projects can be scaled up and applied to other communities. Due to Moçambique's large coast, many villages share the same situation as Linga Linga. Collaborating with KTH students since 2016 as a nonprofit organization, Project Vita bridges the resource and consulting knowledge between Moçambique and the West to help the locals help themselves out of poverty in an environmentally sustainable way. Their three central principles are *community involvement* through local and foreign volunteer collaboration, *knowledge sharin*g by combining Western and local resources, and *pride of ownership* by involving the locals in life-changing projects (Project Vita, no date). In Linga Linga, there is a need to bridge the gap between the capability of the solutions of previous projects and people's awareness of the value of these solutions. Project Vita found a desire to establish a community center in the village, using a holistic perspective, to satisfy needs and to integrate previous projects.

2.2 Community center

A community center can be defined as: "Community centers are a type of community development services that promote social relationships and mutual support, develop a sense of self-reliance, social responsibility, and community cohesion, as well as empowers individuals and families to solve common social problems" (Tony *et al.*, 2019, p. 145). The authors see the purpose of community centers as to enhance the community's quality of life, socially, environmentally, recreationally, economically, and educationally. However, community centers have various functions, for instance, to combine environmental sustainability with social equality (Holcim foundation, 2008), improve local sanitation levels and gender equality (Lee and Na, 2019), or develop social interactions (Colistra, Bixler and Schmalz, 2019). Nonetheless, a shared element of community centers is the adaption to the local needs.

2.3 Rural development

A rural location can be described as: "all the territories that are not urban (or allocated for urban expansion), where the main activities are agro-silvo pastoral and the habitation density is very low*"* (Jones and Kijima, 2018, p. 276). Sustainable rural development can be seen as the process of generating local value in rural communities without negatively affecting the environment.

With urbanization, poverty is drastically increasing, however, poverty is more widely spread in rural areas (Jones and Kijima, 2018). There are issues connected to social and economic perspectives regarding the lack of visibility, equipment, and infrastructure. In addition, rural communities suffer from labor absence, social degradation, and shrinkage of local markets and businesses (Liu and Li, 2017). There is also a tendency for politicians to disregard and being non-engaged to rural areas since there is a negative attitude of what is possible to gain from it (Jones and Kijima, 2018). Therefore, there is a lack of central political trust among the rural population. Additionally, researchers need to shift the focus from urban to rural development (Liu and Li, 2017). However, there are examples of policymakers prioritizing rural areas. In the 1950s Malaysia, the Philippines, and Thailand developed pro-rural policies to enhance public health and education while limiting labor movements to urban areas (Pierskalla, 2016).

Particularly for rural development, where there are limited available resources, social innovation is an approach for better use of them (Hubert, Therace and Dro, 2011). Hubert, Therace, and Dro (2011) raise several opportunities for rural areas to adopt social innovation. This includes basic services for rural inhabitants, for instance, infrastructure connected to child- and elderly care, education, healthcare, and local multifunctional centers that combine culture, leisure, and employment. This aspect also creates business possibilities such as micro-enterprises. For instance, it allows local farmers to diversify their businesses. This opportunity is strongly connected to the project's development of a community center.

Rural communities have been isolated and hold strong traditions which should be seen as valuable instead of an obstacle (Jones and Kijima, 2018). A positive change is created by a strong local connection, being socially inclusive, and building networks to connect social scales such as local, national, and international. The networks should be inclusive, addressing collective learning and being open to sharing success. There is a need for strategies for rural development where improved infrastructure between local producers and the external markets (Özçatalbaş, 2023). The rural and urban linkages should be strengthened with better distribution channels and storage of resources where involving various actors is crucial.

2.4 Social innovation

Social innovation plays a significant role in this project because many challenges rural communities face are socially related. Manzini, a globally recognized leader in design for social innovation, mentions one definition of social innovation: "Social innovation is a process of change emerging from the creative re-combination of existing assets (from social capital to historical heritage, from traditional craftsmanship to accessible advanced technology), the aim of which is to achieve socially recognized goals in a new way" (Manzini, 2014, p. 57).

Hubert, Therace and Dro (2011) identifies three levels of social innovation. Firstly, grassroots social innovations include targeting vulnerable groups in society which the market fails to address. This level can also be the case for inadequate public policies, for instance where governmental authorities are not eager to meet local needs and find appropriate solutions (Jones

and Kijima, 2018). Secondly, broader innovations focus on challenges at societal levels where the social and economic aspects intersect (Hubert, Therace and Dro, 2011). Lastly, a systematic level where fundamental changes occur such as toward attitudes, values, processes, deliveries, and services. This last level includes directing society to become more participating and making people empowered where learning is a crucial parameter. The intention of the thesis targets all three levels to some extent.

2.5 Intercultural awareness

Intercultural knowledge is an important research topic in this project since there are large cultural differences between Sweden and Moçambique. Therefore, the design approach and methods must be adapted to this to understand the context and users further, and generate a solution that is well-suited, for a desired outcome. This chapter presents two cultural models. An important aspect is that they are generalizations and do not represent each individual. Therefore, intercultural theory should be used with carefulness and more as guidance, to avoid stereotyping.

2.5.1 Culture values

Culture has visible and non-visible dimensions (Menzies, 2015). Visual dimensions include symbols, clothing, architecture, language, and traditions etc., while the hidden culture dimension involves values, assumptions and beliefs. The visual dimension stands for ten percent of the cultural identity while the rest belongs to the non-visible. Hofstede's Cultural Dimensions Theory raises initially six dimensions of understanding cultural differences. Figure 1 shows a comparison between Moçambique and Sweden (*Country comparison tool*, 2023). There are many clear differences between the countries as well as some similarities.

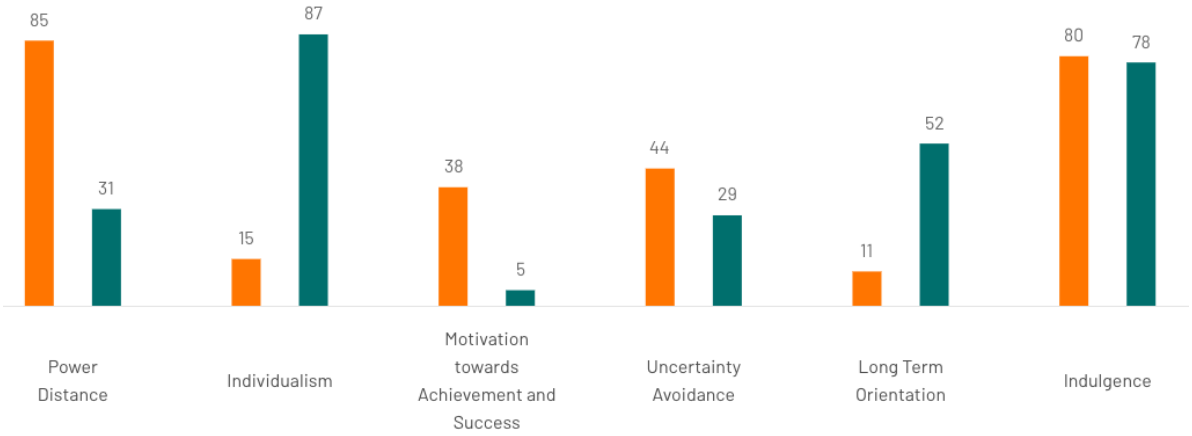


Figure 1. Comparison of the six Hofstede's culture values dimensions between Moçambique (orange) and Sweden (blue) (Country comparison tool, 2023). The scale ranges from 0-100.

1. **Power distance**

Power distance refers to the expectation or acceptance, among the less powerful society members, of how power is unequally distributed. Moçambique has a high score, meaning it is a highly hierarchy society where people accept and respect the order. In high power distance, only limited people have access to resources, knowledge, and skills. Sweden has a low score, indicating a hierarchy based on convenience, more independence, equal rights, and a decentralized power structure. People are comfortable speaking up disregarding hierarchy levels. The communication is direct and participative.

2. **Individualism**

This dimension concerns the degree of individualism in society, whether one see oneself as a "I" or "we". Moçambique's low score indicates a collective society where relationships and loyalty are valued highly, group goals are more important than individual ones, group activities are preferred and social interactions are long and intimate. Sweden's low score means a more individualism society where people take care of themselves and their closest members, the relationships are more loose, individual activities are desired more, and the social interactions are shorter but more frequent.

3. **Motivation towards achievements and success**

A high score means the society is driven by competition, achievements and success while a low score indicates society's value in taking care of others and quality of life. Moçambique has a relatively low score, meaning the aim of working is to live. Sweden scores much lower, indicating care for others and prioritizing a quality life. Compromises solve conflicts and discussions are held until there is a consensus.

4. **Uncertainty avoidance**

Uncertainty avoidance refers to how societies perceive the future, whether it should be controlled or accepted how it turns out. Both Moçambique and Sweden have relatively low scores meaning an openness to change, innovative ideas, and various perspectives.

5. **Long term orientation**

This dimension refers to how the relationship between the past and future should be treated. In Moçambique, with a low score, traditions are valued high, no large focus on saving for the future, instant satisfaction is treasured and there is an unwillingness of societal changes. The middle score for Sweden indicates no strong relation between the two orientations.

6. **Indulgence**

Indulgence refers to the degree people follow their impulses and desires. Both countries scores high which means there is a general positive attitude and optimism, enjoying life is valued high where there is a willingness to spend money.

2.5.2 The Lewis model

The Lewis model categorizes people's behaviors into three distinct types: multi-active, reactive, and linear active, as seen in Figure 2 (CrossCulture, 2015). The first type is a diverse group in regard to religion, beliefs, and values. However, they share characteristics such as being emotional, having feelings before facts, being flexible with truth, being impatient, using body language, and being relationship-oriented. Religion and family bonds are also important. Moçambique can be placed in the Sub-Saharan Africa category, highly multi-active. The second

type is *reactive*, this group is polite and indirect, see statements as promises, diplomacy over truth, is patient, and reacts to a partner's action. Most Asian countries belong to this group except India which is between multi-active and reactive. The third group, called *linear-active*, is polite but direct, sticks to facts, values truth before diplomacy, is sometimes impatient and uses limited body language. The English-speaking countries and Scandinavian countries, including Sweden, are placed in this category. It is worth highlighting that people generally have elements from multiple categories, however, there may be one more dominant one.

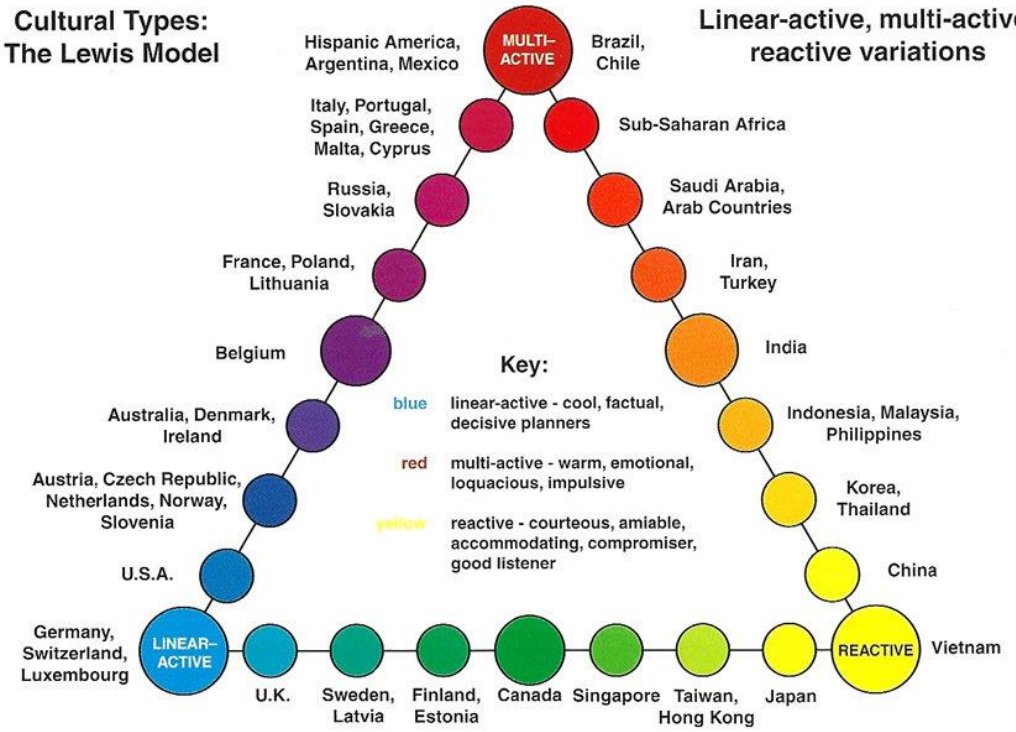


Figure 2. The Lewis model, visualizing the three people categories: multi-active, reactive, and linear-active (Skarbek, 2021).

3. Methodology

Chapter 3 presents the methodology of the project. It involves elements of design for social innovation, a holistic approach, a systemic perspective, and co-design. The chapter also addresses aspects of conducting user studies in indigenous environments.

3.1 Design for social innovation

The thesis project has a strong connection to design for social innovation. The globally regonized leader of the field, Ezio Manzini, defines it as: "Design for social innovation is all that design can do to recognize, stimulate, extend, and sustain, what I call, promising ways of being and doing*"* (Muratovski, 2023, p. 77). This description connects to the essence of this thesis which is enabling a sustained solution that has a long-term positive impact in the community. Similar to other design practices, design for social innovation starts with problem mapping. However, Manzini raises the crucial aspect of also mapping available resources, where social innovation happens when the problem- and resource maps overlap. Moreover, he says social innovation calls for specific targeted solutions rather than generalizations since design contexts vary considering e.g. the people, places, and societies. In addition, systemic rethinking is needed to think beyond previous approaches and viewpoints which can lead to new identifications of problems, resources, and solutions for the future. Manzini (2014) expresses design for social innovation as a collection of design principles to make social innovation more possible, effective, sustained, and able to be spread.

Hubert, Therace, and Dro (2011) mention twelve lessons learned from a previous social innovation initiative, made by the European Commision, called EQUAL One learning highlights the solutions must be for the beneficiaries and be co-created with them. The involvement of local citizens and co-creation is a vital part of this thesis. Another learning indicates a focus on the strengths of individuals and communities rather than their weaknesses. Also, the approach should be holistic rather than targeting individual problems separately. Furthermore, partnerships should be empowered as well as collaborations and networking. Lastly, the solution should be based on the local community instead of a general solution, which Manzini also highlighted.

3.2 Holistic approach

Adopting a holistic approach in the thesis is necessary because, as previously mentioned, Hubert, Therace, and Dro (2011) highlight this approach for social innovation. The relevance of a holistic approach was also strengthened by Stigendal and Östergren (2013) as an example in Malmö, Sweden. They pointed out the need to consider systemic and structural factors rather than facing one isolated problem or perceiving problems as highly grounded in communities and individuals. It was suggested that complex societal challenges in Malmö city should be targeted with a holistic mindset and a merge of different knowledge. The responsible commission in Malmö emphasized the need for sharing knowledge and a more inclusive

governance. In addition, to switch the focus on conditions and results towards continuous knowledge and learning processes to battle many problems simultaneously. This switch requires the process to be more open where starting points and problems are identified by a broad range of stakeholders. The involvement of citizens in the process is also important since it generates the positive aspect of gaining both formal and informal knowledge.

3.3 Systemic perspective

Systemic design plays an integral part of the thesis as it is a well-established approach and process in design practice for supporting social change and solving complex issues, often referred to as "wicked problems" (Jones and Kijima, 2018). Systemic design combines design methods with systemic theory and approaches. Jones and Kijima (2018) separate the design contexts into four domains or design scopes:

D1.0: Artefacts, which are designed objects and communications for limited purposes or support a product and service.
D2.0: Products and services, where the functions are integrated into a larger system and where co-creation occurs with the users.
D3.0: Organizational transformation, considers the design of work practices, strategies, and organizational structures. The co-creations concern all levels of the organization.
D4.0: Social transformation, concerns design for highly complex social systems, policy making, and community design. This includes the management of the participation of multi-stakeholders.

Another approach taken in this thesis is service design. Service design can be defined in various ways, Stickdorn *et al.* (2018, p. 18) present one popular definition: "Service design helps to innovate (create new) or improve (existing) services to make them more useful, usable, desirable for clients and efficient as well as effective for organizations. It is a new holistic, multi-disciplinary, integrative field". Service design has become a recognized practice for transforming social systems (Vink and Koskela-Huotari, 2022). To enable this type of transformation, Vink and Koskela-Huotari (2022) emphasize the importance of reflexivity, which is the awareness of present social structures. This awareness is relevant to the thesis since the social structures in Moçambique are different from those in Sweden. The authors have developed a six-categorical framework for how methods can contribute to people's reflexivity, whereas the relation-, culture-, and corporeal categories are suitable for the context of the thesis. Relation reflexivity involves the definition of social structures through interactions among people (Vink and Koskela-Huotari, 2022). In Corporeal reflexivity, the perception of social structures is addressed by investigating emotions and body expressions, which is relevant for the project since there are language barriers. Culture reflexivity involves the definition of social structures through social behaviors of specific groups of people. However, Vink and Koskela-Huotari (2022) urges practitioners to be aware of the appropriateness of utilizing specific methods in different cultural settings.

Vink et al. (2021) highlight the importance of a systemic understanding of service design, referred to as service ecosystem design, due to the lack of enabling long-lasting innovation in service design. Complex systemic problems, which involve multiple actors, need to be highlighted. Service ecosystem design offers a more systemic and holistic understanding of value co-creation and brings insights into how actors can influence this process within the ecosystem. Vink et al. (2021) emphasize three effects of service ecosystem design. Firstly, the perspective becomes more holistic and includes cultural aspects. Secondly, reflexivity is incorporated to create long-term change solutions. Thirdly, co-design is incorporated into larger collective design processes.

3.4 Co-design

Co-design is a practice that enables users to participate in the design process as an 'expert of their experience' (Visser *et al.*, 2005, p. 132). The practice transforms the traditional design of individual experts towards collective creativity (Steen, 2013). Co-design has an integrated role in both systemic- and service design. Within systemic design, co-design with stakeholders is a vital part and the designer should be seen as a participant rather than an external part (Jones and Kijima, 2018). In service design, co-designing leads to more unique knowledge and experiences which is a key factor of developing successful services (Chang and Taylor, 2016; Trischler and Charles, 2019).

The service design methodology can be adapted to generate innovation by making the involved actors think differently in the design process, which is often lacking in traditional practices (Vink *et al.*, 2021). The creation of innovation within service design involves utilizing the designer's creativity to select appropriate materials and locations, thereby supporting the reshaping of thinking among co-creation participants.

To achieve rural development, there should be a so-called bottom-up approach (Jones and Kijima, 2018). A bottom-up approach means the development is focused on the local's needs, capacities and perspectives (Ray, 2006). The importance of local participation in the design and implementation phases is emphasized, by valuing cultural-, environmental-, and community aspects. This is further elaborated by Barbero (2019) which highlights systemic design principles for economic development in rural areas. The principles cover, for instance, the importance of relationships in rural areas where urban linkages are lacking, the solutions must consider local needs, traditions, and culture, and the design process should involve the local inhabitants. These three aspects are also emphasized by Hubert, Therace, and Dro (2011), who also recognize that a bottom-up approach enhances the sense of project ownership among local stakeholders.

3.5 User study in an indigenous environment

The book *Indigenous Research Methodologies* by Chilisa (2020) addresses emphasis on various research perspectives whereas research in postcolonial areas is included. This perspective is

relevant given Mozambique is a formal Portuguese colony (BBC News, 2023). Regarding conducting interviews, the author criticizes the Western academic approach, for instance, of having fixed rules and codes. The book presents considerations for conducting postcolonial interviews, whereas some are:

- The interview should be guided by the participant's relational way of knowing by celebrating the participant's connection to each other.
- Show respect by presenting yourself and the reason for conducting the interview.
- Enter the interview as a dialogue to make the participant feel comfortable and relaxed.
- Using cultural and symbolic artifacts can help to trigger expressions and reveal hidden aspects such as world views. It is often used in focus groups, for instance, in talking circles, where each participant takes turns expressing themselves and shows respect to one another.
- The participant should decide whether to include their name in the report or not. In postcolonial indigenous research, participants tend to lose their power if their name is not included.
- Conduct the interview at a comfortable setting where the participants feel safe expressing themselves, for instance, having an appropriate distance from one another and avoiding distractions such as direct sunlight or noise.
- Ask simple questions and avoid loaded ones.
- Avoid sensitive questions at the beginning of the interview.
- Be an active listener.
- Give the participant time to answer, where silence is welcomed.
- Watch for signs of embarrassment or boredom and adapt the following questions for avoidance.

In addition to this research, the thesis author has taken two intercultural courses at the Royal Institute of Technology (KTH) to gain more cultural and behavioral knowledge as preparation for this study.

3.6 Design methods

The project was divided into three phases: Pre-study, Field study, and Concept Development, as seen in Figure 3. The Pre-study includes the methods of desktop research and in-depth interviews. The Field study in Moçambique involves another session of In-depth interviews, followed by Pop-up interviews, Observations and A Day in the Life, In-depth interviews session 3, Value network mapping, and Stakeholder analysis. The methods, co-creation workshop, and brainstorming are part of the last phase, Concept Development.

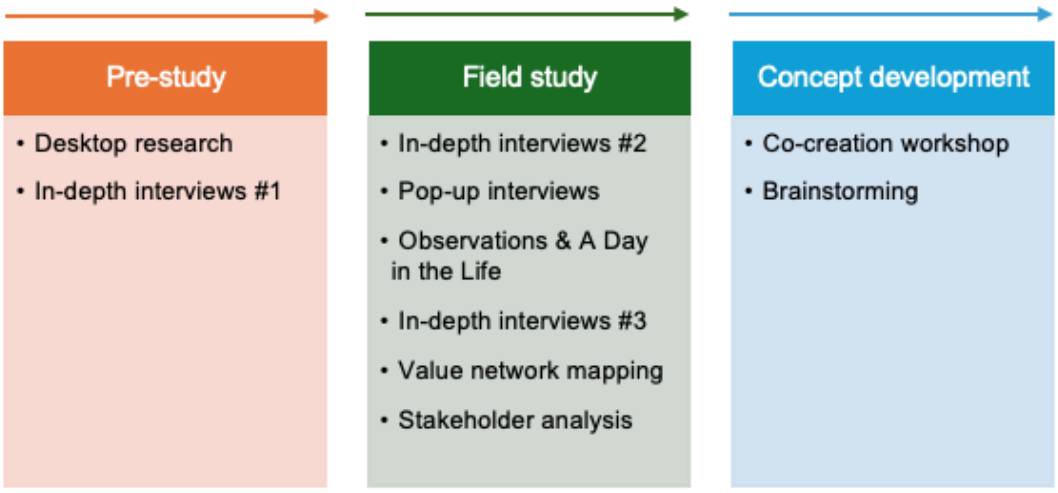


Figure 3. The three phases and involved methods of the project.

4. Pre-study

The following chapter presents the methods, results, and analysis of the pre-study. The study was conducted as preparation for the field study to gain insights on performing projects in developing countries, how to spread projects to new locations, and project-specific aspects to consider.

4.1 Methods

The pre-study was carried out by initial desktop research of relevant topics and thereafter several in-depth interviews with project-relevant Swedish actors with various professions.

4.1.1 Desktop research

Desktop research was done as a way of gaining information on topics based on related secondary data (Stickdorn *et al.*, 2018). This method worked as a starting point of the project, for instance, to avoid doing existing work. It also supported framing the research questions and selecting the methods. The investigated topics were Moçambique in general, previous projects made by Project Vita, and lastly previous community center development projects. The desktop research mainly included sources from the database KTH Primo, along with other search engines, to find relevant scientific reports.

4.1.2 In-depth interviews

To receive qualitative data as preparation for the field study, interviews were held with four various Swedish actors and the thesis's partner organization Project Vita. Information on the five interviews is seen in Table 1. The conducted method was In-depth interviews to get deep knowledge in specific areas (Stickdorn *et al.*, 2018). The interviews were held online since the participants were located all around the world. The structure followed a donut principle, where there is a main question to be answered and the individual questions are divided into themes.

Table 1. Interview information including the interviewee, their role, the duration, and the mode of the interview.

Interviewee	Role	Duration	Mode
Project Vita	Founder	90 min	Video call
ICLD	Research Coordinator	50 min	Video call
SIDA	Case manager	40 min	Video call
Kvinna till Kvinna Foundation	Manager	30 min	Video call
Previous field study student	Engineer	30 min	Video call

Firstly, an in-depth interview was conducted with the founder of the partner organization Project Vita. The purpose of the interview was to gain an initial picture of the situation in Linga

Linga, Moçambique, and the community center project. The interview guide is seen in Appendix A.

Based on the background research, it was desired to get more information about social innovation projects in Africa, community center initiatives, and how to spread projects to other locations. Therefore, an in-depth interview was held with ICLD (Internationellt Centrum för Lokal Demokrati) which is a Swedish non-profit organization working for democratic participation at a local level. The organization sees the local level as the essential starting point for long-term democratic change, by supporting local democracy, local self-governance, and decentralization (ICLD, 2024). ICLD conducts international training programs, funds research projects, and promotes partnerships between Swedish municipalities and other countries. The operating countries in Africa are Botswana, Kenya, South Africa, Tanzania, Uganda, and Zambia, but has previously been active in Moçambique.

The main question of the interview was "How to work with social innovation to accomplish wanted outcomes and expandability of initiatives?". Six themes were explored: social innovation, charity projects, community centers, project expansion, and general questions. The complete interview guide is seen in Appendix B.

The third in-depth interview was conducted with a person working for the Swedish International Development Cooperation Agency (SIDA) at the Swedish Embassy of Moçambique. The purpose of the interview was to gain knowledge of conducting projects specifically for Moçambique, how stakeholder involvement impacts projects, hear about community center projects, and how to do projects that last. This department finances projects while their partners implement the projects. The department follows the Swedish government's directions in the reoccurring five-year strategy program which consists of five sectors, in which the funded projects must fall into: Environment, Human rights, Economic development, Peaceful societies, and Research (SIDA, 2023).

This interview was held to investigate the main question: How does the embassy work with development projects and what are the important considerations? The four themes were: embassy projects, stakeholders, community centers, and development projects. The full interview guide is seen in Appendix C.

An in-depth interview was held with The Kvinna till Kvinna Foundation to understand how to spread projects and how to adapt methods to the context. The interview guide is found in Appendix D. This Swedish, non-profit foundation works for global women's rights by strengthening female organizations in conflict areas by collaboration and support, for instance, by conducting workshops (The Kvinna till Kvinna Foundation, no date). Today, they operate in Europe, the MENA region, the South Caucasus, and Sub-Saharan Africa.

Lastly, a shorter in-depth interview was held with a previous student who conducted their thesis in collaboration with Project Vita earlier. The interview intended to gain insights on how to adapt methods to the local context. The interview guide is seen in Appendix E.

4.2 Results

The results of the desktop research and in-depth interviews are presented in this chapter.

4.2.1 Desktop research

Desktop research was conducted on several topics with high relevance to the project: Moçambique, the village Linga Linga, previous Project Vita projects, and earlier community center projects.

Moçambique

Moçambique is located in South East Africa and has a land area of 799 380 km^2, around twice the size of Sweden (UI, 2022). The country's population is approximately 33.7 million, with a high population increase of 2.85% per year (UN-Habitat, 2023). Moçambique has a 38% urban population, which is projected to rise to 50-60% by 2050. In comparison, the rural population in Sweden is only 12% (*Tätorter i Sverige*, 2022).

Moçambique is a former Portuguese colonial and became independent in 1975 after ten years of armed battles where an almost two-decade-long civil war ended in 1992 (UI, 2022). The official spoken language in the country is Portuguese, but there are over 40 local spoken languages in the country (Sweden Abroad, 2023)

Moçambique is a democratic republic and has been governed by the Frelimo party ever since the country's independence, where the first free election was held in 1994 (UI, 2022). The country is divided into ten provinces and each one has one main provincial city and a governor who is elected by provincial elections (World Bank, 2023). However, corruption is widely spread and the country is ranked 142 of 180 countries in the Corruption Perception Index, a measure of corruption levels in a country (Transparency International, 2023).

Moçambique is one of the poorest countries in the World and the 7th poorest in Africa (UN-Habitat, 2023). 62.9% of the population live below the international poverty line of 1.90 USD per day. The situation has worsened because of the current armed conflict in the Cabo province which has resulted in large population resettlements which has pressured other locations to provide housing and basic services.

The fertility rate has declined from the top noted levels in the 1970s of 6.7 per woman, to 4.6 per woman in 2021 (World Bank, 2022). However, there is an increasing youth population, where children under 15 years old stand for 45% of the country's population, the median age is 17.2 years and the age expectancy at birth is 59 years (CIA, 2024). Among the children, almost 94% enter elementary school (UI, 2022). However, the country's proportion of literate is still relatively low around 63% in 2021.

In 2021, Moçambique was seen as the most vulnerable country in Africa to climate change (UN-Habitat, 2023). Natural disasters are common such as cyclones, floodings, and storms, due to the long coastline, geographical location, and large lowlands whereas 1/3 of the population lives in coastal cities. In 2019 the tropical cyclones Idai and Kenneth struck Moçambique, which is the first time in recorded history two cyclones appeared in the same season where Idai was the largest cyclone in two decades (Unicef, no date). These weather events caused severe damage to communities where 223 947 houses were destroyed and 160 927 people were displaced. The country is increasingly addressing climate change, however, there are still many challenges to building the resilience capacity (UN-Habitat, 2023). These challenges stem from competing matters, limited resources, and technical capacity.

31% of the population had access to electricity in 2021, but only 4% in rural areas, meaning 19 million people were without electricity (UI, 2022). The main occupation in Moçambique is agriculture, covering 70% of the working population. Important export products are shrimp, cashew nuts, cotton, sugar, citrus fruits, wood, energy, and aluminum.

Previous Project Vita studies

Project Vita has since 2018 collaborated with Swedish University students in carrying out field studies in Linga Linga as part of thesis projects. Various topics have been executed, with the common goal of helping people in rural areas. Two previous projects are particularly related to this thesis project.

In 2023, Emilia Säll Magnusson and Melchior Hahn designed an off-grid solar solution for Linga Linga's planned community center (Säll Magnusson and Hahn, 2023). By conducting interviews with the local population, local leader, and the founder of Project Vita, the electricity demand was estimated from identified desires requiring electricity. The expressed desires were a meeting area, a marketplace for fruit and veggies, a study area for children, fish storage, electricity, and computer access. The expressed needs from this project are seen as inspiration rather than fixed user needs since the focus was on identifying the electricity demand.

In 2018, Erica Eriksson and Sofia Johansson Gunnarsson investigated household problems people experience in Linga Linga and potential solutions to them (Eriksson and Johansson Gunnarsson, 2018). The problems were identified based on the people's expressed opinions. The problems were mapped out and weighted by expressed importance among the participants. The largest identified problems were heat in households, lack of light and food cooling systems, unsafe households due to weather, smoke from cooking, limited transport options, and toilets.

Community center projects

Five community center projects at various locations of the world are presented below.

Tsoga Environmental Center, South Africa

In 2006, the organization Tsoga built a community service and recycling center in the poverty-stricken region of Cape Town, South Africa (Holcim foundation, 2008). The project aimed to combine environmental sustainability with social equality for the disadvantaged community. The building offered a meeting area, educational events, recycling facilities, and community services for bringing poor people into society and fighting social inequalities. The building itself was constructed with technical, environmental-, social-, financial-, and cultural value. The center was built with local resources and labor to stimulate the local economy, is energy efficient, brings a local identity, creates jobs, and trains people about sustainable construction in the building process.

The authors point out that previous projects have been met with public resistance when not involving the community stakeholders in the development process (Holcim foundation, 2008). Therefore, by discussions and workshops, the expectations and requirements for the center were determined by involving local stakeholders.

Mymensingh City, Bangladesh

In Gondrapa Village of Mymensingh City in Bangladesh, a community center was designed and built for the urban poor living in a densely populated area (Lee and Na, 2019). The center intended to strengthen the capacity of the community, improve the local sanitation level, enhance gender equality, and develop a better urban environment. The center included a community office, a library for children, a conference area, and a sanitation area for women. In addition, the construction materials were locally sourced.

The design process included three workshops where the first one involved experts and residents (Lee and Na, 2019). The workshop concerned the initial design which was adjusted to fit the needs and culture of the residents. Lee and Na (2019) mean the participatory design and execution with the locals contributed to increased pride and consciousness among the residents.

Xihe Village, China

In 2014, a community center was designed and built in the rural Xihe Village in China (Ren, 2016). The village struggled with poverty, internet access, and infrastructure to the nearest city, meaning there was a rapid rural-urban movement. The project aimed to enhance social sustainability and resilience. The center included a village hall, shop, restaurant, guest room, office, administration space, square, and mini-museum.

The project had a large focus on participation design by involving the inhabitants in the building process. Ren (2016) points out the participatory construction process as the main positive impact on the inhabitants rather than the finalized center. Moreover, the design team engaged with the inhabitants which led to mutual trust and learning between the locals and the architects. The locals felt a belonging to the center since the architecture respected their culture, traditions, and beliefs. A defined issue with the project was the uncertainty of who would own and control the center which led to limited accessibility for the public.

Leeds, UK

Thornham and Parry (2015) investigated the community center's symbolic value, actual use, and social and political role. Three community center examples in Leeds in the UK were compared. In the first one, local entrepreneurship and community activities were endorsed. There was a division between individualism usage and community engagement. The second center focused on individualized activities such as co-working spaces. The third center had multi-purpose spaces rather than focusing on specific activities. Observations suggested a stronger community engagement and feeling, by the collective support.

A department of the Council in Leeds saw local ownership and management of a community center as a way to develop many various activities (Thornham and Parry, 2015). The department called it "…social enterprise and there are lots of spin-off benefits" (page 29).

Hesse, Germany

After World War II, Germany rebuilt war-torn cities, such as Hesse, to improve the damaged infrastructure, and support dislocated people (Hall, 2022). The tough situation in rural areas had led to an unsustainable rural-urban migration where rural community centers contributed to making life easier by enabling access to modern amenities and leisure activities, mechanizing labor, and strengthening democratic principles. The centers reduced the inequalities between the rural and urban areas regarding social, cultural, and economic aspects.

The goal of the program was to give citizens of Hesse a feeling of individual responsibility, while at the same time valuing the community (Hall, 2022). The community centers also supported local participation where the communities customized the centers to meet their own needs. Public meetings created a free space for the locals to meet elected representatives. Thereby transforming the rural citizen's negative view of the government, into a more democratic perspective with a strengthened trust.

4.2.2 In-depth interviews

The results of the five in-depth interviews conducted in the pre-study are presented below.

In-depth interview with a representative from Project Vita

Linga Linga is a rural village on a peninsula in the Inhambane province and is a part of the Morrumbene district. The coastal village has around 1200 inhabitants and the two main spoken languages are Portugese and the local language Gitonga. Linga Linga is a chieftaincy, meaning one local leader represents the village, for instance, in district matters. He talks to a representative in the district who is responsible for all the villages on the peninsula. In addition, certain activities must be approved by the leader, for instance, if opening a new bank account or verifying a person. The local leader also has two assistants, each responsible for one of two zones: Linga Linga A and B, which are divided by the main road.

Some households in Linga Linga are connected to the national electricity grid while more people have various small-scale solar solutions with limited daily electricity access. The necessities are limited within the village. Several small kiosks are selling preserved groceries such as pasta, drinks, and canned food. There is also a school with grades 3-7, a health center, a maternity clinic, several churches, a hotel, and several large house rentals. In the adjacent village, there is a school for students in 8th grade. The closest city Morrumbene, 16 km from Linga Linga, has a wide range of offerings including pharmacies, a hospital, restaurants, grocery stores, marketplaces, gasoline stations, mechanics, and schools including a University. The road connecting Linga Linga and Morrumbene is a dirt road with low maintenance frequency. It is highly exposed to tough weather such as heavy rain, leaving it in an extremely poor condition where it is challenging for cars to drive. Very few people have access to a car, so most people either walk the long distance to Morrumbene or go by Chapas, a shared car service that requires payment, where people stand on the car wreck, sometimes as many as twenty people. The infrastructure of Linga Linga is poor, where the bad road from the city is expressed as the main infrastructure issue.

Linga Linga lays on a sand bed due to its coastal location. This ground makes it difficult to utilize for agricultural purposes. Therefore, other occupations are common. Two common occupations for men are fishery and construction, while the women often carry fish to the Morrumbene market and take care of their children and household. However, in Linga Linga the main occupation for women is bag knitting of palm leaves which is mainly sold in the tourist regions of Tofo, 2 hours from Linga Linga.

The founder of Project Vita, Michel Olofsson, emphasized the importance of having a holistic approach, targeting several problems simultaneously. The aspect of knowledge sharing is also relevant to the project and community of Linga Linga. He mentioned the problems and resources in Linga Linga are shared by other coastal communities in Moçambique because of the country's long coast.

It was pointed out, that charity projects are often not based on the local perspectives, it is rather based on large-scale and complicated solutions which look good in practice but do not work in reality. Michel also emphasized the importance of having micro solutions instead of macro-ones, to have a fast and efficient positive impact on people's life. He said a larger perspective is required and expressed the difference as:

"It is about developing a simple solution to a large problem."
- Michel Olofsson

One of the purposes of the center, which is planned to be built in the fall of 2024, was to integrate previous Project Vita projects into one location so the citizens will see the actual results. Thereafter, other villages will witness the solutions and try to incorporate them. The project should also inspire other NGOs to imitate the community center in other locations.

Concerning the community center, Project Vita will be responsible for the development and building process. It will be built with local actors and be funded by contacts of Project Vita. The center will also be owned by the organization, which was expressed as crucial. However, there is a plan to train some people to operate it.

The main challenge for Project Vita's organization was the financials and the staff. It was desired to have a larger team to expand the marketing and contact network. Project Vita should inspire other organizations and influence them, for instance, communicating Project Vita's projects which may apply to the embassy's ongoing efforts in a similar case. The private sector is easier to cooperate with since their processes are quicker compared to the embassy. Another challenge today is the planning of pre-studies. It could be improved by defining the needs of the citizens, what partners to involve, and securing the financials and the project team.

The information that Project Vita wants to spread is done by talking to the local leader who will bring it forward to the citizens. When spreading solutions to the citizens the instructions are often held to each household separately which is highly time-consuming.

The community center should be a place to create interest in the district and also other actors such as companies. The center could also enable more volunteers to join Project Vita's work, which he calls "sustainable tourism" since there will be more specific projects. Another expressed benefit of the center is gathering people at the same location which makes meetings and workshops easier for students. People may also want to join these activities due to a pleasant meeting area.

The citizens of Linga Linga should feel the center is like their home and it belongs to them, for instance, by involving local building actors, to make it easier for the locals to utilize the center. He also emphasized that the local citizens often help each other within the community.

In-depth interview with a representative from ICLD

For ICLD, social innovation is within the field of local democracy. Social innovation can be developed by citizen assemblies, crowdsourcing, and the use of technology to collect input from citizens which can improve their capacity to influence decisions. ICLD defines social innovation as:

"How to improve citizen's ability to influence the decisions that are affecting their lives."
- ICLD

It was said during the interview many charity organizations want to tackle one certain problem, but these are often complex problems, called wicked problems. Once dealing with these problems, other issues can arise. These problems do not get solved within a short time, there are long-term problems where you need to look at complex issues holistically.

ICLD mentioned today, that it is a huge matter that projects do not last and there may be different reasons for this, for instance, corruption, too short projects, and too top-down approaches. The representative from ICLD expressed the importance of needs originating from the people the project will be affecting, a so-called bottom-up approach. In addition, integrating projects into systems and policies is a vital part of long-term change, to not be dependent on individuals. However, passionate individuals may in some cases also be crucial in making projects last. Furthermore, it is important to show the mutual benefits to the different stakeholders e.g. an enhanced local economy may be appealing to politicians.

It was said that co-creation is important to make things sustainable and helpful, where involving various stakeholders is needed since they all have different perspectives that matter. An accurate quote from the interview was:

"It is difficult to co-create - but it is necessary for sustainability."
- ICLD

ICLD had heard a desire for a community center from three partnered countries. Decentralization is the main focus area by shortening the distances to necessary services which many people access by foot without a certainty of completing the desired service, which damages the trust of the involved actors. It was mentioned, that a community center can save people time and money while developing more trust in other actors. A similar project in Zambia saw the spreading of information, for instance, through computer access, as the main purpose of a community center. An expressed potential challenge with community center projects is the coordination between the center and the outside stakeholders. To face this, it is important to implement a structure by, for instance, weekly meetings with involved stakeholders.

Spreading and expanding projects to other locations are often discussed within ICLD. According to them, it is vital to capture the good results and benefits, make easy visualizations, and develop a summarized one-page policy. In addition, inviting outside people to the developed project's location is good for visualizing the benefits. Moreover, understanding the other location's political culture and structure is important since it varies a lot. Lastly, ICLD often develops toolboxes to be utilized by others, where it is important to be clear what the project is about, why it is useful, how it is done by visualizing a simple step-by-step guide, and finally define aspects to be aware of.

In-depth interview with a representative from SIDA

SIDA, which is heavily working on strengthening rural areas, expressed the importance of local stakeholder engagement for making projects last and establishing long-term change. One raised example was training local technicians to operate the project in the long term. Another important consideration for sustaining projects is the sense of ownership to make local leaders and the community feel it is their project and that they are included. In addition, meeting the expressed needs. SIDA also mentioned there is never a case where all stakeholders are completely satisfied, it is rather a question of balancing the different opinions.

According to SIDA, there are many factors to consider when trying to expand projects to other locations such as differences in religion, culture, and ethical composition. These factors also differ within Moçambique. It was also added that the solution may be identical, but the implementation might differ.

In-depth interview with a representative from Kvinna till Kvinna Foundation

The Kvinna till Kvinna Foundation mentioned it is important to make the participants feel safe and free to express themselves during user studies. Due to societal structures, it is worth considering dividing men and women, and also younger and older people during group sessions. In the methods, oral elements are preferred rather than written, and practical elements rather than theoretical. It was also suggested to map out different groups of the community, for instance, females-, youngsters-, and interest groups, where the needs and desires may be linked to these. On the topic of spreading projects, The Kvinna till Kvinna Foundation expressed it as:

> *"The questions may not differ, however, the answers may be different."*
> - The Kvinna till Kvinna Foundation

In-depth interview with a previous Project Vita thesis student

The previous thesis student suggested separating women and men during workshops to make the women feel comfortable expressing themselves. It is beneficial to make the workshops simple and fun to avoid pressure on the participants, for instance, by using visuals and including drawing elements.

The interviewee experienced that participants felt that they had to find a "right" answer. Therefore, it is good to try to avoid this when structuring the interview. It was also recommended to start with social questions and ask short, simplified, and open questions to make the participants feel comfortable. Due to the language barrier, some answers will be lost via the interpreter. Therefore, observations of behavior and facial expressions can generate valuable insights. Lastly, it was suggested to have a planning meeting with Project Vita to structure the user study.

4.3 Analysis

Here, the results of the desktop research and the in-depth interviews are discussed and compared to the theoretical framework. The key takeaways, which are brought to the upcoming project phase, are also presented.

4.3.1 Desktop research

Connections can be seen between the practical community center projects and the theory regarding rural development. The scholars Jones and Kijima (2018) pointed out the lack of infrastructure which was also the case for the rural Xihe Village in China. In addition, the negative attitude towards the government and lack of democracy were both expressed by Jones and Kijima (2018) and the practical case of Hesse in Germany.

Among the previous community center projects there is a general participation of locals, however, there is a lack of transparency regarding how the needs of the center were defined. This means there is no evidence of a complete bottom-up approach. The development of the centers in Tsoga, Mymensingh City, and Xihe Village, only considered participational design concerning the architecture and using local labor in the building process.

Takeaways desktop research:

- The main purpose of all the community center projects was mainly socioeconomic strengthening.
- It is important to involve the local stakeholders in the development and operation of the center.
- All projects share a common issue: a lack of transparency of what the stakeholders' needs were and the methodology of how they were defined.
- Previous offerings of community centers have been:
    - Meeting area/hall
    - Educational space
    - Library for children
    - Shop
    - Restaurant
    - Office
    - Multi-purpose space
    - Access to modern amenities

4.3.2 In-depth interviews

Project Vita mentioned citizens of Linga Linga often help each other, which can be connected to the intercultural awareness research where Moçambique is defined as a country with collective societies. Another aspect that can be linked to intercultural awareness research is the use of a leader of the village, which is a sign of a hierarchical society.

Both Project Vita and ICLD mentioned the importance of facing complex issues holistically where they also saw the issue of other organizations tackling one certain problem.

Project Vita, ICLD, and SIDA expressed how crucial it is to have a bottom-up approach, facing the local's needs and utilizing participation in the process, for making projects work in reality

and last in the long term. This can be linked to the desktop research, where several community center projects also emphasized this.

Project Vita and SIDA both mentioned the importance of making people feel ownership of the center and that it belongs to them, to make the project last and have the desired outcome. This consideration was also seen in the desktop research regarding previous community center projects.

SIDA and The Kvinna till Kvinna Foundation had slightly different thoughts regarding applying solutions to new contexts. SIDA mentioned the solutions may be identical while the implementation approach differs. In contrast, the Kvinna till Kvinna Foundation said the solution may differ whereas the problems may be similar.

Takeaways in-depth interviews:

- A potential challenge with community centers may be the coordination of stakeholders which can be resolved by regular meetings.
- To face highly complex problems, a holistic perspective is required.
- To create long-term solutions, it is important to: have a bottom-up approach, co-create with the stakeholders, meet the expressed needs, local stakeholder engagement, create a feeling of ownership, integrate the project into the system, and express the various benefits to the different stakeholders.
- To spread a project to other locations it is important to invite external people, understand the political culture of the other location, and develop an easily understandable toolbox. The toolbox should express the benefits, use visualizations and step-by-step guides, and define what to be aware of.
- Many factors are to be considered when implementing a project in a new location. However, the solution or problems might be identical.
- Find ways to make the participants feel comfortable expressing themselves during the user study.
- Map out different community groups since the needs might differ between these.
- Make the workshops simple and fun. Consider dividing men and women to enable free speech.
- Ask short, simple, social, and open questions during the interviews with the locals.
- See observing as an opportunity to find insights during the interviews.

5. Field study

This chapter includes the methods, results, and analysis of the 8-week-long field study in Moçambique. This phase was about understanding the users and other stakeholders of the community center deeply, which should be the base for the upcoming concept development phase. The insights from the pre-study were brought into this phase.

5.1 Methods

This section presents the methods of the field study which can be visualized in Figure 4. The methods are connected to service design and systematic design. To address reflexivity in the relational-, corporeal-, and culture categories, Vink and Koskela-Huotari (2022) identify several applicable methods. Incorporated methods connected to relational reflexivity are Interviews (In-depth and pop-ups), Stakeholder maps, Value network maps, Ecosystem maps, and Co-creation workshops. The in-depth interviews were divided into two sessions, firstly with Linga Linga citizens and secondly with the leaders. Methods within corporeal reflexivity include Observations. The authors point out that some methods within cultural reflexivity may not be culturally appropriate. Therefore, the suitability of the methods below has been assessed from the intercultural awareness research and by consulting with the local contact person in Moçambique. In the mode of culture reflexivity, the utilized methods are a Field study and A Day in the Life.

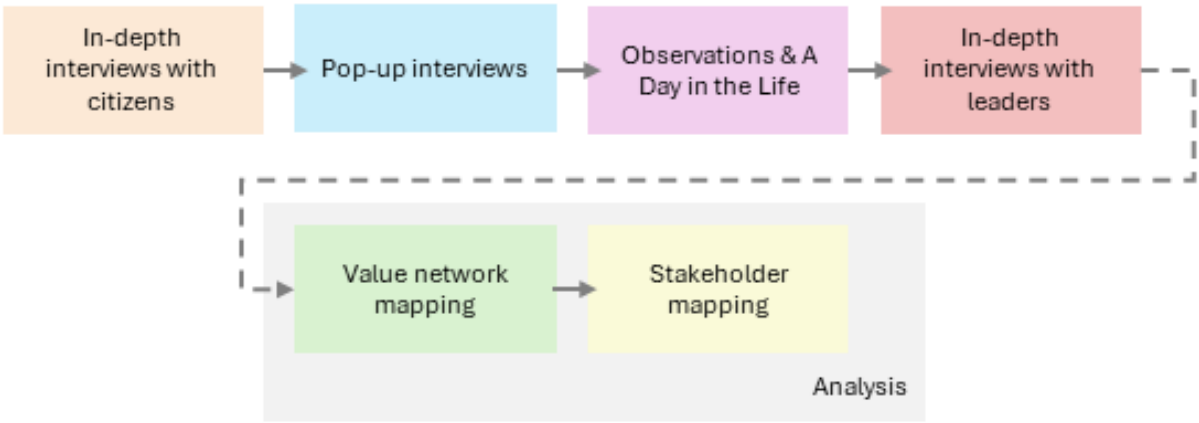


Figure 4. The design process and methods used in the field study.

5.1.1 In-depth interviews with citizens of Linga Linga

In-depth interviews were conducted with four citizens of Linga Linga to understand how it is to live in the rural village. The aim was to understand their perspective about drivers and challenges in their everyday life, common activities, their movement, communication channels, and identify key actors in the village. The data was analyzed through clustering into themes and color coding. The participants were two men and two women, with various ages ranging from 19 to 45 years old, as seen in Table 2. An interpreter translated the asked questions from

Swedish to Portuguese and the interviews took place at people's homes and workspaces. The main question to be answered was "How are people's daily lives and how can they be improved*?"*. The interviews consisted of six themes: demography, daily activities, communication, ease activities, Linga Linga community, and community center. After the interviews, the answers was clustered to analyze the results, find patterns, and insights.

Table 2. Information about in-depth interviews including the interviewee, their role, the duration, and the mode of the interview.

Interviewee	Gender	Age	Occupation	Duration	Location
P1	Women	41	Housekeeper hotel	30 min	Their home
P2	Man	20	Seamster	20 min	Their home
P3	Women	45	Housekeeper, bag weaver	30 min	Their work
P4	Man	19	Janitor	30 min	Their work

5.1.2 Pop-up interviews

Due to the broad project topic, pop-up interviews were performed to cover a larger quantity of people living in Linga Linga. The purpose was to understand certain people deeply, become aware of networks and actors, and comprehend people's experiences (Stickdorn *et al.*, 2018). The goal was to interact with a wide range of people, of various ages and occupations. But also to talk to the key people identified from the in-depth interviews with Project Vita and the Linga Linga citizens. Similar to the in-depth interviews, the data was analyzed through clustering into themes and color codes. Interviews were held with people carrying out the main occupations in Linga Linga, to get diverse perspectives which can be seen as an overall representation of the whole village. Eleven people with various occupations participated, 3 men, 4 women, 2 boys, and 2 girls, as shown in Table 3. The interviews were 10-15 minutes long and were held at people's households or workspaces.

Table 3. Information about pop-up interviews including the interviewee, their role, the duration, and the mode of the interview.

Interviewee	Gender	Age (~)	Occupation	Location
P5	Women	30	Kiosk seller, bag weaver	Their home
P6	Man	30	Teacher	Kiosk
P7	Man	20	Unemployed	Their home
P8	Girl	17	Student 7th grade	Their home
P9	Boy	15	Student 7th grade	Their home
P10	Girl	17	Student 7th grade	Their home
P11	Women	27	Baker, hair braider, bag weaver	Their home
P12	Man	35	Administrator hospital	Their work
P13	Women	20	Bag weaver	Their home
P14	Man	50	Kiosk owner	Their work
P15	Women	42	Fish carrier, bag weaver	Their home

The questions which gained interesting inputs from the in-depth interviews formed the questions to the pop-up interviews. However, the questions varied depending on the participant and the obtained answers. The general asked questions were:

- Where do you live?
- What do you do for a living?
- Do you go to other places outside of Linga Linga?
- Is there something you would like to learn?
- Do you miss something in Linga Linga that you would like to access?
- There is a plan to have a place in Linga Linga to offer different services. What would you like to have and do there?
- Do you have any thoughts about your future?
- Lastly, what do you think is fun to do?

The household locations of the participants in the pop-ups and in-depth interviews are seen at a map of Linga Linga in Figure 5. The figure also presents the locations of the available facilities including the kiosks, bar, school, hospital, and the planned community center location.

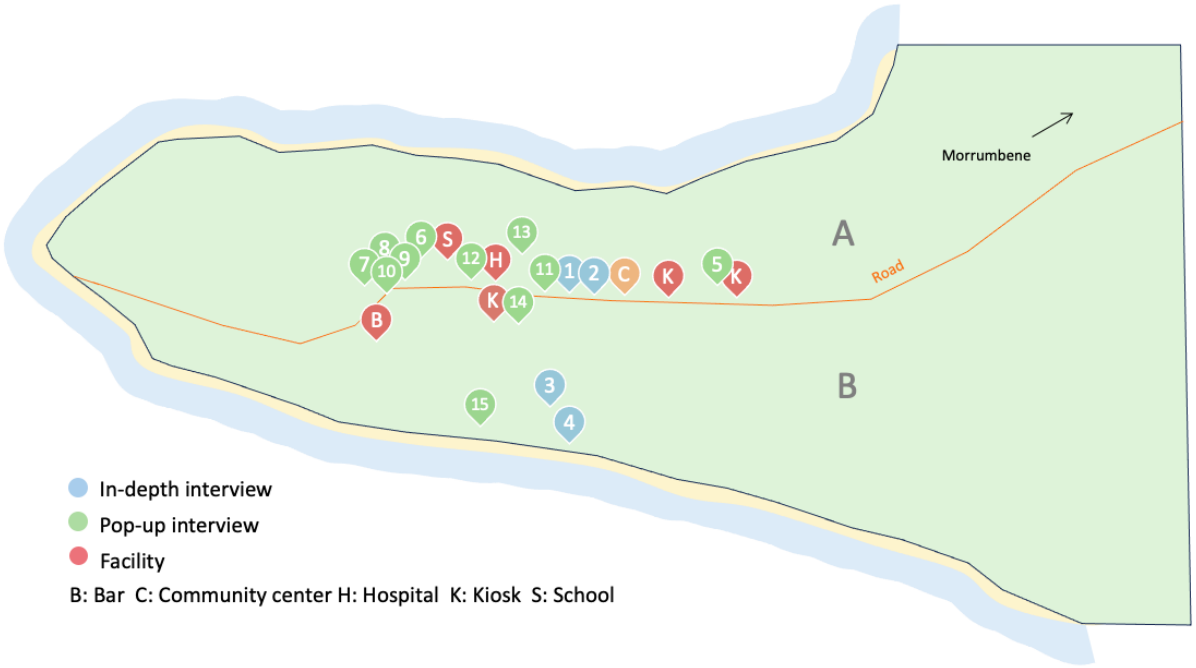


Figure 5. Map of Linga Linga, the facilities and the locations of the interviews.

5.1.3 Observations & A Day in the Life

Observations were done at various locations and occasions to get insights based on non-verbal actions (Stickdorn *et al.*, 2018). Observations were conducted at locations of interest based on the interviews and had various durations, see Table 4. The first locations were at people's homes and other locations in Linga Linga to gain a more developed picture of the living conditions, daily activities, and movements in the village. Secondly, participation and observation of a local community party, to see people's behavior during non-work-related activities. In addition, a

community meeting with the mayor was observed to see how people behave to the authority and the other way around. Moreover, a market in Morrumbene and Maxixe was observed to further understand the most desired element from the interviews.

A more extended observation method was executed, called A Day in the Life. This method was a way to reveal hidden issues, behaviors, and activities associated with people's daily routines (Milton and Rodgers, 2013). Activities, experiences, and environments were captured. In A Day in the Life, a person is followed during a whole day or at least parts of it. In this case, a woman carrying fish for a living was the main person being followed since this is a common profession among women in Linga Linga. In addition, the fisherman was a sub-target of the method. The activity lasted for five hours with the woman at where she buys her fish. The method was combined with an interview to gain insights that observations alone may not cover.

Table 4. Information of the observations including the observees, duration, and location.

Observee	Duration	Location
Homes & facilities	Several occasions	Around Linga Linga
Community party	3 hours	Kiosk
Community meeting	1 hour	Under a large tree in Linga Linga
Markets	Several occasions	Morrumbene & Maxixe
Fish carrier	5 hours	Work

5.1.4 In-depth interviews with leaders of Linga Linga

An additional three in-depth interviews were conducted with people with various leadership positions in Linga Linga: the leader assistant, local leader, and the head of locality at Morrumbene district, as seen in Table 5. The reason was to gain other perspectives, the leader's needs, and needs which the citizens may not be aware they have. The interviews with the local leader and his assistant were held in Linga Linga while the interview with the head of locality of Morrumbene district was done via voice call. The data was analyzed through clustering into themes and color codes.

Table 5. Information of the in-depth interviews with the leaders.

Interviewee	Gender	Duration	Mode
Leader assistant	Man	45 min	In-person
Local leader	Man	45 min	In-person
Head of locality Morrumbene District	Women	25 min	Phone call

5.2 Results

The results from the in-depth interviews, pop-up interviews, and observations are presented here.

5.2.1 In-depth interviews with citizens of Linga Linga

All four participants travelled to the nearest city Morrumbene, to buy food but also to buy material for work as the seamster said. They generally travel further to the even larger city Maxixe or even Inhambane, around 2 hours away. The trip to Morrumbene occurs once a month for the women while it happens more often for the two men. The transport mode is by Chapa. The complete responses of the interviews are seen in Appendix F.

Women 45 and Man 19 shared things with others. The woman sometimes shared rice or kitchenware with her neighbors while the man only lent out things to people he trusted. Concerning information about events in Linga Linga, the sources often derive from contacts, but the delivery varies between personal meetings or through the phone. When it comes to information regarding Moçambique, Woman 41 and Man 19 heard it from a local radio channel while Woman 45 used Facebook. They did not see gaining any additional information as relevant except Woman 45 who wanted more information from her family.

All interviewees shared ideas with close ones. Women 41 interpreted the question differently and said she would go to an older man in case she had a problem with her sister. If the interviewees need assistance with something practical, such as renewing an ID card, they go to the local leader's house.

Concerning what was difficult or not pleasant to do in their lives, Woman 45 mentioned having problems with someone and talking bad about people. Man 19 said fishing because he did not know how to swim. All participants clearly said they wanted to learn something, where two interviewees mentioned English while the other two said sewing and music.

Women 45 and Man 19 met people in larger groups, for instance in church, parties, and football games. Man 19 also mentioned the occasions when people steal something. In this case, a group of people is gathered to discuss and solve the situation with the involved members where the local leader and his assistant are often included.

All except Man 20 missed a market in the village where they wanted to buy groceries which they can mainly access in Morrumbene, for instance, vegetables, cava leaves, and tomatoes. Man 20 and Women 45 expressed the need for a marketplace as:

> *"A market so I do not need transport. To make life easier."*
> - Man, 20

"It would not be as much suffering."
- Women, 48

The interviewed women would also like to sell goods themselves. Women 41 had leaves in her garden she wanted to share and she was willing to pay to be able to sell her resources. She also mentioned a market is a fun place to socialize. Women 45 saw the possibility of selling clothes at a market.

The participants also desired better electricity access. The reason for it differed, where Man 20 wanted electricity for a music studio, and Woman 45 thought other people needed an incubator and a freezer. Women 41 even possessed a fridge and a freezer, however, not the electricity. She also mentioned if electricity was more accessible her sons from the capital could work here. Other missing elements in Linga Linga were a school above 8th grade, an emergency hospital, and good water sources. Man 19 also mentioned a small school for computer access to be able to get information. He had never used a computer before, but he had seen his boss concentrate on the computer and therefore he thinks computers do important tasks.

All except Man 20, thought working was fun to do since you earn money, while he mentioned singing and playing music as fun. Man 19 said he wanted to buy a house in the future.

5.2.2 Pop-up interviews

All participants quickly and distinctly answered yes to the question about if they wanted to learn something. However, the answers varied. The most common answer was to learn English, which the teacher also said would be good for people to learn. Others wanted to learn French, carpentry, teaching techniques, and better cooking. The teacher mentioned learning sewing as a good work opportunity for people. Woman 27 wanted to learn how to use a computer since many jobs require this. Women 42 got the question if she had any knowledge that she could teach others. She brought up bag weaving since many people in the village do not know this craftmanship. She saw teaching this as a favor and did not want to get paid.

All participants except one, clearly stated they missed things in Linga Linga, where the main need was a marketplace to buy groceries. In the pop-up interviews, 7 out of 11 desired this place, where the hospital administrator expressed the cost aspect as:

"It costs me 120 MZN to go back and forth to Morrumbene and then I buy food for 500 MZN. I would rather buy food for that 120 MZN here. More practical."
- Man, 35

It was also desired to sell things at a market. Woman 20 mentioned she would like to sell her bags personally to the customers instead of using an intermediary because there are benefits such as earning more money. She also wanted to sell clothes. The fish carrier said she would rather like to sell her fish in Linga Linga where the buyers from Morrumbene could buy it from her. The reason why she currently does not sell fish in the village is the absence of a market.

The second most expressed need in the village was electricity. The participants expressed various electricity needs, for instance, Women 27 wanted to use a hairdryer and baking machine for work. However, access to a freezer was the most cited purpose among the ones who expressed a reason. The kiosk owner wanted electricity for his business to cool down drinks, ice cream, meat, and fish. The fish carrier also wanted a freezer to store her fish at the market to prolong the lifetime and sell the fish both frozen and fresh. Today she owns a freezer where she stores fish and she even rents space out where people pay per stored kilogram. However, she would like to extend her storage to also freeze other goods such as chicken and ice cream and she was willing to pay to be able to sell her fish there. She mentioned that in the market in Morrumbene, it costs 10 MZN per day. She also saw an opportunity to sell tomatoes, onions, and cakes which she could buy in Morrumbene and thereafter sell for a higher price in Linga Linga. Lastly, she mentioned she sells a full bucket of fish for 1000-1500 MZN and earns 200-300 MZN per bucket including the cost of transport and paying the fishermen.

The third most expressed need was a school above 8th grade. Woman 30 and Man 35 were concerned that children must travel the long distance to Morrumbene to study in 9th grade and above. The man said:

"Some children cannot afford the transport to Morrumbene for studies. The ones who can, think it is hard with the transport, for example when it is raining their clothes and books get wet."
- Man, 35

Woman 27 and Woman 42 saw a kindergarten as necessary. The older women thought it was needed because many children do not have much knowledge and Woman 27 expressed it as:

"Children can learn from the age of three, however, in Linga Linga children start school when turning six years, meaning they miss several years of learning things."
- Women, 27

Other mentioned missing things in the village were a better road, a dentist, and good water sources. The teacher also desired computer access in the village to gain knowledge for adults and children. The main reason was 7th grade students only learn the theory of how to use a computer and the students become unprepared for grade 8 because there they use actual computers.

Regarding the question about what they like to do, the answers differed. The teacher and administrator thought teaching children was fun. Other responses were going to church, football, playing music, baking, weaving, farming, playing games, eating, sleeping, and going to the beach.

Regarding thoughts about the future, all participants answered from an occupation perspective. The teacher and administrator wanted to continue their studies to develop within their working field. The kiosk seller, bag weaver, and baker, all women, had future desires to change their

occupations. The students in 7th grade all said they wanted to become teachers when they grew up. All three students said they liked school and that they wanted to study more.

5.2.3 Observations & A Day in the Life

The initial observation was concerning the public areas of Linga Linga. The village consists of one main sand road where people walk and occasionally cars drive, as seen in Figure 6. These cars are mostly chapas. The kiosks are located along the road since that is the main walking path. Figure 7 shows one of the kiosks. The main travel method is walking, and bicycles are rare due to the sand surface. People also walk on paths ranging between people's yards while avoiding walking inside the private property, and often exchange greetings with each other. Women, and some men, often carry heavy items on their heads, such as wood, tools, and groceries, while walking. Women with babies carry the baby on their back, also while doing an activity such as working. This is also often done by the sister of the baby. Young children often follow their mothers to work, for instance, during farming next to the road.



Figure 6. Main road in Linga Linga.



Figure 7. A kiosk next to the main road.

The people tend to be active and outdoors during the daylight, typically from early morning to early evening. However, some people walk in the darkness without any light source. Another observation regarding time, is people's flexibility towards determined meeting times, in other words, people can be several hours late to a meeting. Concerning free time, the men seem to socialize with their friends more than the women, based on several observation occasions. Another time aspect is that the students in Linga Linga attend school for half days, either in the morning or afternoon session.

People's homes are widely spread out and there is often a substantial distance to the neighbors. The households usually include several buildings and a central element, often a tree or larger plantation to create shadow which is used as a seating area for work and socialization. The houses vary between a cement construction (Figure 8) or natural materials (Figure 9) such as weaved palm leaves, which is the most common structure. People cook their food over open

flames both outside and inside the house. Many people own chickens and ducks which are raised for eating and they are walking freely in people's yards.



Figure 8. A cement house in Linga Linga.



Figure 9. A house made of natural materials in Linga Linga.

During an open community party where food was served, music played, and 360 people attended, several observations were noted. The main participants were children since they stand for the majority of the village's population. They were dancing to the music in an engaged and expressive way, which was DJ'd by local teenagers. There was a highly joyful environment. The women were also dancing while the men did not. There was a clear division between men and women because they were separated and people were not next to their partners.

The community meeting with the mayor took place under a large and isolated tree. The women and men were sitting separately, the women sat on the ground while the men sat on the available chairs. However, some men who were late sat on the ground next to the women. There were around 50 attendees and they were all dressed up. The mayor and his subordinates were 1,5 hours late. They sat on a line of chairs on one side while the participants were on the opposite side. A picture of the Mozambican president and flag was also placed. Communication between the mayor's staff and the attendees was done by standing up and without being interrupted. The staff and the attendees acted respectfully to the mayor by not interrupting him or saying something against him. The mayor and his staff began their expressed voice by cheering "Oya Linga Linga" multiple times and raising their hand while the attendees followed this gesture. When the meeting ended the women started dancing and singing a community song in the local language Gitonga.

The marketplace in Morrumbene consisted of a long street of people selling various groceries outside. Figures 10 and 11 show the market stands along the road. There was a designated inside market space, however, it was empty. The sellers were exclusively women. Some sellers had a table and a sunshade above, while most people sat on the ground where the goods were placed in buckets or on fabric. Some sellers also walked around along the street while carrying the

groceries on their heads. Each person sold a certain grocery, for instance, solely oranges. However, many people sold the same grocery. The sellers helped each other by exchanging cash. Mainly the seasonal veggies and fresh groceries were sold on the market. Fish was also sold outside in the sun where the seller occasionally sprayed some water and removed the flies with a broom. The processed groceries were sold in the inside shops. In these shops, a person stands behind a counter and picks the desired items.



Figure 10. An outside market stand in Morrumbene.



Figure 11. Outside market stand selling veggies.

In a market in the larger city Maxixe, both fresh and processed food was sold outside at the market, as seen in Figures 12 and 13. The processed food was both sold in packages and loose weight. The market was more structured where every seller had a table and roof.



Figure 12. Outside market in Maxixe.



Figure 13. Goods sold at one market stand in Maxixe.

The common transportation called chapa can be highly crowded, reaching approximately 20 people based on observations. People sit on the side racks of the truck's bed and the rest stand

and hold on to each other. Some people bring a lot of cargo which should also fit. People seem to help each other with finding a relevant place for the cargo. Sometimes there is also a person working at the chapa, apart from the driver, with placing the cargo and notifying the driver when someone wants to go off. People are standing along the main street, usually under a larger tree, waiting for a chapa to arrive. Therefore, the chapa makes many stops to pick up and drop off people, to and from Morrumbene.

In the long-lasting observation method, A Day in the Life, a woman with the profession of carrying and selling fish was followed for half a day. She started her work in the morning at 9 by walking from her home to the fishing site at the nearby beach, carrying two buckets, as seen in Figure 14. She walked for 20 min where she sat down with other women waiting for the fishermen to get their catch, which is made along the coastline by releasing a large net and pulling it towards the beach, sweeping a half-circle motion around the shore. It was a good atmosphere between the women. Some people were also selling fruit at this meeting area. At 11 o'clock, the fishermen caught fish and the women hurried to get a good position near the fisherman in charge of delivering the fish to the women. Different kinds and sizes of fish got caught. The women observed the fish and the fisherman's actions closely. It was a small catch, so only a few women were able to obtain fish, while others, including the observed woman, did not. The ones who got fish paid the fisherman and placed their fish bucket in the shade, as seen in Figure 15. Thereafter, they sat down and waited for the next catch which occurred 1,5 hours later. At this catch, when the fish was delivered to the women, a short conflict arose. The women who got fish the first time had an empty bucket this time, meaning they used two buckets. Again, the observed woman did not receive any fish. It seemed some women were prioritized. The next catch occurred after another 1 hour and 20 minutes, where the woman got left out again. There was a total of 11 women waiting for the fish. This was the last catch of the day so the observed woman went back to her home empty-handed.

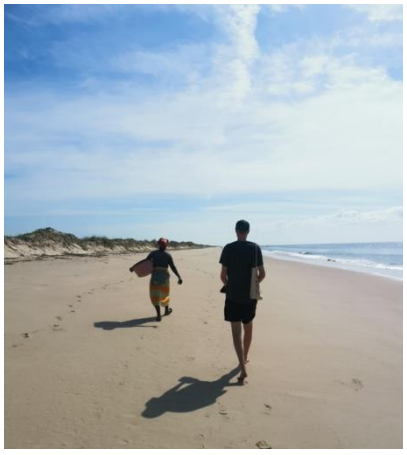


Figure 14. The women carrying fish, as part of A Day in the Life, walking to the fishing site.



Figure 15. The fish are stored in the shades between the fish catches.

During the interview afterward, she mentioned her other source of income which was weaving hats. When she receives fish, she travels to Morrumbene and sells it to people who pay per bucket. She had noticed the number of fish declining over time and the price per fish bought at the fishermen increasing. Sometimes she is not able to sell all of the fish so she brings it back home where it often rottens. She does not dry it to keep it eatable because she dislikes the taste. She found it difficult to sell fish in Linga Linga because there was no designated place for it so she must walk around in the village. She thought the main obstacle was during the days when she received many fish since she did not have a storing solution, meaning the fish rotten. This was also an improvement she desired, a cooling solution to make the fish stay fresh, which she wanted to keep in her house.

5.2.4 In-depth interviews with leaders of Linga Linga

The in-depth interviews with the local leader and his assistant led to an understanding of their work, how communication occurs in the village, confirmation of citizens' needs, new perspectives of how their work can be improved, and the linked benefits to the community. The complete responses to the leader's interviews are seen in Appendix G.

The local leader looks over the citizens, solves local problems such as family conflicts, and forwards messages from the district to the citizens and vice versa. The assistant helps the leader with these tasks. A monthly meeting is held with the leaders of Linga Linga, leaders of other nearby villages, and the district representative. Events and matters in the villages are expressed and forwarded by the district representative at a monthly meeting with the mayor.

There are different communication channels present depending on the case. Firstly, a person who has something to say to the leader and his assistant meets them at their houses. If the matter requires a larger meeting, it is held under a designated tree with the involved people. If the matter is not solved, it is forwarded to the district. Secondly, the leaders arrange community meetings regularly under a tree where information from the district is transferred, for instance, regarding election cards and events in Linga Linga. Thirdly, various departments of the district arrange community meetings, walk around on their own, or talk to the leaders who forward the information, for instance about malaria spread. Few people attend these larger community meetings, around 10-30 attendees, and the mayor has been complaining about this. The leaders wanted more people to attend since it is good for the people to learn and receive the information. The local leader sees the reason for the few attendees being people are busy with their lives and it is difficult to gather people at the same location since people live in different places. However, the leader's assistant described the meetings as less organized compared to previous meetings when more people attended. The local leader said these meetings occur twice a month while his assistant says only three times per year. They spread information regarding the time of the meeting and other important information to the people by visiting churches, the school, and walking on the street. The leader assistant previously used a microphone and hit a bucket to get people's attention when walking around. He also mentioned that some individual citizen meetings could be combined into a larger meeting to be more efficient.

The citizens of Linga Linga received information about upcoming extreme weather from a committee from Morrumbene which walked around in the village with a megaphone and informs about the approximate start date. However, people cannot prepare themselves well. People wait for the weather to be over while hiding in concrete buildings such as the school, the hospital or at neighbors, which the leader assistant did in 2017. He expressed his worry which many people experienced on days like this:

"I was scared, I was wondering if we are going to survive this. I survived but not the houses."
- Leader assistant

The leaders missed elements in Linga Linga which the other interviews also included: the absence of a market, electricity, storage of fish, a good road, and access to flowing water. A market was necessary because people did farming and fishing but there was no place for trading. Electricity was desired so the wealthier people could purchase freezers that other people could use for fish storage. The storage was desired since the fish turned bad for the women carrying the fish while waiting long times for chapas. A better road makes more transport options available for the people and lowers the price because today's bad road makes the chapas charge more and pick up many travelers per journey. The leaders agreed this influences the children studying in Morrumbene and many parents cannot afford 900 MZN per month for travel costs. The local leader said the children in Linga Linga need access to necessities that students in the cities have, for instance, computers. He also expresses education as:

"Everyone needs studies, to improve the life – more benefits."
- Local leader

The leaders thought that a community center could help them in their work by conducting the community meetings there because they do not have to borrow neighbor's chairs for seating and would not have to clean up the ground before and after the meeting. In addition, interactions could be held there instead of walking around to spread information. Moreover, the center could be a place where people know where to meet a certain person and where meetings conducted at various places can be connected to one location. Furthermore, a person in control of the center can inform the local leader of relevant happenings at the center. The local leader expressed, as the teacher also mentioned, that school activities could be arranged at the center.

The head of locality of Morrumbene district was responsible for the villages surrounding Morrumbene, including Linga Linga. The district has broad responsibility areas such as education, security, agriculture, health, and communication to the citizens. The communication is done through community meetings which according to the interviewee are held 1-3 times per month. She emphasized the importance of having a meeting at least once a month. The communication with the citizens could be improved by decentralizing the district's work by having an arm in Linga Linga representing her, for the citizens to talk to regarding certain matters, and for her to receive information more easily since today it occurs only through the local leader. Another expressed improvement would be to keep statistics of the citizens.

The district saw the largest problems for Linga Linga being the lack of a market, transport options, police station, and the bad road. A community center could mitigate these problems by introducing a market, so people do not need to use the road, establishing a police station to make Linga Linga safer, and offering energy to enable a fishing market. In addition, a retirement home would be relevant. The distribution of the market goods should be controlled by a district representative. She also said increased tourism in the area opens up more opportunities for distributing goods.

Regarding the road, she expressed the reason for the poor quality being a financing issue. Castello, a hotel in Linga Linga, has desired a meeting with the other investors of Linga Linga to solve the road matter. She saw this meeting as an opportunity to hire people from the villages to implement the construction instead of hiring an external company that is maintaining the road today.

Regarding education, she saw it important for students to learn vocational skills such as carpentry or electronics since they can practice this after school hours. A secondary school was also desired to shorten the distance for the students. She thought adult education could be improved by having a designated place for it with teachers and appliances if needed. Lastly, she had a desired vision of Linga Linga to be more developed for tourism:

"I want Linga Linga to become a point of reference in the Inhambane province, regarding tourism."
- Head of locality Morrumbene district

5.3 Analysis

In this sub-chapter, the results are analyzed and the key takeaways are brought to the next stage of the project.

5.3.1 In-depth interviews with citizens of Linga Linga

The analysis of the answers and observations during in-depth interviews are presented here.

During the in-depth interview with the janitor, it was brought up that he talks with his friends about their futures. This answer was interesting and a question about future thoughts was added to the pop-up interviews to understand how they picture their futures.

The contrasting perspective to Sweden was seen when the participants answered what they thought was difficult in their lives. No person mentioned the aspect of living in poverty or complaining about their job or the hard work involved in earning enough money for food. Two participants did not even answer the question after thinking for a long time. The participants also found working as the main fun thing to do. In Sweden and other Western countries, people may bring up difficulties such as being stressed, working too much, and the current tough

financial times. It seems it should be the other way around. However, people in Linga Linga have another mindset and are grown up in an environment where you as a child need to take responsibility at a young age.

There seems to be a hierarchy between men and women in Linga Linga because Woman 41 mentioned she solves problems concerning her sister by talking to an older man. This corresponds to the cultural value in the intercultural awareness theory in chapter 2.5, where Moçambique is defined as a highly hierarchical society.

The women travel seldom to places outside of Linga Linga such as Morrumbene, meaning they must purchase large amounts of food to sustain themselves for a long period. These occasional travels mean the food has to be stored long, which lowers the quality, or people eat the fresh groceries quickly, which means they are without fresh groceries for a period before the next travel.

Linga Linga seems to be highly built on trust, relations, and a feeling of community, which was also the case in the theory of intercultural awareness in chapter 2.5 of Moçambique concerning the cultural values and the Lewis model. Information that happens within the village are spread via personal contacts. Woman 45 did not like to have a problem with people and also wanted more information from her family. Many participants had a perspective of what others wanted rather than their desires of missing elements in Linga Linga. One example is the answer from Man 20 who missed a school above 8th grade and he pointed at the younger kid next to him. In addition, Woman 45 said she missed electricity in the village even though she had it in her home. Also, trust becomes evident as a strong factor in Linga Linga. Man 19 mentioned that he only lends out things to people he trusts and people generally solve matters in large groups.

There were various responses to what the people think is fun to do. However, they all expressed something they enjoyed doing. This aspect may be an important factor in developing engagement among the people to use the community center.

Takeaways in-depth interviews with citizens:

- The most desired thing in Linga Linga is a marketplace for local trading, both buying and selling.
- Electricity is desired to have access to a fridge and freezer.
- Travels outside of Linga Linga occur seldom.
- People see working as fun since you earn money and people's future visions mainly circulate their occupation.
- Linga Linga is built on trust, relations, and community feeling.
- To be explored further:
    - Further needs.
    - Need for computer access.
    - Future visions.
    - Hierarchy in the society.

5.3.2 Pop-up interviews

During the in-depth interview with the janitor, he brought up his perspective on the future. It was believed this question could reveal hidden insights. This idea would turn out to be right because this question showed all participants in the pop-up interviews answered concerning their occupation. These answers show that work and earning money are drivers in people's lives and this was also highlighted when several people expressed working as fun. This is interesting because if this question was asked in Sweden people would probably answer anything other than working as fun to do. Citizens of Linga Linga have a different life perspective and the main driver is often to secure daily necessities such as food, water, and an income.

Society seems to have a clear hierarchy where there is no gender equality. The women in the pop-up interviews pictured their future with a new job which differed from the men who desired to expand their business or continue their studies. This might show that women do not have the same conditions for work opportunities and reaching their desired futures.

No interviewee said they would like to buy fish at a potential market in Linga Linga and only one person said they buy fish in Morrumbene. There might be several reasons for the lack of interest in fish, for instance, people catch the fish by themselves, or the fact that fish is more expensive than other groceries which means veggies may be more coveted. Another interesting aspect is that the kiosk owner did not mention a market as a missing element in Linga Linga, which may be because such a location could be a competitor to his business. Today he buys his groceries in Morrumbene and sells them in his kiosk in Linga Linga for a higher price.

The previous identification of a strong community presence from the in-depth interviews was also highlighted by the answer from the fish carrier who was willing to teach others how to weave bags and do it without charge. This means she was not concerned about the potential negative effect on her business which can be connected to Moçambique's collective society, where common goals are valued higher than individual ones. Alternatively, she may not have considered this aspect.

The two questions that received the most engagement were, firstly, concerning if something is missing in Linga Linga and secondly, if they wanted to learn something. All participants except the youngest person answered "yes" or "much" is missing in the village. Another observation occurred during the pop-up interview with the teacher where he was keeping an eye on the road for available chapas to travel with to Morrumbene. This behavior showed the problems of uncertain transportation options to the city.

Takeaways pop-up interviews:

- The need for a market and freezers from the in-depth interviews was strengthened by the pop-ups.
- A poor road and inadequate transport options make grocery shopping and studies difficult.
- Transports outside of Linga Linga are done seldom due to cost and inconvenience.

- A computer is a source of information, a knowledge- and work opportunity, and a way for students to prepare for further studies.
- All participants wanted to learn things and the most common skill to learn was English.
- Earning money is a driver in the community.
- There is no complete gender equality in society.
- To be explored further:
    - How do markets function today?

5.3.3 Observations & A Day in the Life

The sense of a clear relational environment and feeling of community from the previous interviews were also witnessed during several observations. This can be connected to the theory of intercultural awareness in chapter 2.5. Firstly, people respect each other's yard boundaries and greet one another when passing by their household. Secondly, the expression "Oya Linga Linga" while raising a fist during the community meeting is interpreted as a way of emphasizing the strength and importance of the community. Respect for one another was further observed when a person who wanted to speak, whether it was a person from the mayor's staff or a citizen of Linga Linga, stood up and was not interrupted. Moreover, the meeting took place under a majestic tree which may be symbolism for community strength, or on the other hand, only to access shade. Thirdly, respect and helpfulness were displayed during the observation of the chapas, where people helped each other with placing the cargo in suitable areas on the car wreck. Lastly, a feeling of community was observed during the A Day of the Life session, when there was a friendly atmosphere among the women buying and carrying the fish. This is interesting because, despite being competitors when the fisherman delivers the fish and when selling it further to the buyers in Morrumbene, there is still a sense of friendliness and community.

Another aspect that strengthens the insight from past interviews where the non-gender equality. This was shown during the community meeting where the women and men were separated and the women sat on the ground while the chairs were occupied by the men of the village. This separation was also seen during the community party. The gender inequality is further evident when the men socialize more with their friends than the women, who are assumed to have less free time since they work while taking care of the household and children. These observations show the community feeling is strong among the women and men separately.

A level of authority was shown when the mayor sat in a large chair, his staff sitting in front of the people, and the placement of a picture of the president next to the Moçambican flag. These actions show the importance of strengthening the government. It can be connected to the cultural value in intercultural awareness theory in section 2.5, where hierarchy's in Moçambique are a part of society.

During the community party, it became evident music and dancing were a way people express themselves and find joy in their lives. This activity seems to be an important aspect, for instance,

due to people's demanding lives concerning work, and care of household and family. It was interesting to see this joyful behavior even though their lives are tough, and they live in poverty. The use of singing and dancing during the community meeting showed another usefulness of the activity. In this case, it was interpreted as a gesture of showing respect to the high-end guest and also to the community since Linga Linga was mentioned in the song.

The reason why people are active during the daylight rather than the darker hours is assumed to be because of the absence of light sources on the streets and in people's homes. Many citizens of Linga Linga do not have access to electricity or portable lamps.

The reason why the market of Morrumbene is located on the street instead of the designated inside area is assumed to be because of the cost to sell there, which was expressed by Woman 42 in the pop-up interviews. It is interesting how people avoid the low cost of 10 MZN (1.7 SEK) per day and prefer doing their sales at a prohibited location.

During the A Day of the Life, the women who received fish kept it outside for 3 hours at the beach and thereafter they transported it to Morrumbene which is at least an additional hour depending on the waiting time for the chapa. This lowers the quality of the fish and thereby the income for the women when selling to the buyer in Morrumbene. Once arrived at the market, the fish is kept outside. This must become a health hazard to the people eating the fish later.

People's lives are often built on uncertainty concerning income. The woman in A Day in the Life waited for 5 hours without receiving any fish, meaning no income from this activity for the day. People are often dependent on others since they get income per sold item and not a monthly salary. Therefore, it is assumed people prioritize living and planning for the short term which can also be seen from the intercultural awareness research where Moçambique scored low on a long-term orientation. Another sign of living for the short term is assumed by the declining number of fish at the shore of Linga Linga. The fishermen caught all sizes of fish, even the young ones, meaning there is no long-term sustainable fishing technique.

Takeaways observations and A Day in the Life:

- Music and dancing are used for various reasons and are an integrated part of people's lives, a way to express themselves, show respect to others, and find joy.
- Women carry various items and also their babies.
- There is an authority presence among the citizens to the mayor and his staff.
- The largest problem for fish carriers, is the lack of prolonging the fish's quality, meaning less income and a possible health hazard.
- People live for benefits in the short term.
- The previous sense of community and gender inequality is further strengthened.

5.3.4 In-depth interviews with leaders of Linga Linga

It was interesting hearing different answers from the local leader and his assistant regarding the frequency of the community meetings. The assistant said he had been telling the leader many times to conduct more meetings than three each year, while the leader claims meetings are held two times per month and Morrumbene district said it is conducted 1-3 times per month. The leader of Linga Linga was not confronted to prevent any conflict between him and his assistant, so the actual frequency is unknown. The assistant was more outspoken than the local leader, presumably because the leader is hired by the government and has direct contact with the district.

The system for spreading information seems inefficient and a risk of missing delivering information to people since it is based on walking around in the village. It must be highly time-consuming to cover 1200 people and people may be in places outside Linga Linga at the time, therefore, missing the information. Moreover, since the information is spread from mouth to mouth there is a risk of it being misinterpreted. The lack of substantial information regarding expected weather hazards causes worry and danger to the citizens.

The few attendees at the community meetings must look bad when the district or the mayor is present. It makes the community also less democratic since people do not raise their opinions.

Regarding the meeting locations, there seems to be no natural space for people to see it belonging to them, since the expressed areas from the local leader are owned by local investors.

The local leader and the head of locality at Morrumbene district have the same desire to increase tourism in Linga Linga since it introduces job and logistic opportunities. The reason for the aligned opinion may be they both are hired by and have the perspective of the district. Project Vita also saw the benefit of more tourism, however, by increased number of volunteers.

Linga Linga's strong relational and trust foundation was further understood by the leader interviews since the citizens are responsible for the external people they bring into the village. Also, local conflicts are often solved in large groups by involving the citizens to express their views on the matter.

The head of locality at the district had a clear view of the benefits of having district representatives in the community center and also managing the deliveries of goods to a potential market in Linga Linga. This would increase the power and influence of the district in Linga Linga and decrease the independence of the center, as well as potentially interfere with democracy and the citizen's perception how who the center belongs to. The district's presence in Linga Linga could be further strengthened by the desire for a police station in the area. The district saw a present security issue in Linga Linga, however, this was not expressed as an issue by the citizens.

Takeaways in-depth interviews with leaders:

- A proper meeting area would be beneficial for the leader's work since it potentially attracts more attendees.
- The leaders want more people to attend the community meetings.
- District representatives and keeping statistics were desired to improve the receiving of information to the district.
- Coordination of Linga Linga investors is an opportunity for developing a solution to the road issue. It is also a way of not becoming dependent on external actors.
- Increased tourism makes Linga Linga more attractive to investors and creates job opportunities for the residents.

5.3.5 Value network map

Value network maps were done to understand the service dynamics further of the current systems, identified from the in-depth and pop-up interviews. It was also a way for visualization to detect gaps, opportunities, and hidden need groups. The involved stakeholders and their exchange of value with each other were visualized. In addition, the identified pain points from the interviews were positioned on the map. Six value network maps were created: Communication, Trading, Fish carrying, Bag weaving, and Education for students in 3rd-7th grade and 8th grade and above. These systems were analyzed due to their scale in the community.

Communication

This value network map visualizes the system of how communication occurs regarding matters originating from the citizens of Linga Linga. The placement of the stakeholders onto the circles in Figure 16, represents their close relation to the central stakeholder, the citizen. The message that the citizen wants to bring forward can occur in three directions toward the local leader. The blue arrows in Figure 16 represent direction 1, by firstly speaking to the leader's assistant, where the value becomes "being heard", and "power" to represent the people, for the leader. Thereafter, if the leader's assistant cannot solve the matter himself, he brings forward the message to the local leader, where the value of the assistant is "influence" and the local leader gets "power". The second direction is represented by the green arrow in Figure 16, where the citizen raises their matter at the community meeting where the value is "being heard" and the community meeting gains "power". Thereafter, the matter is brought forward to the local leader, who possibly also participated during the meeting. The value exchange is "influence" for the community meeting and "power" to the local leader. The third direction is direct communication to the local leader, where the values again are "being heard" and "power".

The citizen also communicate their opinion to their friends and family members as represented by the circulating arrows in Figure 16, where the value exchange is "being heard". If the received message to the local leader is not resolvable, it is brought forward to the district's head of locality during the monthly meeting, where the value is "influence" for the leader and

"power" for the head of locality. The last communication channel is between the head of locality to the mayor which occurs during the monthly meeting, with the values "influence" and "power". The response to the message follows the same process back to the citizen. All communication in the system is done orally.

There are two related pain points, seen as red faces in Figure 16. The first pain point is at the community meetings, which includes the participants sitting on the sand, few attendees, and inefficient communication about the time and place of the meeting. The second pain point is the inefficient forwarding of messages by using individual meetings with the local leader and assistant in the first step, which could be done through community meetings instead.

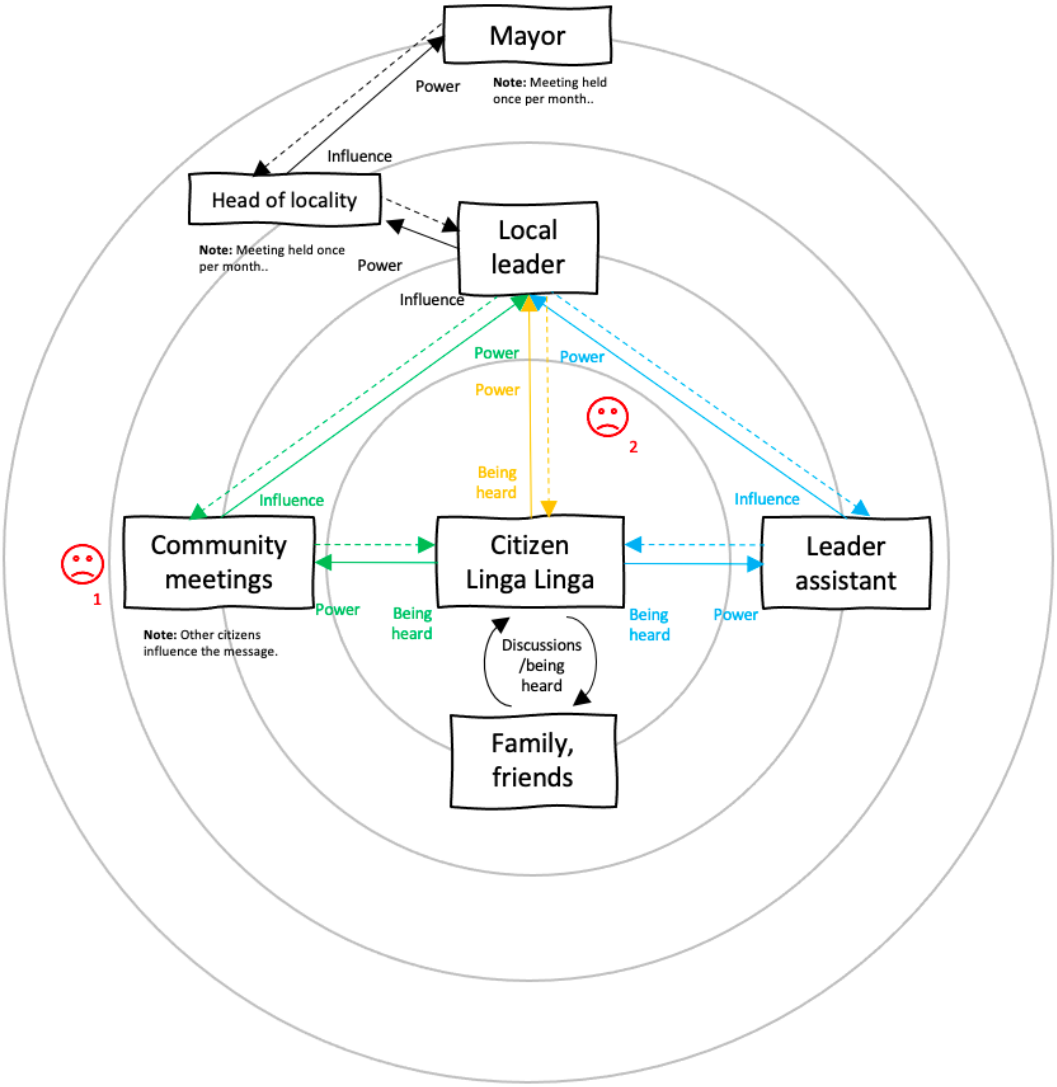


Figure 16. Value network map of communication from citizen to the district. The circles show the stakeholder's level of close relationship to the citizen. The color of the arrows indicates one communication channel. The red faces show where the pain points occur.

Trading

The value network map considers the trading that occurs by the citizens of Linga Linga, most often the women. Three different channels were identified from the interviews. Firstly, the large trading, shown in blue arrows in Figure 17, occurs by the citizens using chapa for transport to Morrumbene to buy groceries. The value to the citizen is firstly the “transport” service while for the driver it is “payment”. The value to the market seller is “payment” and to the citizen, it is received “groceries”. Afterward, the citizen uses chapa for transport back to Linga Linga. The market seller in Morrumbene buys the goods from another seller possibly from the producers or distribution partners.

The second channel is the small trading occurring within Linga Linga via the kiosks, seen as orange arrows in Figure 17. The citizen receives the goods through a payment to the kiosk owner, which are the involved values in this channel. The kiosk owner buys the goods from the sellers in Morrumbene and uses their car or pays for a transport service. The values are then “transport” and “payment”, between the kiosk owner and the driver. The value to the market seller in Morrumbene is “payment” and for the kiosk owner, it is the “groceries”.

The third channel considers the trading of seasonal goods where large quantities occur, as shown in green arrows in Figure 17. Here the citizens access groceries from a seller walking around in Linga Linga who obtained the groceries from Morrumbene or a producer in Linga Linga, who can also be a walking seller. The values involved are “groceries” and “payments”.

The cost of the groceries is estimated to differ between the channels. The highest cost for the citizen is buying through the kiosk owner, the cheapest is by buying from the walking seller or the market in Morrumbene, however, it also involves the transport expense.

The first pain point involves the chapa transport. It is perceived as expensive, time-consuming, with unpredictable departure times, often crowded, and lacking protection against rain. The second pain point is the inefficient distribution by the producer of goods in Linga Linga, meaning some citizens with resources do not sell them. In addition, there is a risk citizens who live further away may not receive goods if the seller sells out before reaching them. The third pain point is the unavailability of cooling groceries at the kiosk.

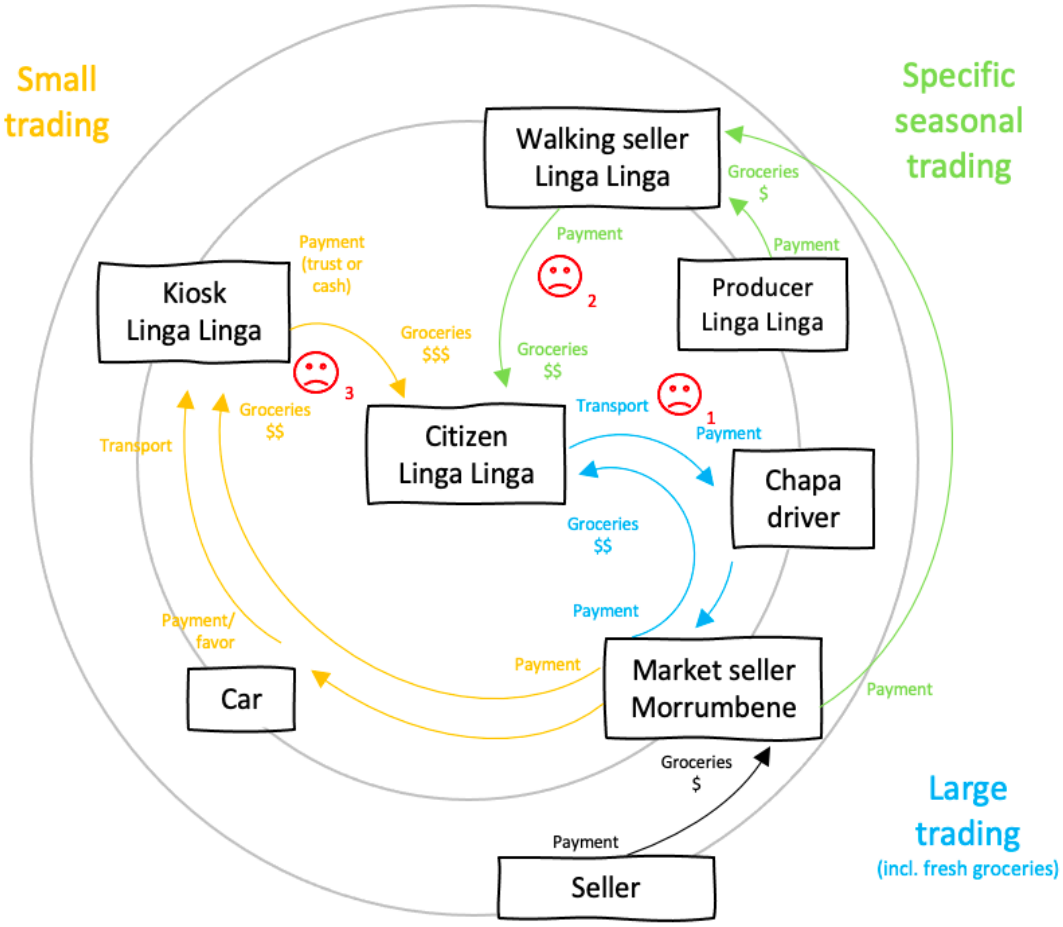


Figure 17. Value network map of local trading for a Linga Linga citizen.

Fish carrying

This value network map, shown in Figure 18, represents the system of women carrying fish from the source in Linga Linga to sales in Morrumbene. The women buy the fish from the fishermen in Linga Linga, where the value arises in receiving “fish” and for the fishermen, it is “payment”. Thereafter the women travel to Morrumbene using a chapa, where the value is “transport” and for the driver, it is “payment”. The value to the market buyer is “fish” and to the women, it is “payment”. An additional involved actor is other women with the same occupation, the values between them and the fishermen are “fish” and “payment”. There is also a mutual “socializing” value between the fish carriers. The last involved actor is the fruit seller at the fishing site, where the exchanged values are “payment” and “fruit” to the fish carriers.

The first identified pain point for the fish carriers is the long waiting time for the catches and the unpredictive amount of caught fish. The second pain point is the exposure of sun to the fish, affecting its quality. The third point is the expensive, unpredictable, and crowded transport to Morrumbene. The last pain point involves the competition among the fish carriers for receiving fish from the fishermen.

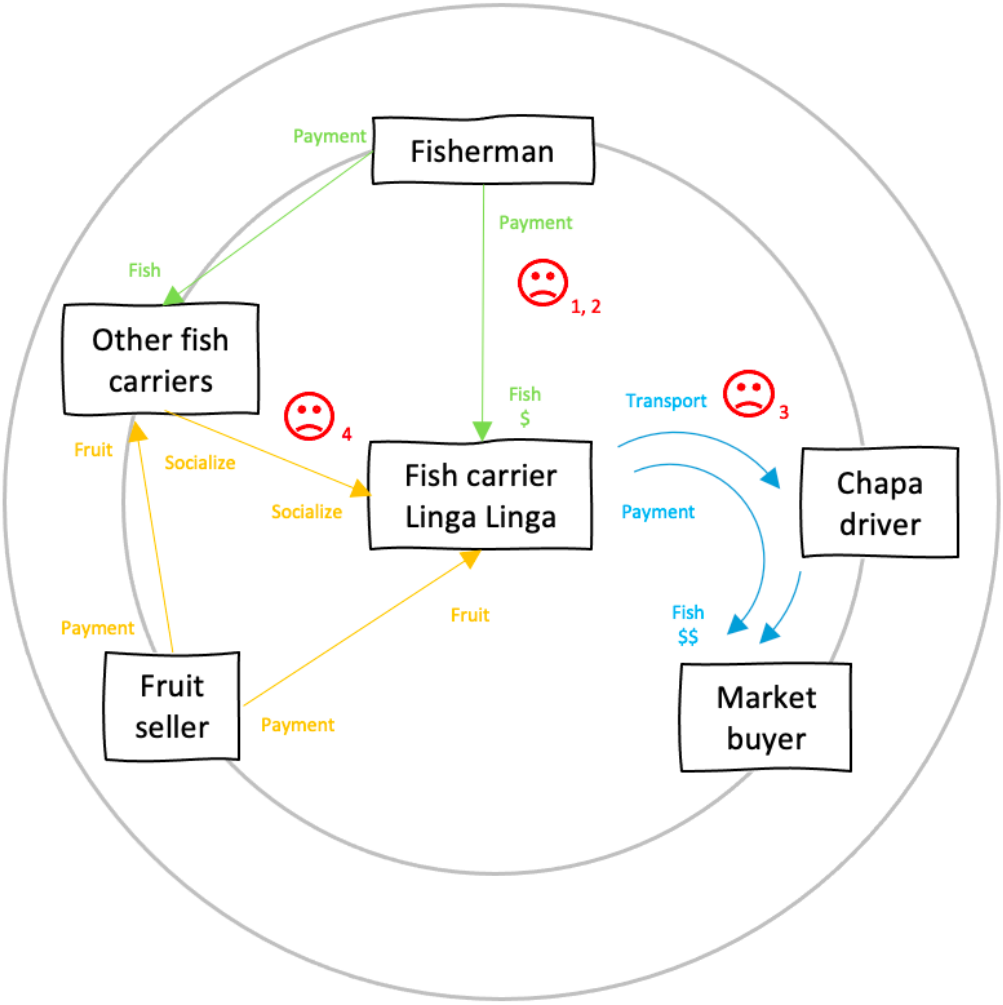


Figure 18. Value network map of the fish carrier's occupation.

Bag weaving

The value network map in Figure 19 shows the system concerning the women weaving bags in Linga Linga. The women sell their bags to several buyers in Linga Linga, where the value is "payment" and the" bag" to the buyer. Thereafter the buyer transports the bags to the market seller in the Inhambane city, where the exchanged values are again "payment" and "bags". Lastly, tourists purchase the bags on the market, where the same values are included. There are three levels of the selling price to make the actors gain profit for each sale. Another actor is the children of the weaving women, who help with weaving bags. In return, they get various values such as a household, food, and education.

The two pain points here are firstly, the impersonal sale to the end-customer. Secondly, the inadequate profit for the first sale to the Linga Linga buyer.

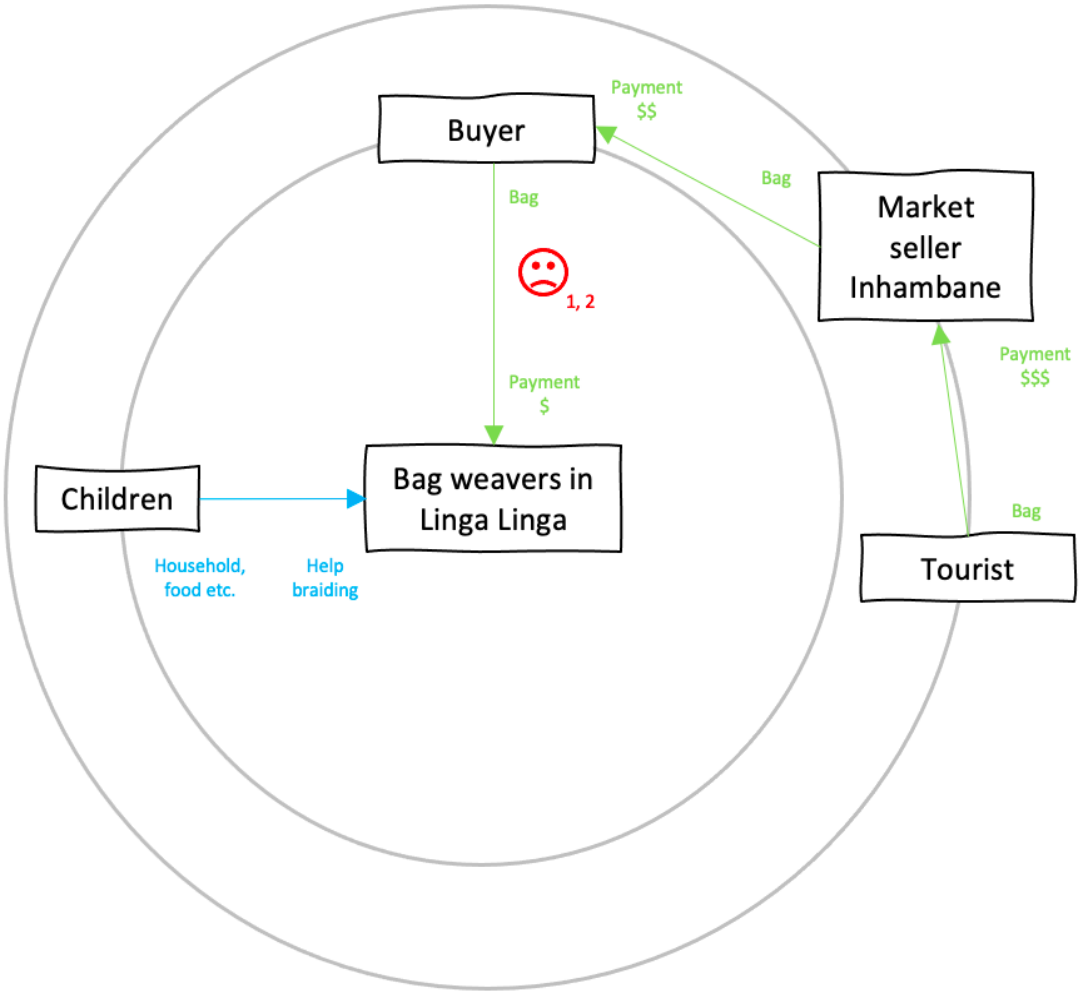


Figure 19. Value network map of the bag weaver's occupation.

Education

The value network map in Figure 20 represents the education system of students in 3rd to 7th grade. The school in Linga Linga offers "education facilities" to the students while the school receives "students". The school receives monetary "support" from the government while the government gets "influence". The government pays the "salary" to the school's teachers and the government's value becomes "employees". The teacher gives "education" to the students and their value is "students" to teach. Another involved actor is the student's parents who pay for the education to the government. The value for the parents of their child is "help at home". Another actor is student's school friends, providing "socialization" and learning support.

The value network map in Figure 21 shows the education system for students in 8th grade and above. The channels are similar to the map in Figure 20, however, an additional actor is included, chapa drivers. The students travel to Morrumbene to access the secondary school where the transportation occurs through a chapa, which is reserved for students. The exchanged values are "payment" to the chapa driver and "transport" for the students.

The identified pain point for the students in 3rd to 7th grade is their non-access to computers at school, making them unprepared for further studies. The pain point for the students in 8th grade and above is the expensive, crowded, and possibly rainy ride to Morrumbene.

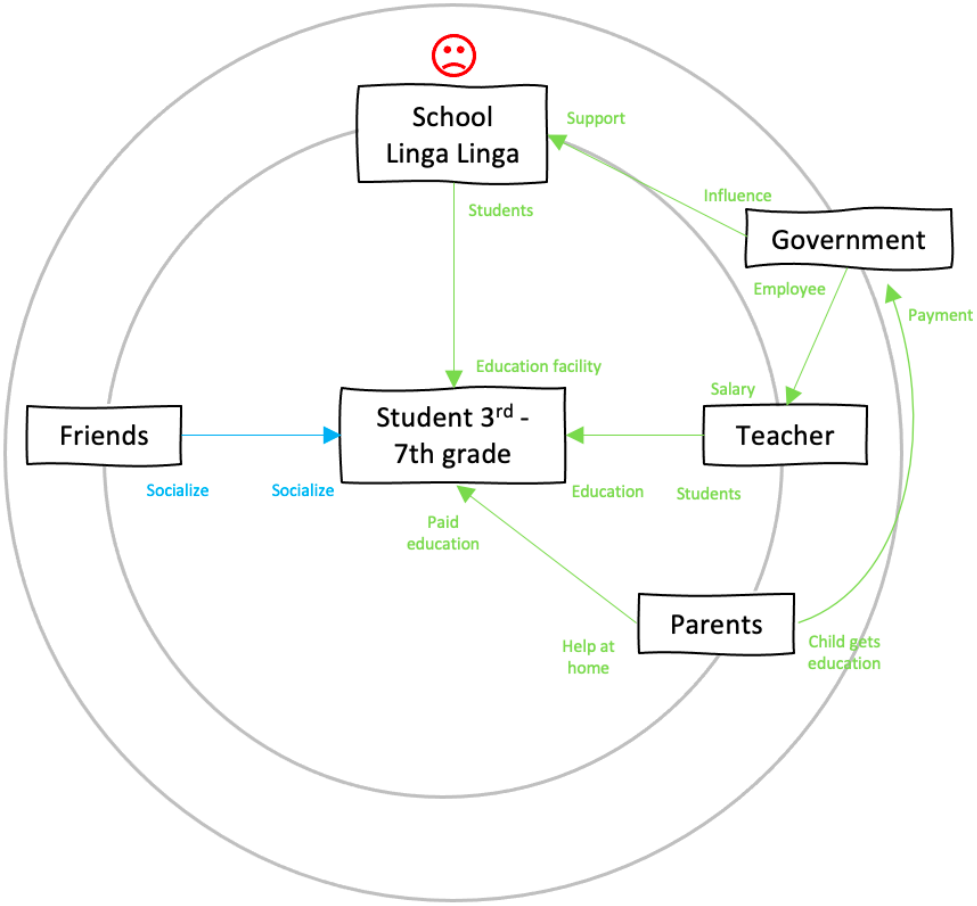


Figure 20. Value network map of students from 3rd to 7th grade.

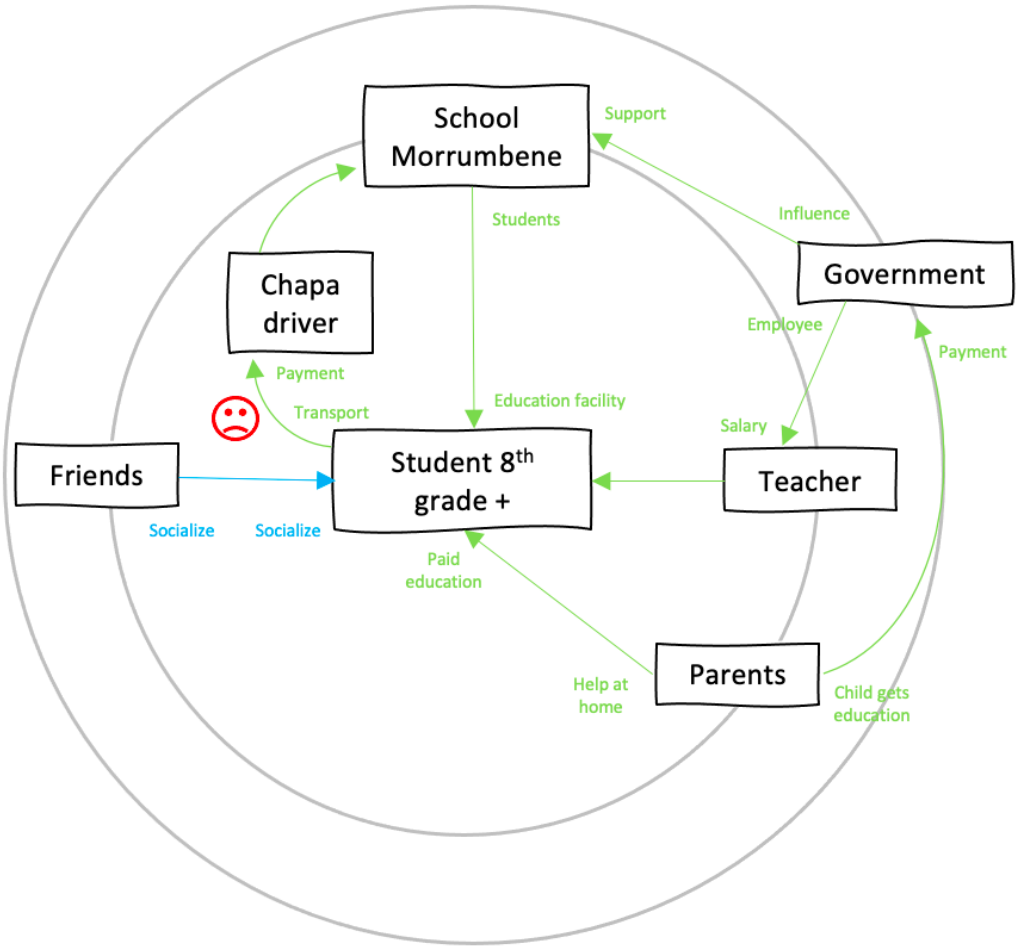


Figure 21. Value network map of students in 8th grade and above.

5.3.6 Stakeholder analysis

The clustering of data, analysis, and value network maps, were followed by identifying the need groups, as seen in Table 6. The stakeholder's experienced problems and needs were collected from the previous interviews and observations. The needs of each need group were assigned a descriptive element. Thereafter, the needs with the shared element were collected, seen in Table 7. The need groups, or stakeholders, were analyzed by doing two stakeholder maps. In the first one, the stakeholders were placed in a Power/Interest grid (Figure 22) to analyze their power to affect and interest in the project (Mindtools, 2022). It was used to understand what stakeholders to prioritize satisfying. The second stakeholder map (Figure 23) analyzed their relations with each other.

Need groups

13 need groups, which are also the stakeholders, were identified. In Table 6 below, the description of the stakeholders, their experienced problems, their needs and the corresponding element, are presented.

Table 6. Description of the stakeholders/need groups, their experienced problems, their needs, and the corresponding elements.

1 **Market buyer** People, mainly women, travelling to Morrumbene by chapa to buy groceries which cannot be found in Linga Linga.	**Problems:** Transport • Expensive • Time-consuming • Crowded (uncomfortable, not guarantee travel, issue cargo space) • Unsafe • Unpredictable times • Rain Supply • Cannot find needed groceries in Linga Linga • Fish sold outside (health and quality hazard)	**Needs:** • Access to certain groceries locally to avoid transport **Elements:** • Local trading
2 **Women weaving bags** Women work with weaving bags, generally from home while taking care of their children. But also, at various places where the women move.	**Problems:** • Children often follow and is a distraction Sales • No personal sales to end customer • Much lower income than the middle person	**Needs:** • Become prepared for starting school • Personal connection to the end bag customers • An adequate income **Elements:** • Local trading
3 **Women carrying fish** These women buy fish from the fishermen, store it in buckets, and travel to Morrumbene by chapa and sell it to market sellers.	**Problems:** Transport • Expensive • Time-consuming • Long wait time • Crowded • Unsafe • Unpredicted times • Decreased fish quality • Not adapted for the cargo Sales • No place in Linga Linga to sell fish • Lower income when fish turns bad • No good storing solution to prolong fish's lifetime Fish source • No guarantee of fish (poor supply, competition) • Occasionally too much fish, cannot sell all with good quality • Decreasing amount of fish in the sea	**Needs:** • Being able to sell resources efficiently • Prolonging good's quality **Elements:** • Food quality • Local trading

4 **Students in 3rd to 7th grade** Children in the ages 8-12 years old studying in the school in Linga Linga in 3rd to 7th grade.	**Problems:** Education: • No access to computers • Unprepared for 8th grade	**Needs:** • Become prepared for 8th grade **Elements:** • Education
5 **Students in 8th grade and above** Children older than 12 years old travel to Morrumbene by chapa to study 8th grade or above.	**Problem:** Transport: • Time-consuming • Crowded • Unsafe • Unpredictable travel duration • Rain damages books and make clothes wet • Expensive for parents	**Needs:** • A reliable, cheap, pleasant travel to school **Elements:** • Transport
6 **Children under 8 years** Children in Linga Linga below 6 years old, often following their mother or siblings.	**Problems:** Education • No education • Unprepared for 3rd grade	**Needs:** • Become prepared for starting school **Elements:** • Education
7 **Children above 13 years old** This group involves children older than 13 y.o. who are not students.	**Problems:** Transport • Expensive Education • No education passed 7th grade • Less future opportunities	**Needs:** • Education/work opportunities • Cheaper transport **Elements:** • Transport • Education
8 **Citizens with resources** Citizens in Linga Linga have various resources e.g. crops, meat, and fruits.	**Problems:** Sales • No designated place to sell resources • Time-consuming to walk around in Linga Linga • Some citizens do not get the chance to buy the goods	**Needs:** • Being able to sell resources efficiently **Elements:** • Local trading

9 **Chapa drivers** Men driving chapas to and from Linga Linga to Morrumbene for a living.	**Problems:** Customers • Many stops to pick up people (time-consuming, higher fuel consumption) Road • Damages car (costs, meaning a higher price for customers) • Travel is time-consuming • Only some cars manage the conditions	**Needs:** • A safer, faster travel time to/from Morrumbene **Elements:** • Transport
10 **Leaders of Linga Linga** The local leader and the assistants of Linga Linga inform, solve issues, and represent the citizens.	**Problems:** Information spreading • Inefficient (time consuming) • May miss people Community meeting • Few participants • Sit on the sand, borrow chairs • Individual meetings are held to solve issues even though it can be more efficient as a larger meeting	**Needs:** • Efficient information spreading • Appropriate community meeting location **Elements:** • Communication
11 **Project Vita** Project Vita conducts projects in Linga Linga to help the citizens living in poverty.	**Problems:** • Difficult to communicate the value of their projects • Difficult to spread donations • Lack of relevant and clear projects for volunteers • No designated location for workshops and meetings for students	**Needs:** • Efficient information spreading • Relevant projects for volunteers • Appropriate workshop area. **Elements:** • Communication • Tourism
12 **Morrumbene district** The Morrumbene district is a governmental institution and is responsible for the city Morrumbene and the surrounding villages in areas such as education, agriculture, access to food, health, and communication with the population	**Problems:** • Lack of receiving information from Linga Linga • Lack of tourism in Linga Linga (lack of income) • Few participants at community meetings (looks bad to the district/mayor visits and citizens miss information) • No appropriate community meeting location	**Needs:** • Efficient information receiving • Increased tourism • Appropriate community meeting location. **Elements:** • Communication • Tourism

13 **Citizens of Linga Linga** This group involves all residents of Linga Linga, excluding the leaders and Project Vita.	**Problems:** Community meeting: • May miss information about meeting time, place, content • Lack of time (busy citizens) • Miss important information, e.g. health and election • Less democratic, opportunities to speak up, if people miss meetings Information spreading: • Inefficient information spreading, may miss information from leaders • Information can be wrongly interpreted if spread from mouth to mouth • Not enough information on weather hazards (lead to worry, danger) Women • Women have worse conditions than men • Mothers have to take care of the household and the children while working. Other: • No good conditions to educate themselves or learn new things. • No good conditions for doing what they enjoy and not many fun activities to do in Linga Linga • Limited access to electricity, affecting necessities such as charging phones, light access, and storage of goods. • Poor opportunities to prolong resource quality	**Needs:** • Appropriate community meeting location • Efficient information spreading • Prolonging good's quality • Education/work/ hobby opportunities **Elements:** • Food quality • Transport • Communication • Education

Elements

The needs sharing the same element are presented in Table 7, including the need group numbers. The 17 needs were distributed into 6 elements.

Table 7. The elements, the needs (N), and the corresponding need groups (G).

Element: Food quality		**Element:** Local trading	
G3, 13	N1: Prolonging good's quality.	G1	N2: Access to certain groceries to avoid transport
		G2	N3: An adequate income.
		G2	N4: Personal connection to the end bag customers.
		G3, 8	N5: Being able to sell resources efficiently.
Element: Transport		**Element**: Education	
G5	N6: A reliable, cheap, pleasant travel to school.	G4	N9: Become prepared for 8th grade.
G7	N7: Cheaper transport.	G2, 6	N10: Become prepared for starting school.
G9	N8: A safer, faster travel time to/from Morrumbene.	G7, 13	N11: Education/work/hobby opportunities.
Element: Communication		**Element**: Tourism	
G10, 11, 13	N12: Efficient information spreading.	G11	N16: Relevant projects for volunteers.
G12	N13: Efficient information receiving.	G12	N17: Increased tourism.
G11	N14: Appropriate workshop area		
G12, 13	N15: Appropriate community meeting location.		

Power/Interest grid

The Power/Interest grid in Figure 22 shows the stakeholder's interest and power to affect the project. The grid is divided into four zones which describe the stakeholders within. Morrumbene district has high power since they must approve the community center project to be implemented. Their interest is the lowest because the center is for the citizen's best, meaning the district should be kept satisfied with the project. Another stakeholder in the same zone is the leaders of Linga Linga. They have less power than the district, but more interest because they live in Linga Linga and work closely with the citizens. Project Vita has much power since they initiated and controls the center's development. Their main interest is to help the Linga Linga citizens rather than their operation, placing them as "manage closely". The stakeholders "Citizens of Linga Linga" include all the need groups who live in Linga Linga. They are in the

same zone as Project Vita and have the highest power since the center is based on their needs and without them using the center, the project is irrelevant. They also have the highest interest because the center is developed for them to help in their daily lives. Lastly, the Chapa drivers have relatively little power to affect the project, as well as their interest since the project has a mediocre effect on them, placing them in the "keep informed" zone.

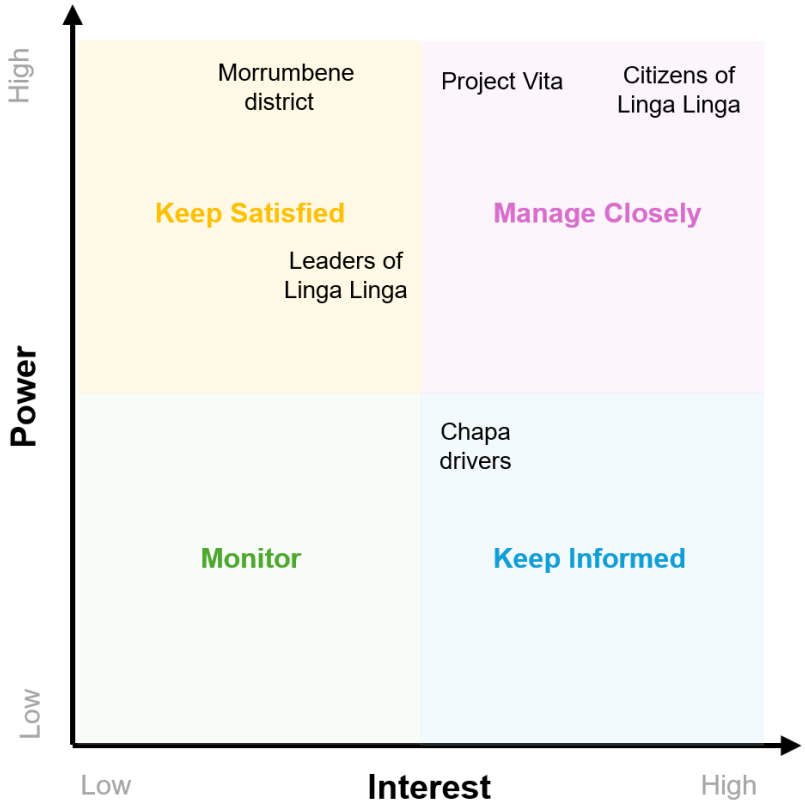


Figure 22. Power/interest grid to the project.

Relationships between need groups

The need group's relations are visualized in Figure 23 and explained below in the numbers 1-12. The size of the need groups in Figure 23 is determined by the estimated number of people it belongs to. There are also need groups which belong to several groups. Three groups are within the "Market buyer group": "Women weaving bags", "Women carrying fish", and "Citizens with resources". These three groups are often intersecting since women carrying fish often also weave bags and may have other resources. The "Market buyer" group is included in the "Citizens of Linga Linga group" which also consists of "Children below 8 years", "Students 3rd – 7th grade", "Students 8th grade and above", and "Children above 13 years".

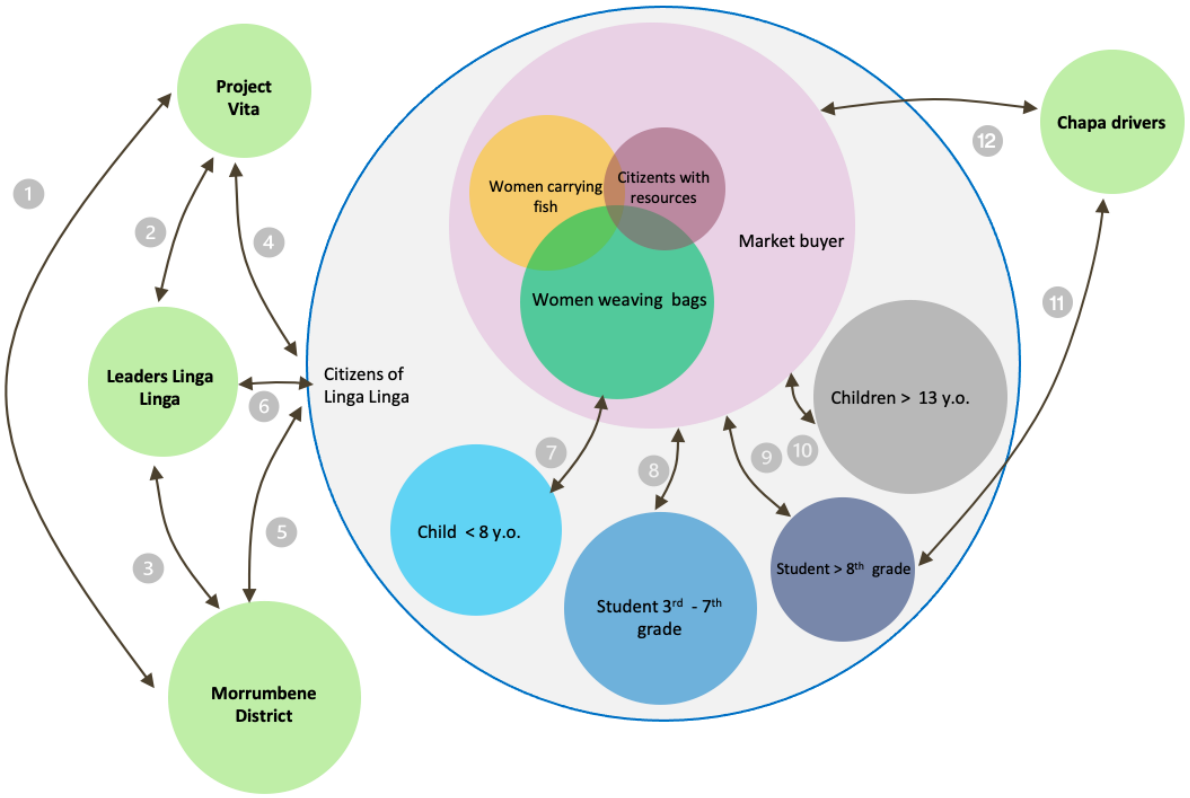


Figure 23. Stakeholder's relationship to each other.

Relationships:

1. Project Vita influences the district due to their strong position in Linga Linga and the people's trust and support. The district also impacts Project Vita since their decisions influence Project Vita's operations e.g. the district must approve the projects.
2. Project Vita has a mutual relationship with the leaders of Linga Linga. Project Vita helps the village to develop which is appreciated by the leaders whereas the leaders support Project Vita in district matters and community involvement.
3. The leaders of Linga Linga both inform and receive information from the district, while also implementing directives from the district to the citizens.
4. Project Vita is a strong actor in Linga Linga since they have helped the citizens and are trusted and supported by them. Therefore, the citizens are willing to support Project Vita projects.
5. The district has a direct influence on the citizens of Linga Linga due to the arranged department meetings. The citizens sometimes has a direct influence on the district during the community meetings. However, oftentimes the influence is passed by the local leader.
6. The leaders of Linga Linga support the citizens in multiple ways, for instance, resolving conflicts, matters must pass by the leaders, and they forward the directives from the district to the citizens. The citizens bring matters to the leaders which impacts the leader's work.

7. The women who weave bags often take care of their young children as they work while the children help the mothers to some extent. However, the youngest children are only taken care of.
8. The parents of the students pay their tuition fees and the students help their parents in the household or take care of their younger siblings.
9. Again, the parents of the students pay their tuition fees and travel costs to Morrumbene while the students help their parents in the household or take care of their younger siblings.
10. The parents supports their children while the children help with taking care of their siblings and in work such as weaving bags.
11. The chapa drivers transport the students to and from Morrumbene while the student's parents pay the travel costs.
12. The chapa drivers transport the citizens to Morrumbene where the travellers buy groceries and pay the driver for the service.

6. Concept development

The concept development chapter is divided into the conducted methods. Firstly, an initial ideation session was done in a group through a Co-creation workshop. In the next part, a following ideation session is presented which was conducted by the author themselves to generate ideas for each need of every element, identified from the Field study. Lastly, ideas were chosen from each need in the Concept selection sub-chapter.

6.1 Co-creation workshop

This sub-chapter is divided into three parts. Firstly, a description of the conducted Co-creation workshop. Followed by the results of the session and lastly an analysis of the results including the takeaways from the second ideation session.

6.1.1 Method

Co-creation workshops can be used in the early research stage of a project but also the ideation phase (Stickdorn *et al.*, 2018). In this project, a co-creation workshop was conducted to involve the potential users in the concept development process and support the ideation by receiving insights on the elements of most interest, which were education and local trading. The workshop had three female participants the ages 27, 30, and 45 years. The first participant was part of a pop-up interview while the third participant participated in an in-depth interview. The second participant was not a part of the previous interviews. Table 8 displays the information of the participants.

Table 8. Information on co-creation workshop participants.

Participant	Gender	Age	Occupation
P1	Women	27	Baker, hair braider, bag weaver
P2	Women	30	-
P3	Women	45	Housekeeper, bag weaver

A simple and small-scale model of a building was placed on the table to be referred to during the activities to assist the participants in imagining the center and understanding the instructions. An interpreter was involved in the preparation by supporting the translation of the Portuguese names in the activities. The interpreter also participated during the workshop by explaining the activities and reading the alternatives aloud to help if someone had limited reading skills. In addition, the interpreter assisted in making the workshop into a friendly environment and conversation to make the participants feel comfortable expressing themselves. The participants got the time they needed to complete the activities to create a non-stressed environment. After each activity, the participants shared and explained their reasons for their choices. The workshop took place outdoors, in the shadows, and in an isolated area to reduce distractions.

The workshop started by letting the participants share their names and ages. Thereafter, the designer explained the reason for conducting the workshop and the plan of the community center. Seven short activities were done with the participants. The exercises invovled drawing elements and placement of paper labels which had various expressions (a happy face, a sad face, and a question mark) to represent their attitude towards various alternatives. The activities were:

1. To circle the services they would like to access in the community center by placing the attitude labels on 12 alternatives, determined by expressed missing elements from interviews. They also had the possibility to add new services. The reason of the exercise was to make them understand the potential of a community center.
2. To present what days of the week they would like to visit the center by placing the paper labels on the seven days of the week.
3. To present what times of the day they would like to visit the center, using the labels on the four alternatives: morning, lunch, afternoon, and evening.
4. To pretend the building model is a market and draw how they want a potential marketplace to look like. Three questions were also asked:
    - What would you like to buy at a market and a shop?
    - Would you like an inside or outside market?
    - If you had something to sell, would you like to sell it yourself at the market or sell it to a person working at the market?
5. To pretend the building is a place to learn things and choose how they would like to learn by placing the labels on eight alternatives: communication with people, reading, listening, learning, using a tv, using a computer, working with the hands, learning at home and in a classroom.
6. To present who they would like to learn with by placing the labels on seven alternatives: with friends, their kids, partner, parents, men, women, and by themselves.
7. To determine who they would like to teach them by placing the labels on three choices: a person from Linga Linga, a person outside of Linga Linga, and a foreigner. A question was also asked:
    - Do you know something which you would be willing to teach others?

6.1.2 Results

Regarding the first activity, the desired services varied slightly between the participants. The complete results of the workshop are visualized in Appendix H. The services were a market to sell and buy veggies, a shop, a place to learn English, computer access to learn how to use a computer, a meeting place to have a designated location, a preschool to make children learn things before the age of eight, and a secondary school for students to avoid the transport to Morrumbene.. Also, a freezer to store fish and other food, where one expressed reason was to buy much fish once and then store it, this was further expressed by another participant:

"A freezer, to store food so I do not have to eat 3 kg every time, I can divide it over a longer time."
- Women, 45

In the second activity, the answers varied. Women 27 saw Tuesday, Thursday, Friday, and Saturday as relevant opening times of the center since those are the days where there are more movement. Sunday was not chosen since it is the day to go to church. Women 30 wanted the center to be open Monday to Friday to leave the weekend only open for a market and not education service because during the weekend people can practice what they have learned. Women 45 saw every day as a possibility to visit the center, even on Sundays when she go to church.

The third activity showed consistent choices since the participants wanted the center to be open to the public at all times of the day. However, the 27-year-old herself wanted to visit the center during the mornings and afternoons.

In the fourth activity, all participants interpreted the instructions as drawing the whole center rather than just the market. The drawings are seen in Figure 24. Women 27, pictured the center as one building, starting with a pre-school, then a market, thereafter a fish storage, and lastly a general storage. Women 30 drew several buildings including a secondary school, a preschool, a learning and reading building, an inside and outside market, and a building for freezing fish. Women 45 drew a market building, a freezer, a shop, and a secondary school.

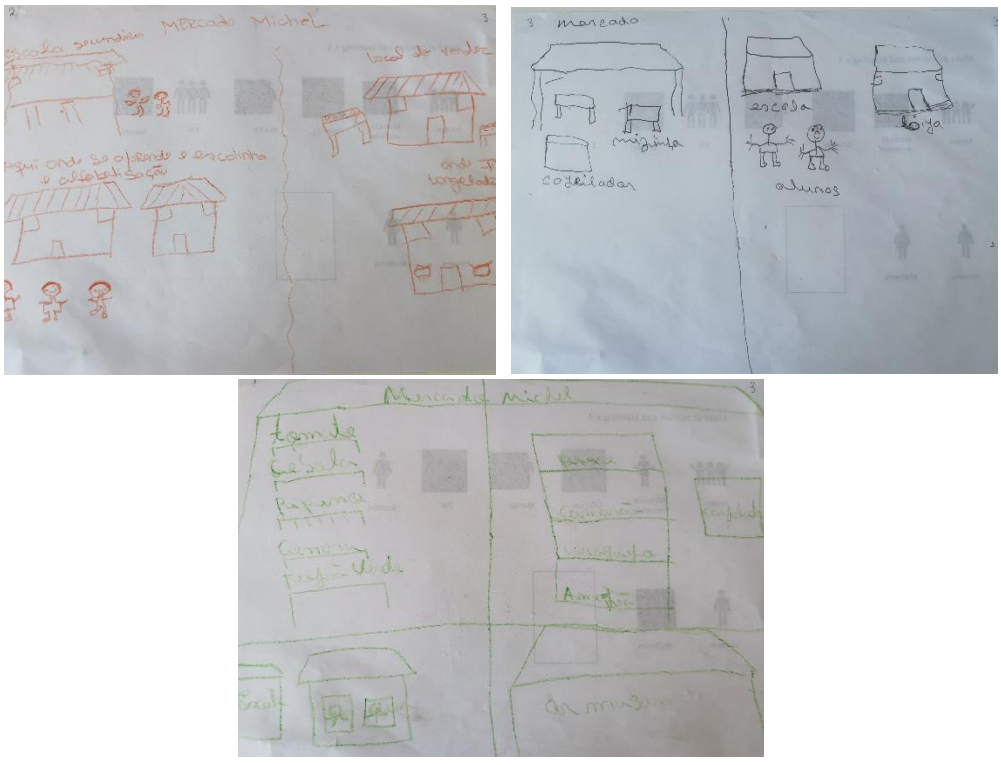


Figure 24. Workshop activity four, drawings of the center.

During activity 3, when asked about grocery access, all participants expressed there were essentially no differences between accessing groceries at a market or a shop, except that rice should preferably be bought in a shop. They desired the shop to be outside and the market to be inside. They all saw a benefit of selling their goods themselves at a market rather than selling it to someone working there, due to the larger earnings.

In the fifth activity, all participants saw communication with others and being in a classroom environment as a good way to learn. Listening was seen as a way to receive news. Certain activities such as learning how to cook, weaving bags or sewing were described as good to learn through actively working with their hands. The 27 and 30-year-olds said it would be difficult to learn from their homes since there are other tasks to do there. Women 27 also selected the computer as a learning tool because she wanted to learn how to use it. Women 30 saw the TV as another way to get the news.

The sixth activity was the quickest where all the participants saw no issue of learning with anyone. The last activity showed that the participants would not mind being taught by a person outside of Linga Linga or a foreigner. However, a person from Linga Linga was not desired since this person knows the social structures and norms in the village and may therefore isolate some people. To the question, if they would not mind teaching others they all said yes. Women 27 saw an opportunity to teach bun baking and weaving bags. Women 30 wanted to teach how to do business and sales while Women 45 saw cooking and weaving bags as teaching topics.

6.1.3 Analysis

At the beginning of the workshop, the participants were unfamiliar with the symbols of the attitude labels. However, after explaining their meaning the participants felt comfortable using them.

The participants laughed from time to time and discussions arose, especially during the drawing activity. This is interpreted as a relaxing and comfortable atmosphere which was desired.

It was not clear why they desired the shop outside and the market inside, they became quiet when this question was asked. Generally, based on the previous observations, markets are often held outside while shops are inside. Also, the reasoning of the differences between a market and a shop regarding goods access could suggest that the shop's importance is relatively low. The expressed reason for having a shop was to avoid the travel to Morrumbene to access the groceries.

The expressed need for a freezer is interpreted as saving money and time by buying many fish at once and thereafter storing it. Additionally, a freezer can enable eating various meals without the necessity of consuming the same food repeatedly to prevent spoilage. A various diet may also make the meals more enjoyable.

The participants all wanted to sell goods themselves at the market since it is a good way of earning money. However, Women 45 had stable work and still saw a possibility to work as a market seller apart from that. During the workshop, she was concise in her explanations compared to the others. She also changed her answer in the last activity when hearing the other's explanations. This behavior can be interpreted in various ways, possibly she was uncomfortable due to a more limited vocabulary, generally being more shy, or the fact the other women where close friends.

Women 27 and 30 expressed learning as challenging from home because there are many tasks to do there. This can be connected to the responsibility women have to take care of the household and children. There are also potential distractions such as their husband demanding things or people passing by.

The most interesting response during the workshop was the participant's choice of teacher. Before the workshop, a probable answer was someone from Linga Linga. However, it was seen as a problem since this person may isolate certain people due to the social structure in the village. Another interesting aspect of this is their willingness to teach others while being against being taught by other Linga Linga citizens. Their view of receiving a contribution of this service is seen to be needed since Women 27 and 45 charge tourists a certain amount when teaching bun baking and bag weaving.

Takeaways Co-creation workshop:

- Storage of fish and other goods is good from a cost and enjoyable perspective.
- It is desired to have the center open to the public from the morning to the evening.
- A shop can be unnecessary in the presence of a market since the same groceries are sold there.
- If having resources to sell, it is desired to sell these at a market themselves due to the higher earnings.
- There are no limitations to people learning with others.
- There are various ways of learning things which depend on the topic.
- Potential teachers should not be from Linga Linga to avoid isolating certain people.

6.2 Idea generation

The idea generation sub-chapter includes the second ideation session, where the method of brainstorming was conducted. The chapter is divided into a description of the method, followed by the results.

6.2.1 Method

Brainstorming was conducted to generate many ideas to fulfill the needs (Stickdorn *et al.*, 2018). The process included looking into every element individually, meaning six sessions, and creating solutions for each of the element’s needs. The sessions began with determining the

overall question of the element on which the ideas should be based on. The ideas were written on physical and digital post-its and visualized on a billboard to get an overview.

6.2.2 Results

The generated ideas of every need, as part of the six elements, are presented below.

Food quality
N1: Prolonging good's quality.
The first idea concerns offering access to freezer space in the center. People would be able to rent a freezing compartment to store various goods such as fish and cooked dishes. The second idea is similar, however, accessing a fridge instead. This enables people to store goods in the short perspective but possibly more types of groceries, for instance, prolonging vegetables and fruit.

Local trading
N2: Access to certain groceries to avoid transport.
The first idea was a local market in Linga Linga where people can access groceries which today are found in Morrumbene. The distribution of groceries can occur through various channels, for instance, the market sellers can transport the groceries from Morrumbene, one determined person can transport groceries from Morrumbene, the district can control the delivery, market sellers from Morrumbene can transport and sell groceries to the market sellers in Linga Linga. The second idea considered people ordering groceries from a person who buys and transports the groceries from Morrumbene and distributes them at the community center.

N3: A higher income for weaving bags.
The first idea, concerning generating more income for the bag weavers, involved the women selling their crafts directly at a market in Linga Linga, meaning today's distribution person is avoided. Another way the distribution person is avoided is if the women sell their bags to a person working at a shop, which is idea 2. Idea 3 is about having an online platform, e.g. Facebook marketplace, where the women post their craft and sell directly to the end customer. The logistics could be controlled by one specific person who gives a personal description of the buyer to the women. Idea 4 considers having a collaborative business between bag weavers where one person in the center buys the bags in large quantities and sells them further. Another collaborative solution is idea 5, where the center is used as a place for gathering the weavers to establish shared principles, for instance, regarding what price to set per bag to avoid competition among the women. The sixth idea is about using the center as a place for educating the weaving women on how to sell their bags for a larger profit e.g. demanding a certain price from the distribution people.

N4: Personal connection to the bag end customers.
This need can firstly be fulfilled if the women sell their bags at the market. The second idea is if the sales occur via a shop where the women do not directly see the end customer but can receive a personal description from the person working at the shop. Idea 3 is about using the

online platform to get a personal connection with the end customers. In idea 4 the customers buy the bags directly from the women, at their homes.

N5: Being able to sell resources efficiently.
In the first idea, people sell their resources at a local market by themselves. This involves the women carrying fish and other people with diverse resources. The second idea is similar except they sell it to a person working at the market. Idea 3 considers people from Morrumbene buying the resources from the people in Linga Linga and selling them further in Morrumbene, where the community center acts as a first distribution spot. In idea 4, a decided chapa transport desired resources to the market in Morrumbene, mainly considering the fish resources.

Transport
N6: A reliable, cheap, and pleasant travel to school.
One idea is to use the community center as a location for coordination of local investors who are interested in creating and maintaining a good road for their businesses, a so-called road association. The center can offer a meeting space for the investors to organize their efforts and resources. Also to connect with the local leaders along the road and thereafter deliver a suggestion to the district. In this way, if the maintenance plan is approved by the district, the road is improved, meaning other more comfortable transport options may become available such as small busses. More transport options would potentially lower the travel price, which is also done by a better road due to the less car damages. In addition, a good road makes the travel time more predictable. The second idea is to introduce a secondary school in the center. In this way, the students in Linga Linga do not need to travel to Morrumbene for their studies. The third idea is adjusting the educational system to allow students to study at the center a few days of the week, meaning saved travel expenses which may make it affordable for more parents to send their children to study above 8th grade.

N7, N8: Cheaper transport. A safer, faster travel time to/from Morrumbene.
The first idea for needs 7 and 8 is the same as for idea 1 in need 6, a road association. The second idea is to have a designated station for the chaps at the community center. It would potentially attract many travelers to the same place and therefore they could offer a direct ride to Morrumbene from the community center. This would enable a faster travel time, meaning a possible lower travel price. In addition, travelers do not need to search for a long time to find a chapa.

Education
N9: Become prepared for 8th grade.
This need concerns offering computer access for students to become prepared for 8th grade. The first idea is to have a computer hub at the center where the students share computers where the teacher teaches them. The second idea is about distributing computers to the school, but the center owns and stores them. In idea 3, computers are rented out to students over a longer period, for instance, a few weeks, where the center owns the computers. Here, the computers are used in school and at home. Idea 4 concerns volunteers coming a few times per year,

bringing computers and teaching students how to use them. The last idea concerns having a computer hub as part of a multifunctional space where other activities are also done.

N10: Become prepared for starting school.
In the first idea, children under 8 years can be left at a kindergarten at the center to learn basic things as preparation for 3rd grade. The second idea concerns the children who are brought to an education hub with their parents where both parties learn things simultaneously but about different things. Idea 3 also concerns the multifunctional space again. In this case, it can be used as a kindergarten during certain times of the day when their mothers are working.

N11: Education/work/hobby opportunities.
In this need, the education aspect concerns education for older children and adults. The first idea concerns having two designated places, one for education and the other for practicing hobbies, for instance, singing, dancing, and playing board games. The teaching can be done in various ways, for example, by paid teachers, volunteers, citizens themselves, or self-teaching by digital or non-digital tools. The second idea is once again having a multifunctional space for hobby and education purposes, where education can lead to work possibilities. The last idea is a better road which can improve the local work opportunities since it would attract investors such as hotels.

Communication
N12: Efficient information spreading.
The first idea is to introduce a billboard at the center to showcase relevant information to the people passing by, for instance, community meeting times, district department visits, fun activities, and other events. In this case, information spreading from citizen to citizen is avoided. The second idea is to have a visual place where activities are taking place, so it easily catches people's attention and thereby receives information about it. The third idea is the presence of certain people at the center providing information such as the leaders, Project Vita, and the district departments. The fourth idea is to have a person working at the center who gives information about activities to the people using the center or passersby. The fifth idea concerns the need for Project Vita to show the value of projects, where the center can be a place to incorporate them for people to see. This would catch people's attention and they may ask for further information.

N13: Efficient information receiving.
The first idea is connected to idea 4 in need 12, where the person working at the center is in contact with the district or the local leaders and supplies them with relevant information. The second idea is the presence of a district representative at the center who people talk to about certain matters and then pass it forward to the district. In idea 3, statistics are tracked at the center and relevant topics are sent digitally to the district.

N14: Appropriate workshop area.
This need concerns Project Vita's desire to have better workshop spaces. The first idea concerns the center providing a building for this purpose. In idea 2, these workshops take place in a multi-functional area at certain times. The last idea is about having an outside area for this purpose.

N15: Appropriate community meeting area.
Idea 1 is about offering a designated building for meetings with necessities such as chairs and a stage. The second idea is today's solution of having the meetings outside, however, on the community center's grounds where, for instance, chairs can be provided. The last idea is to have the meetings at the multifunctional area at certain times.

Tourism
N16: Relevant projects for volunteers.
In the first idea, volunteers participate in the building process of the center. The second idea is about volunteers engaging for educational purposes to the citizens where reoccurring labor is needed. The third idea concerns external researchers and companies seeing potential in testing their projects and products at the center.

N17: Increased tourism.
The first idea is connected to the ideas in N16, attracting volunteers, researchers, companies, and students to the center for various projects, meaning more external people in Linga Linga. The second idea correlates to the ideas for improving the road. A better road would appeal to more tourists to visit the established hotels in the area. In addition, it would attract more investors to Linga Linga, for instance, hotels, meaning even more tourist opportunities.

6.3 Concept selection

In this section, the selection of ideas is presented, which were based on the level of fulfillment to the need and its feasibility to implement.

Food quality
N1: Prolonging good's quality.
To fulfill the need to prolong food quality, idea 1 is seen as the most appropriate solution. Offering freezing spaces eases the fish carrier's job and potentially increases their salary since the quality persists. Offering fridge spaces is not as relevant for the fish carriers since the fish would not stay cool until reaching Morrumbene, meaning the quality can be affected. In addition, vegetables and fruits may not be as desired for people to store in a cooled environment since these goods can be stored in a more chilled area at people's homes, as today. The decrease in quality is not as evident and important for these goods. However, if the desire rises, fridges can be implemented at a later stage in the center.

Local trading

N2: Access to certain groceries to avoid transport.

To fulfill need 2, idea 1 was chosen which is a marketspace at the center since this would enable people to access desired groceries. The distribution to Linga Linga is done by a chapa driver buying the groceries from Morrumbene because it is perceived as the most financially feasible solution compared to each market seller having a chapa travel cost or making the individual market sellers of Morrumbene drive to Linga Linga, who probably do not have direct access to a car. During the rest of the day, the grocery-delivering chapa can transport travelers as today. The reason for not involving the district in controlling the deliveries is to prevent the income from going to the district instead of the local actors of Linga Linga. The second idea did not seem as feasible since it would require much planning and logistic work, which would risk errors and a higher cost.

N3: A higher income for weaving bags.

To meet the need for an increased income for the bag weavers, idea 4 was chosen. The women should be able to sell their bags for a higher price to an employed person at the center who controls the sales to the distributors and makes sure the fair and agreed upon payment is done. In this way, the women would gain more money per sold bag since the employed person is not driven by profit due to their fixed salary. Also, competition would be avoided. In addition, the women should use the center as a location to meet and discuss their business together with the employed person to improve the operation for all involved, as part of idea 5.

The first and second ideas did not seem reasonable since there are not enough bag customers in Linga Linga to satisfy all women's creations, meaning the bags must keep being distributed to other locations where the buyers are. Idea 3 seemed unreasonable since it probably would require a full-time working person in the process other than a distribution person, meaning high expenses. Idea 6, regarding educating the women seemed to be met by combining ideas 4 and 5.

N4: Personal connection to the bag end customers.

Idea 2 was perceived as the most relevant idea to give the women a personal connection to the end customers, even though they may not hand over the bag directly to the end customer. However, the person working at the shop could take orders from the customers which is then distributed to the weaving women. Thereafter the women can choose if they want to deliver the bag directly to the customer. The bags sold at the shop should be combined with the sales channel of idea 4 in N3 since there are not enough customers in Linga Linga. This is the reason why idea 1, regarding the women selling the bags at a market, is not reasonable. As mentioned in N3, an online sales platform, as idea 3 considers, is not financially feasible. Lastly, idea 4 is a way of establishing direct contact with the end customer. However, it would become difficult for the end customers to navigate to the producers and this idea would also create competition among the women.

N5: Being able to sell resources efficiently.
To fulfill this need, idea 2 was chosen where people sell their resources to people working at the market, who sell them further to Linga Linga customers. In this way, people can also have another occupation. However, people should have the opportunity to sell their resources themselves at the market. This may be relevant for people who have many resources and see a financial possibility in doing it full-time to the Linga Linga customers. In addition, idea 4 is incorporated, where the women with fish resources can sell it to an employed person at the center who stores it in the freezer. Thereafter the fish is transported by one designated chapa to Morrumbene where the sales to the market sellers occur. This solution provides an incentive for the women with the fish to use this service since they save money due to the removal of transport to Morrumbene. The other resources that have been provided to the Linga Linga market can also be distributed to the market in Morrumbene if there is a desire from there.

Idea 1 is not plausible since there are not enough customers in Linga Linga to make it financially feasible for all people to do sales at the market full-time. Idea 3 is interesting since there is a shift of transport from the Linga Linga producers to the Morrumbene buyers. However, the market sellers of Morrumbene most likely do not own cars, meaning they have to go by chapa. This means the problem is shifted towards them instead, which is not a reasonable solution.

Transport
N6: A reliable, cheap, and pleasant travel to school.
To meet this need, idea 1 was chosen, meaning the center acts as a coordination location for the local investors to plan and maintain the road, a so-called road association. This idea would give the students more transport options, lower travel expenses, and offer a more pleasant travel. The second idea of having a secondary school at the center seems irrelevant since there is no reason to locate it there, compared to other places of Linga Linga. It would require a large facility and may also generate too much movement of people in this limited area. However, a school in Linga Linga may still be relevant, but in another location. The third idea is less feasible since it requires large changes to the educational system and approval from the district.

N7, N8: Cheaper transport. A safer, faster travel time to/from Morrumbene.
Need 7 and 8 are met by a road association, presented in the previous need. Idea 2 regarding a chapa station in connection to the community center also seems relevant for travelers. If the road association works and the road's quality is improved, the station could be used for other transport options such as busses.

Education
N9: Become prepared for 8th grade.
Idea 5 was chosen to fulfill this need, meaning computers would be accessed in a shared computer hub as part of a multifunctional space. In this way, the computers are controlled within the center that owns them. If the computers were brought outside the center, as in ideas 2 and 3, it would reduce the center's control and increase the risk of damage to the computers. Idea 4, of involving volunteers in bringing and teaching about computers, is assessed to become too seldom to satisfy all students in 7th grade. The first idea also has a computer hub, but a

stand-alone one. This would probably result in too much non-usage of this facility during the days, therefore a space used for multiple activities seems more relevant.

N10: Become prepared for starting school.
The tenth need is satisfied by idea 3, by offering a kindergarten in a multifunctional area at the center. The motivation for having the kindergarten here instead of another location is because the mothers can leave their children there in connection to another activity at the center, for instance, buying groceries. The reason for not offering a space only for kindergarten purposes is that it would probably be unused parts of the day. The second idea is also discarded since it would not give the mothers a chance to conduct their work without their children.

N11: Education/work/hobby opportunities.
Need 11 is faced by offering older children and adults education and hobby opportunities in a multifunctional space. The reason for having one space with multiple activities rather than several spaces is to optimize the usage of the space. Regarding the lead of education or the hobby activity, several options are seen as relevant. Teachers and other actors can get paid by the participants, volunteers can educate people for free, also self-teaching solutions, such as using computer programs can be useful. The idea of citizens teaching others was questioned during the co-creation workshop due to the risk of isolating and favoring certain people. However, the citizens are perceived as valuable assets to include in education activities due to their knowledge. Therefore, it could be tested at the center with an additional person, perhaps an employed person, to prevent this issue from happening. The last idea, of improving the road's quality to attract more investors, is also seen as a way to give people employment chances. The already mentioned road association is a possible solution to this.

Communication
N12: Efficient information spreading.
To meet this need all presented ideas are chosen since they do not compete against each other. Also, it is good to have various channels for spreading information to reach out to many people. Incorporating a billboard at the community center area, as idea 1, allows for spreading information to people walking by. However, reading skills are required which everyone does not possess. Therefore, it is complemented by oral communication in this area by actors who want to spread certain information, which ideas 3 and 4 involve. The community center is a preferable place since it would become a natural meeting point for people. The second idea is to strategically place the activity spaces visually to people passing by to catch their interest and thereby spread information. The last idea also considers spreading information through vision, where projects are showcased at the community center to spread information about project value.

N13: Efficient information receiving.
Need 13 is met by combining ideas 1 and 3. The first idea makes the local leader receive useful information from the employed person at the center which can support his work and where he can forward it to the district. Idea 3 seems also relevant since keeping statistics is useful for various reasons, for instance, to assess the solutions at the center and adjust them if necessary.

Idea 2, of involving a district representative in the center did not seem appropriate since it would give the district more influence on the community. The local leader is a more relevant person where information can be spread forward since he is perceived as the representative of the citizens of Linga Linga.

N14: Appropriate workshop area.
To satisfy Project Vita's desire for a workshop area, the second idea is chosen, meaning workshops take place in a multifunctional space. This choice is reasonable since in idea 1 the use of one designated workshop space would not be optimized and in idea 3 an outdoor space would not be optimal for all workshop activities. A space with the possibility to remove distractions such as weather events or people passing by is preferable.

N15: Appropriate community meeting area.
Since community meetings only occur a few times per month today, and will not be changed to a daily basis, it seemed relevant to conduct these meetings in a multifunctional space, as idea 3 suggests. This solution would attract more participants since it is a determined location, the meeting is seen by people passing by, and it is a place adapted for this occasion with seats and weather protection. In idea 1, regarding having a space only for community meetings, the facility would be unused for the most part. The second idea, about keeping the meetings outside, would not be a significant improvement from today's meetings.

Tourism
N16: Relevant projects for volunteers.
To meet need 16, all three ideas are seen as applicable and do not interfere with each other. In the development stage of the center, volunteers can participate in the building process to an extent that does not interfere with the involvement of local actors since it is essential for establishing a feeling of ownership for the locals, as heard from the Pre-study. Idea 2 is also relevant because involving people in the education hub is needed. Idea 3, involving researchers and companies to do user studies at the center, is seen as a good opportunity for these actors to develop products and help people in similar situations as the Linga Linga inhabitants. An important consideration is to only allow this to actors with good intentions.

N17: Increased tourism.
Both generated ideas are viewed as ways to increase tourism in Linga Linga. Involving volunteers, companies, researchers, and students in various projects creates "sustainable tourism" in the area. A better road would also attract more investors and tourists. Tourism may create more job opportunities for the citizens, however, it may also lead to a population increase. Therefore, an important consideration for the district is to allow a sustainable expansion of tourist operations to not interfere with the current citizens.

7. Final concept

This chapter presents the final developed community center which offers seven services that together fulfill the needs of the stakeholders. The services are access to freezers, a market, a shop, a road association, a multifunctional hub, a communication hub, and a transport station. An overall system of the services and their connections is seen in Figure 25.



Figure 25. Visualization of the community center services and their connections to each other.

Service 1 – Access to freezing spaces

The offer

The community center has freezing spaces for two main purposes: to ease the work for the fish carriers and to prolong the food's quality for the general public. Therefore, the center has two freezing spaces, one for the fish delivery operation and one for the public.

How it works

The fish carriers buy the fish from the fishermen and transport it to the center once a day where it is sold to the center employee for a pre-determined price, based on two factors: kilogram and fish size. The fish is then placed in the freezer by the center employee.

The citizens of Linga Linga can rent a compartment in the freezer and monthly pay per stored liter. The center employee makes sure the service is handled smoothly.

People

- Women carrying fish – sell the fish to the community center.
- Citizens – store food in the freezer.
- Center employee 1 – controls that the freezer service runs smoothly.

Opportunities

- Fish carriers gain more income.

The service enables more income for the women carrying fish for several reasons. Firstly, the women save daily expenses by avoiding the travel to Morrumbene. Secondly, the several hours saved by not doing this travel can be utilized to work more, for instance, weaving bags and thereby gain more income. Thirdly, fewer fish are thrown away due to poor quality since the walking time from the fishing site to the center is shorter than the travel to Morrumbene. This effect is most visible during the days when many fish are caught. Lastly, the women get a fixed payment per kg fish, meaning the days when many fish are caught they do not receive a lower income per fish, as today.

- Less dependency on the daily fishery.

The daily variation of fish-catching success among the fishermen does not impact the women's long-term income, as a stock of fish is established in the freezer, and ready to be sold further.

- Healthier eating.

The risk of eating poor quality food is lowered for the citizens who choose to store their goods in the freezer. This means there is less chance of getting sick.

- More varied diet.

Citizens who freeze their goods have the opportunity to vary their diet, as cooked meals can be stored for the long term.

- Saved time for citizens.

Time will potentially be saved for women who cook food since the cooking time is reduced when heating chilled pre-made food rather than cooking new dishes daily. The saved time can be utilized for working more or participating in any of the activities at the center.

- More sustainable fishing.

The higher value of every fish can influence the fishermen's fishing operation to become more sustainable to secure fish in the long term.

Challenges

- Risk of losing the women's fish-carrying occupation.

There may be a risk the job of the women carrying the fish disappears if the fishermen see the shorter transport distance as an opportunity to take over the women's responsibility for delivering the fish to the center. However, the center can establish a policy to prevent this.

- Expense for the citizens.

To run the service it is estimated to require a fee for people using the freezing storage, which some people may not be able to afford. However, the service aims to help the people, so the fee for using the service should be as low as possible and only cover the necessary expenses of the operation.

- Fishermen keep the same fishing amount

There is a chance the fishermen maintain their fishing amount since they see an opportunity to earn more money, meaning the fish population is not strengthened. However, this risk can be mitigated if they prioritize a secured fish population in the long term.

- Fishermen raise the fish price.

If the women gain more money per sold fish, the fishermen most likely raise their prices, meaning less income for the women. However, the new pricing model could benefit all parties.

Societal impact

- Improved life standard.

The increased income for the fish-carrying women means more opportunities to improve the household and enhance the life standard for themselves and their children, such as affording education expenses.

- Healthier population.

The lowered risk of consuming poor-quality food makes the population healthier.

- Strong fish population.

Potentially a strong fish population, meaning secured fish access for generations of fishermen to come.

Service 2 – A market

The offer

The community center offers a micro market for local trading, where people can access necessary groceries and sell their own resources. It is also a distribution space for transport of resources to the market in Morrumbene.

How it works

The citizens access groceries, including frozen fish, at an inside market which are distributed by a pre-designated chapa who purchases the groceries at the Morrumbene market. The chapa and center employee 1 daily check the inventory and determines the amount to deliver the next time.

The citizens with resources make the sales to the center's market, for the other citizens to purchase. The chapa also distributes desirable resources, such as frozen fish, further to the Morrumbene market. This means resources are delivered back and forth between Linga Linga and Morrumbene, on a basis to please the demand in Linga Linga and Morrumbene.

People

- Market buyers – citizens who want to access groceries.
- Market sellers – sell the groceries to the citizens.
- Center employee 1 – controls the market's inventory and delivery.
- Chapa driver – distributes the resources between Linga Linga and Morrumbene.

Opportunities

- Citizens in Linga Linga access their desired groceries.

A market in Linga Linga enables the citizens to access the groceries they normally purchase in Morrumbene. Moreover, a market gives all citizens, including the ones living furthest away, the opportunity to access the groceries since the previous distribution of sellers walking around is avoided.

- Market buyers saves money and time.

By utilizing the local market, citizens can access their desired groceries without travelling to Morrumbene, meaning saved travel expenses. The price of groceries in Lings Linga will be slightly higher than purchasing it directly from Morrumbene. However, not as high as the round-trip expense, 120 MZN (20 SEK). The avoided trip also means saved time, approximately two hours, excluding the waiting time for the chapa, which can be long. The saved time can be utilized to work more, meaning more income, or perhaps use the center's services.

- Socialization and enjoyment.

A market is a natural meeting spot for socialization and enjoyment for the sellers and buyers.

- Citizens with resources earn money.

Citizens who possess resources but cannot sell them due to the lack of a trading place will now be able to sell them at the market and earn money. The sellers who currently walk around selling goods can distribute them to one place instead, meaning saved time.

- Improved job for the market sellers in Morrumbene.

It is probably preferred by the market sellers in Morrumbene to receive the goods once a day at a recurring time rather than in smaller amounts during an uncertain time.

- Comfortable environment.

The market will be indoors to be less affected by weather, such as rain and strong sunlight, which can attract more customers, even during bad weather.

- Healthier eating.

The Linga Linga or Morrumbene buyers of the fish will have a lower risk of eating a bad quality fish when it is distributed frozen, and likely, also sold frozen.

Challenges

- Much logistic planning.

The distribution of goods calls for more logistic organizing due to the inventory checking and involved actors. Regular meetings, open communication, and experience can mitigate the risk.

- Risk of fewer customers for the chapa drivers.

When the citizens purchase their groceries locally and the fish-carrying women sell their fish to the center, the chapa drivers get fewer customers. Therefore the chapa drivers may have to compensate for this by increasing the travel price.

- Risk of fewer customers for the kiosk owners.

A market in the village creates competition for the kiosks. The market may offer groceries for a lower price meaning the kiosks would receive fewer customers. Therefore, the kiosks may have to adjust their business, for instance, selling groceries that cannot be found at the market.

Societal impact

- Healthier population.

The lowered risk of consuming bad fish in Morrumbene makes the population healthier.

- Improved life standard.

The sales of resources, and saved time and money for the market buyers, mean a better financial opportunity to improve their household or afford education for their children.

- Strengthened community.

When the citizen's trading occurs in Linga Linga, and the Morrumbene market becomes dependent on the chapa transport, the Linga Linga community gets strengthened.

Service 3 – A shop

The offer

The center includes a shop for sales of local crafts, for instance, weaved bags, honey, and coconut oil. The bag-weaving women also have the possibility to sell their bags to the center.

How it works

The bag-weaving women get paid per sold bag to the center employee at the shop, who places the bags in a storage space. Thereafter, the center acts as a distribution space because the employee sells the bags further to the various buyers of Linga Linga, who transport and sell them at tourist locations. In addition, bags and other local crafts are also directly sold at the shop to tourists. The bag weavers should also receive a description of the end-customer from the center employee if they desire it. The tourists can also order personal bags, which the center employee contacts a relevant bag weaver, who delivers it personally to the tourist when finished.

People

- Bag-weaving women – sell their bags at the center.
- Other crafters such as Linga Linga's bee farmer – sell their crafts to the center.
- Center employee 2 – controls the bag weaving sales operation and the shop.
- Distributing bag buyers – purchase the bags from the center in large quantities on single occasions.
- Tourist – the potential buyer of the crafts at the shop or personally orders bags.

Opportunities

- Higher income for the bag-weaving women.

The service allows the women to sell their bags at a pre-agreed upon price which should be higher than the current sale price to the distributing bag buyers in Linga Linga. The service makes sure the bags are sold in large quantities on single occasions to the distributing bag buyers, meaning the bags can be sold for a higher price to them.

- Local community identity.

The shop expresses the community's identity by showcasing and selling local crafts. Therefore, the shop becomes similar to a local museum for the citizens to be proud of, by recognizing and valuing local artists. It can also contribute to a feeling of ownership of the center among the citizens, which means they may be more eager to use the center's services.

- Personal connection for the bag-weaving women.

The bags that get sold to tourists at the shop are communicated to the bag weaver who can observe the usage if the tourist stays in Linga Linga. The shop visitors who order bags enable the women to deliver it personally to them.

- More secure business for the bag weavers

The women who sell their bags to the center leave the responsibility of delivering and contacting the distributing bag buyers of Linga Linga. Once built, the center will become a strong actor in the community. This position means the distributing bag buyers have less power to influence the price, which they may do today when purchasing the bags from individual women.

Challenges

- More logistic planning.

This service requires logistic planning between the three key actors: bag weavers, center employee 2, and the distributing bag buyers. Regular meetings with the involved actors can mitigate the risk or errors.

- Bag weavers critical of changing their business.

Some bag-weaving women may be initially critical to a new business strategy. However, if they understand the financial benefits and improved income security, it may convince them.

- Unsatisfied distributing bag buyers.

The distributing bag buyers may be critical to a higher price per bag. However, they may see the value of their work being simplified due to the change of having one pick-up spot and one contact person, compared to today's several pick-up points and contact people.

Social impact

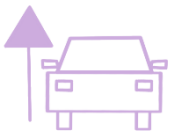
- Improved life standard.

The higher income for the bag-weaving women opens possibilities for improving their life standards, for instance, securing electricity for the household.

- Proud community.

The shop's recognition of Linga Linga's crafts and artists can make the citizens strengthen and feeling of proudness.

Service 4 – Transport station



The offer

The center has a designated station for chapas along the road to pick up travelers who want to go in the Morrumbene direction.

How it works

Citizens who want to travel wait for chapas at an allocated spot with benches next to the community center. The chapas pick up the travelers and if enough people are traveling to Morrumbene, the trip goes directly to this destination.

People

- Travelers – citizens who want to travel.
- Chapa drivers – transport the citizens.

Opportunities

- Potentially cheaper and faster travel.

The community center is a meeting spot for potential travelers. If enough people join the chapa there, it can offer a direct trip to Morrumbene, reducing the travel time since travelers are avoided along the way, meaning potentially a lower travel price.

- Clear departure location

Travelers do not have to search for locations where chapas will stop. It becomes security for the traveler that the chapa departs from the center's station.

- Simplified job for the chapa drivers.

The station makes the chapa driver's job easier since many people join at one location rather than needing to frequently stop along the trip.

- Comfortable waiting area.

A station next to the center, equipped with a bench, lightning, and possibly a roof makes the waiting area more comfortable for the travelers.

Challenges

- Travelers further away are left out.

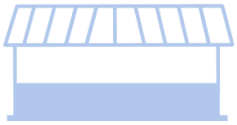
The direct trips to Morrumbene can disturb the other Morrumbene travelers along the way. Other chapas would still transport the other travelers, but it may be fewer available ones.

Societal impact

- A more developed public transport system.

If the station becomes accepted, similar stations can be spread along the road where travelers can gather to ease the transport further. In addition, some approximate timetables at the stations can be set to ease the travel.

Service 5 – Multifunctional hub



The offer

The community center supports various activities by providing spaces for people to gather in larger groups. The service uses multifunctional buildings and includes a computer hub, education hub, kindergarten, hobby space, workshop space, and meeting area.

How it works

The multifunctional hub will include two spaces: one larger than the other. The large space will be used for activities with many participants, for instance, community meetings. The space must have the capacity to accommodate over 100 people. The smaller space is offered for activities which are preferred to be done in smaller groups, for instance, workshops. It should have a maximum capacity of around 20 people. Both spaces are multifunctional, meaning they are flexible to handle various activities, from music events to formal meetings with the mayor. The multifunctional spaces are also flexible to be able to meet unforeseen needs and future needs. The activities are carefully planned to fit the schedule of the two spaces. To avoid conflicts due to society's structure, education, and hobby activities are divided into men and women groups and also younger and older groups.

Activities large space:

Meetings

Meetings involving the local leaders and Linga Linga citizens, occasionally also district departments and the mayor, sitting on benches under a roof to create a comfortable space. Other relevant meetings are with the road association, center personnel, bag weaving operation, and other community leaders.

Kindergarten

The larger space requires adaptive elements to be suitable for a kindergarten. This service is open from the morning to the afternoon, to secure time for the other activities in the large space. To handle as many children as possible, the kindergarten will be divided into two-time sessions: before and after lunch, similar to the school in Linga Linga. The kindergarten will have some employees whose salary is financed by a service fee to the parents.

Education hub for large groups

The large space is used for education events that attract and can handle many participants. The time of the activities is in the afternoon to evening, to not interfere with people's work and other duties. The teaching is handled by volunteers and citizens with knowledge of the area. The education hub should offer long-term teaching solutions, for instance, self-learning, which is developed by the volunteers. The teaching techniques can vary depending on the topic, for instance, reading, listening, and using computers. A small fee per participant is paid to finance teaching citizens. In addition, an available projector supports the teaching and can be used to show news broadcasts or documentaries. Some relevant education topics are:

- Learning how to read
- English, French and Portuguese

- Farming
- Female entrepreneurship

Hobby activities for large groups
Hobby-related activities attracting many participants or requiring a large area, such as singing, dancing, and movie nights, are held in the large space. These activities are run by the citizens themselves and the center employees make sure it runs smoothly.

Larger gathering events
Occasionally, large events such as local traditional parties are held at the center's facilities, where other surrounding communities are allowed to join.

Activities small space:
Computer hub
The teachers of Linga Linga's school will use the small multifunctional space to access and teach 7th graders computer usage in small groups. The walk from the school to the center is not long. The computers will also be available to the citizens for a small fee.

Workshops
Students and researchers conduct workshops with people in the smaller space due to the fewer attendees.

Education hub for small groups
The small space can be utilized for education where smaller groups are more applicable, for instance, sewing, computers, and cooking healthy, cheap, and nutritious food.

Hobby activities for small groups
The hobby events that do not attract as many people or do not need much open space, for instance, board game activities, are held in this smaller space.

People

- Local leaders – their work is involved in the center.
- Morrumbene district departments – conduct meetings at the center.
- Bag weaving women – utilize the meeting area for improving their business.
- Citizens of Linga Linga – all ages are invited to participate in the various activities.
- Road association – utilizes the meeting area for coordinating their operation.
- Volunteers – their projects are connected to the various spaces.
- Researchers – use the center as a pilot study.
- Center employee 2 – makes sure the activities are working smoothly.
- Kindergarten personnel – are responsible for the kindergarten service.

Opportunities

- 7th-grade students become better prepared for 8th grade.

The access to computers means the students in 7th grade learn the practice of computers which makes them better prepared for starting 8th grade.

- Young children become prepared to start 3rd grade.

The children who go to kindergarten will learn basic elements to prepare for starting school. They will also socialize and have fun with other children.

- Mothers can focus more on their work.

A kindergarten means that mothers with young children have more time by themselves where they can focus on work or other duties such as purchasing groceries. Thereby the possible higher income means they can, for instance, afford the kindergarten fee.

- Adults are educated.

The education hub enables adults to develop and learn new skills which can create work opportunities. By showing news, people become aware of national and international events. By having a designated location for education, the women avoid distractions compared to learning from the household.

- More volunteer opportunities.

The education hub creates more projects for volunteers and the workshop space supports researchers and collaborating students in their studies.

- Uplifting local interests and traditions.

The service endorses people's interests and local traditions which brings joy to the community. The group activities strengthen the community feeling within the village.

- Bringing communities together.

The occasionally larger events, where other villages are invited, bring communities together and welcome more engagement and empowerment. Previously, a large festival event was held in Linga Linga, where local actors were engaged, which attracted people from Morrumbene.

- Community meetings better organized.

A designated location, regular events, and more efficient communication make the community meetings more organized.

- More attendees at community meetings.

A designated and well-adapted area for community meetings, means more attendees are likely to join. This makes the citizens more aware of the events in the village and empowers democracy since more people learn how to vote and get the chance to raise their voices. It is also appropriated by the leaders, the district, and the mayor since it shows respect and increases their influence.

- People feel ownership.

Meetings are held with the local citizens to hear their thoughts on how the center can be improved. This creates a feeling of ownership and power to impact.

Challenges

- Logistics.

A challenge with multifunctional spaces is the planning of activities. There are many potential activities and actors involved that require logistic planning by the center employees. However, having regular planning meetings and starting with a few activities, can ease the planning and prevent bigger errors.

- Communication.

It is important to communicate the information and times of the activities so it is understandable to the public. If not, it will lead to confusion, few attendees, and the center will lose its credibility.

Societal impact

- More educated citizens.

This service makes the citizens more educated, opens more work opportunities, and potentially increases their income. The community's standard will be strengthened and valued higher in the district. For young and older students, the service lays a better groundwork for further studies.

- A happier people.

The endorsement of hobbies and local traditions gives the citizens more joy and well-being in their daily lives.

- Spread of projects.

The meeting space and the center in general, provide an incentive for other community representatives to visit to become inspired and see the value of developing a similar location elsewhere. This means the community center opportunities will reach more people.

Service 6 – A road association



The offer

The center offers a meeting space for the local investors to plan and coordinate the efforts to maintain the road. The center's role is to assist in the development of a road association.

How it works

The investors who have an incentive for a well-functioning road regularly meet at the center's meeting space in the multifunctional hub. They plan and coordinate how the district's monetary contribution and local knowledge can be utilized to improve- and thereafter maintain the road's quality. The center's role is to provide a meeting space while the investors are responsible for the meetings and execution of operations.

People

- Local investors along the road – plan and assist in the execution to improve and maintain the road's quality.

Opportunities

- More investors.

A proper road opens many possibilities. More investors, such as hotels, see the opportunity to establish businesses in Linga Linga and the nearby villages due to the location's tourist potential. More tourists means the current investors can expand their business, more job opportunities for the citizens, and the money is potentially kept within the villages. In addition, more tourists mean more customers to the center's market and shop, where the bag weavers and fishery can earn more money if the resources are sold within Linga Linga, compared to transporting and selling them further.

- Wealthier district.

Morrumbene district would also benefit from a better road since more investors and tourists generate income for them by selling land and receiving taxes. The increased income can be invested in developing necessary fields, for instance, schools and health care.

- More transport options.

A good road opens additional transport options, other than pick-up trucks with 4-wheel drive. The mobility options can be better adapted to the traveler's purposes, for instance, offering cargo spaces when buying large items. In addition, biking can become a cheap and flexible transport alternative for students and others.

- Cheaper travel price.

A road where cars do not take as much damage and where there are several transport options decreases the price for travelers. This means more people can afford the trip to Morrumbene and more parents can send their children to school.

- The investments are kept within the community.

If the Morrumbene district chooses to contribute to local actors for maintaining the road instead of external companies, the money is kept within the community and it also creates local jobs. In addition, the local actors have an incentive to maintain the road well since they are the users of it, which the external companies do not.

Challenges

- Large investments for the road association.

Initially, there may be large investments needed for the road association to start improving the road, for instance, concerning tools and filling materials, which an external company already possesses. This cost may become an obstacle to the district to see the possibility of utilizing local actors. However, when passing this initial cost it would be cheaper in the long term since maintenance is more economical rather than the current quick fixes happening on too rare occasions.

- Inexperienced local actors.

Even though the villages around the road possess much construction knowledge, the planning and coordination of how to operate on a large scale may be uncommon ground. However, it may be a learning process that can be assisted by the district and volunteers.

- Too high immigration.

If too many investors and tourists enter the area in a short time it can damage the local identity. The job opportunities may not be accessed by the current population if outside, more educated people see the option to relocate and work there. However, the district can manage a sustainable expansion where the locals can adapt and access the opportunities a better road offers.

Societal impact

- Improved life standards.

If the travel costs decrease and more people get employed, it leads to more income for the residents where they can, for instance, improve their household.

- More developed community.

A better transport system and more local investments means the community can develop in several aspects, for instance, electricity and water access.

- More educated citizens.

If parents can afford the travel expenses for their children to study in Morrumbene and further studies, their more educated children get more life opportunities.

Service 7 – Communication hub

The offer

The community center is a point of communication and information spreading. It is a location where people meet and where local leaders and the center personnel provide necessary information to the people. The center also includes billboards on the property to spread information about local events. In addition, relevant and appropriate statistics are being communicated to various actors.

How it works

People who visit the center for various purposes, gain information through different channels. The local leaders are stationary in the area occasionally communicating about happenings such as the time for the next community meeting or upcoming weather hazards. The center personnel are also a source of information to the visitors. The billboards communicate information through vision and the public is allowed to publish their flyers on it, for instance, about local party events. The visitors also gain information by visualizing solutions at the center that attract their interest, for instance, the previous Project Vita thesis project about glass recycling (Ramm and Sjöstrand, 2023). The solutions are equipped with a descriptive sign regarding the benefits, the cost of implementing it, and who to talk to if interested. The statistics are kept by the center personnel, for instance, regarding the flow of visitors to the various activities.

People

- Local leaders – communicate and spread relevant information to the visitors.
- Center employee – communicate, take statistics, and spread relevant information to the visitors.
- Citizens – visit the center, express themselves, and receive information.

Opportunities

- Information is reached to more people.

The center makes information more available for the people since it is spread at the potentially largest meeting spot in the village. In addition, the information is provided through various channels. This makes the citizens more knowledgeable about community events.

- Incorporate sustainable solutions.

When people see the value and potential of solutions at the center, it may lead to incorporating them into their household.

- Communication of important statistics.

By keeping statistics on relevant topics, it can be used to communicate the benefits to relevant actors such as other community leaders and charity organizations, to implement similar projects. Statistics are also beneficial for analyzing and improving the center's operation.

Challenges

- Statistics handled poorly.

One potential issue is if the statistics are handled poorly and distributed to people with bad intentions. Therefore, only non-sensitive statistics are kept, and the information receiver is carefully selected.

Societal impact

- More community centers.

Communicating the statistics and utilizing the meeting space for other leaders, can inspire other communities and organizations to implement similar centers, adapted to their needs. This would mean a larger scale of positive impact.

- More democratic society.

If information is delivered to more people in the village and people act upon it, it creates a more democratic community.

Community center overall

Other than the seven services, the community center offers electricity access for charging smaller devices such as phones, for a fee per hour. The center also includes toilets for the users. In addition, a lighted environment including streetlights is provided, to allow activities in the afternoon and evening. The light should also make the visitors feel safe and comfortable inside and outside the center. Another safety aspect is the center's contribution during extreme weather such as cyclones. In this case, the center's role is to create a secure space for people who cannot access cyclone-safe buildings.

The center and its services require several actors to function:

- **Center-employee 1.** The first employee of the center is a person who is responsible for the fish storage operation, paying for and bringing the fish to the freezer. The person is also involved in the inventory of the market goods and controls the resource delivery by the chapa driver. This person should preferably be a man to possess authority in some situations.
- **Center-employee 2.** The second employee is the person responsible for the weaved bag operation, by being in contact with the collaborating weavers, bringing the delivered bags to the storage, and selling them further to the buyers. The person is also in charge of the shop, as well as the planning of activities and ensuring they are working properly. This also involves keeping simple statistics of the participants and services. This person should preferably be a woman to endorse women's power and to create a comfortable relationship with the bag-weaving women.
- **Market sellers.** Several designated sellers provide groceries to the citizens.
- **Kindergarten employees.** A few people are needed to take care of the kindergarten service.
- **Chapa driver.** The chapa driver transports the Linga Linga resources to Morrumbene and handles the payments, as well as purchases and delivers groceries from Morrumbene to Linga Linga.
- **Project Vita.** The organization is the owner of the center. They control the overall operation and make the larger decisions. Project Vita also coordinates the volunteers, researchers, students, and external companies.

Financials

The center has expenses and income to function. The expenses involve the salary for center-employee 1 and 2, and kindergarten personnel, as well as maintenance and cleaning. To finance this, several channels generate income for the center. Firstly, people must pay a fee to access the computers, consume electricity, access freezer space, and participate in certain educational events. Secondly, there should be an added percentage to the price of the sold groceries at the center's- and Morrumbene market. There is also a percentage of the fish and bags bought from the citizens. The objective is not to gain as much profit as possible, it is rather to finance the

center's operation and make life cheaper for the Linga Linga citizens. Donations are a way to provide the services with the necessary appliances to function, such as computers, internet, and a solar power system.

Service Importance

The importance of each service was estimated based on the expressed frequency of the needs and the level of contribution to the user, as visualized in Figure 26. The service with the most importance is the multifunctional hub since it covers several areas and needs which was expressed in the interviews. The most valued services within the group are the computer hub, education hub, and meetings. The kindergarten, hobby activities, and workshop space are lower since they do not have an equally large potential contribution to the citizens. The second most important service in the center is the market and road association. A market was highly desired by the interviewees and it would have a great positive effect on the citizen's daily life and the community. The road association would be highly impactful if it worked in reality and reached the goal of a better road. Thirdly most important is access to electricity, freezers, and the communication hub. Electricity would ease people's lives but would not have a huge effect on the community. Access to freezers would simplify the women carrying fish and other citizens, but it would not be used for all. The communication hub would make information reach out to more people and ease the leader's job, however, it does not affect as many people as the market. Fourthly, the shop, mainly contributes to the bag-weavers. Lastly, the transport station simplifies the traveler's and chapa driver's journey but has a limited impact compared to the other services.

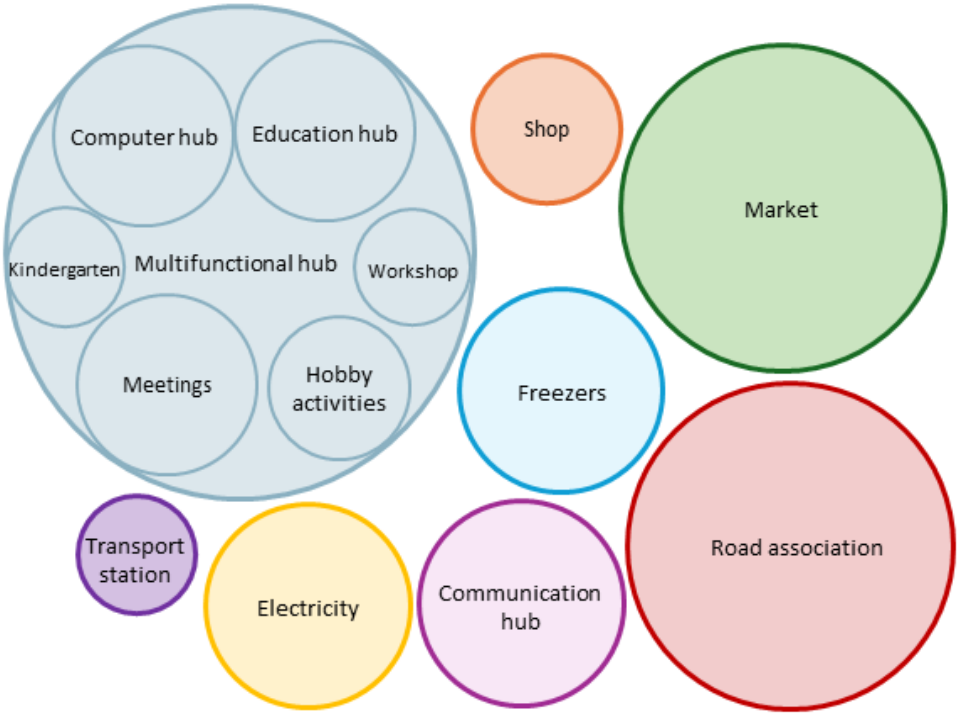


Figure 26. Visualization of the service importance in the community center where a larger size means a higher importance.

8. Discussion

This chapter discusses the study regarding the developed concept, the answer to the research questions, and the execution of the project.

8.1 Concept discussion

The developed community center concept is estimated to have a positive impact in various ways on the citizens of Linga Linga. Even so, some challenges in the community are highly complex and rooted in society, also referred to as wicked problems, which require large-scale societal changes to be possible to resolve. These challenges include for instance corruption, gender inequality, and non-democracy, which falls into the D.40 design context of systemic design by Jones and Kijima (2018) as presented in chapter 3.3. Due to the project's limitations, the scope is within the D.20 context, meaning the wicked problems are not the targeted areas. Nevertheless, the concept is still useful since it benefits the community economically, socially, and environmentally, and empowers the community which may support the resolving of certain wicked problems. An example is gender inequality, which can be addressed if the men see the value of more income to the household when the women educate themselves further. Another example is the empowerment of the rural community which can strengthen democracy by influencing the district and other governmental arms. Furthermore, the concept targets large-scale challenges by adopting a holistic approach to solving many problems simultaneously, which was seen as important by scholars from Chapter 3.2 and the practitioner ICLD.

The answers from the user study are based on the current situation in Linga Linga, for instance, the presence of a poor road. If the situation was changed, the role of the center would therefore also change. The need for a local market would change if the road got improved since the travel to Morrumbene would become more economically and time-pleasant. However, it is assumed people prefer to access goods locally to avoid transport even if it is more expensive overall. In addition, the center's facilities and services should be adaptable to changed and future needs, caused by external factors or by the center itself.

During the concept development, there has been a following dilemma regarding how much to influence the local culture. Bringing a Western culture and values to a different context challenges the designer's work. Perspectives about gender equality, animal welfare, and hierarchy are contrasting, meaning it becomes a question about designing to the local cultural values or bringing a Western influence. In the final concept, both perspectives were considered, for instance, empowering women was seen as important to not only supporting those individuals but also the community. Another dilemma has been the district's level of influence. While increased tourism is seen as an opportunity for job creation and the sale of local crafts, it is crucial for the district to ensure any expansion is sustainable and does not interfere with the local community.

The community center concept is built upon changing the current practices for various occupations. The bag weavers and fish carriers sell their resources to the center which distributes them further. These system changes can be challenging for them since they have done their work in a certain way for a long time, meaning they have experience and knowledge in this area. In addition, the practice can originate back several generations. However, if the individuals see the value of changing their business, for instance, gaining a higher income which was an identified driver, they may be willing to change their work process.

The expressed purpose of community centers by Tony et al. (2019), presented in chapter 2.2, is found to apply to this project's concept. The expressed purpose was to improve life quality, socially, economically, environmentally, recreationally, and educationally, all of which are addressed in the concept. Similarities can also be seen when comparing the previous community center projects with this developed center. The centers in Hesse, Germany, strengthened the democratic principles by offering public meeting spaces to make people heard, which this project's concept also considers. Strengthening the community and enhancing gender equality were similar to the center in Bangladesh. However, a difference is the focus on sanitation level which was a local need for the Bangladesh community. Moreover, poor infrastructure to the nearest city was a shared issue between Xihe Village in China and Linga Linga. The centers also have the same services as a shop and mini-museum, which is combined in Linga Linga's case. Lastly, one center in Leeds in the UK also had multi-purpose spaces, which have led to increased community engagement.

Similarities between Linga Linga and other rural coastal communities in Moçambique can probably be drawn, for instance, regarding long distances to necessities, lack of local trading, poor education opportunities, and where fishery is a common occupation. However, there are also many differences such as power structures, local knowledge, cultures, and religions. The community center concept is based on the case of Linga Linga. To make the project applicable to other communities it is essential to develop solutions to the local context. This was also highlighted in the In-depth interviews in the Pre-study and by scholars such as Manzini (2014). However, there are general factors from this thesis study that are relevant to be investigated for other communities, such as defining what is lacking in the area, the power structure, the local occupations, and how the solutions can be designed to the local culture. In addition, parallels can be drawn between this particular case and rural Western communities regarding certain needs. Long distance to necessities, high transportation costs, lack of local businesses, and high rural-urban movement, are some examples. The solutions to these needs may be completely different, however, the questions may be similar, as the Kvinna till Kvinna Foundation also said during the In-depth interview.

8.2 Research questions

The section below discusses the project's outcome concerning the two research questions.

RQ.1: What needs do the stakeholders of Linga Linga have that can be met by a community center?

The project featured an 8-week-long field study to map out the needs of various stakeholders in Linga Linga, as seen in Chapter 5.3.6. The field study involved 20 people with diverse backgrounds, including the perspectives of the local leaders and the district to gain a comprehensive mapping of the needs. Different methods were conducted to gather expressed needs from the stakeholders themselves and to interpret their actual needs. The need study laid a stable groundwork for the concept development to create solutions that meet the stakeholder's needs and create long-term community impacts.

RQ.2: What services could the community center offer to meet the identified needs?

A community center concept, involving seven services, was developed to meet the stakeholder's needs, which is presented in Chapter 7. However, the extent to which the needs were met varied, as some were constrained by the community center's ability to face them. Nevertheless, the solutions have the potential to enhance the quality of life of the citizens of Linga Linga in several aspects while supporting other important stakeholders such as the district. The concept also considers how the center should practically function financially and in terms of labor.

8.3 Method discussion

The project had a service and systemic design approach, by co-designing with the users, and having a user-centered and bottom-up perspective while valuing multiple actors with various backgrounds. This provided insights from the actors and gave a holistic view of the current system in the community and the corresponding needs, contributing to a systemic perspective. The project may have benefited from utilizing a more iterative process, especially in the ideation phase where ideas could be tested in several iterative loops with the users as a trigger material to gain further insights and contribute to a further implementable final concept.

In the 8-week field study, 20 people were involved in different user study methods. There was a saturation of answers on certain topics for instance missing elements in Linga Linga and desired things to learn. However, this was not the case for other topics such as desired recreational activities. Also, the large target group limited the number of interviewees of each need group, for instance, the bag weavers. Therefore, conducting more interviews may have provided better coverage of the study's broad target group and generated a saturation of answers, enriching the ideation phase with further supported insights. Moreover, conducting additional A Day in the Life sessions with more stakeholders could have provided other interesting insights. Furthermore, this study involved people between the ages of 15-50 where further interviews would be necessary to address the needs of the older population. Furthermore, it may have been rewarding including the citizens of Linga Linga's neighboring

village since they are also potential users of the community center due to the shorter distance to access the services in Linga Linga compared to Morrumbene.

The user study methods were adapted to the indigenous environment by following the considerations by Chilisa (2020) in chapter 3.5, as well as, by the Kvinna till Kvinna Foundation and the previous Project Vita student from the Pre-study. Despite active listening by the designer, the dialogues were mostly between the interpreter and the participant. If the designer had talked more in Portuguese during the interactions, a better dialogue and comfortable environment may have been created, which potentially would have led to additional insights. In addition, distractions could have been further avoided during the interviews, for example in the second interview, where the participant was distracted by two people beside him who listened and reacted to his answers.

The use of an interpreter in the user study affected the results. There is a higher risk of misinterpretations since there is an additional link the answers go through and expressions in languages differ which cannot be translated completely accurately. In addition, the designer cannot combine verbal and body expressions which may leave interesting insights unseen. This can also occur when the interpreter summarizes the answers, meaning certain content is disregarded since it is perceived as non-relevant, however, it may still be of interest to the designer. Furthermore, the appropriateness of using the founder and CEO of Project Vita as an interpreter can be debated. The interviewees know the interpreter well which makes them feel comfortable answering questions. On the other hand, the participants know he is a powerful actor in the village who can make things better in the community, and therefore the answers may be adapted to this. Another interpreter from Linga Linga who speaks the local language may have been relevant to preventing guided answers and making it easier for some participants to express themselves.

During the interviews, participants frequently had difficulty expressing themselves, particularly with the more complex questions. Other user study methods may have assisted the participants, for instance, using visual aids and physical models. This would have been especially relevant in the interactions with the younger population who may express themselves better in other forms than question-and-answer interviews. Examples of revealing unspoken insights could be using games or other fun activities.

The bottom-up approach and co-creation workshop were important for social innovation based on the research and Pre-study interviews. The workshop assisted the second ideation session with a deepened understanding of how the services should practically work. The workshop was only conducted with women to prevent the community's power structure from influencing the female participants. Adopting a more iterative process and testing ideas in additional workshops, also involving men, could strengthen solutions, simplify concept selection, and enhance their feasibility. Another benefit of testing ideas with the users is that it also becomes a trigger material for the designer. This would be particularly relevant to this project led by a single designer, as it allows receiving of feedback, a process which naturally occurs in group settings.

9. Conclusion

The project presents solutions and an approach to tackle problems a rural community faces. There is a wide range of issues and needs in Linga Linga, but also opportunities for change in the village. The purpose of the study was to expand the knowledge of community center development and build more sustainable rural coastal communities in Moçambique. It is met by conducting a case study in a rural village, complimentary by research and involvement of Swedish actors. The project's result supports the citizens of Linga Linga since the solutions target their needs to improve the quality of life in the long term. Among the citizens, the center should be perceived as a location with possibilities, positive energy, and security. The center has the potential for socio-economic contribution through enhanced democracy, information spreading, city accessibility, and increased income through education opportunities and system change to benefit female entrepreneurs. The center also strengthens the rural community while valuing its traditions and culture. In addition, the transparent and considered approach to finding local needs fills the gap in community center development projects, thereby being relevant for other project leaders in similar initiatives. Furthermore, the defined needs, discussion of study improvements, and solutions based on the community's opportunities provide further knowledge to community center development. In a larger context, the project, and the upcoming implementation process, can have a larger impact because it can act as a guideline to other Moçambique locations to develop similar projects where 2/3 of the population lives in rural communities and 1/3 in coastal areas. The project can also inspire communities outside Moçambique since several aspects apply to rural areas all around the World. The overall conclusion of the project is that a community center can improve the quality of life for rural citizens, by building on enhanced community engagement rather than large-scale investments.

There are relevant areas in the project to be explored further. Firstly, the architectural considerations to be adapted to the culture, to create a local identity, and to incorporate the services to reach their fullest potential. This particularly involves the multifunctional spaces designed for flexibility to enable various activities. Another area to investigate further is the financial aspects of the center, for instance, assessing the feasibility of operating the center and appropriate payment models for accessing certain services. An additional subject for further work is how the 16 km long road between Linga Linga and Morrumbene can be improved since this would greatly impact the community. A possible solution identified in this project is engaging local actors by establishing a road association. Relevant topics could include mapping the relevant actors and tools, analyzing the financial parameters, and defining the impact on the citizens. This potential work would also have an impact on the community center's role. Another potential area to be explored further is what to consider when spreading projects to other locations. This study would increase the feasibility of spreading this community center project and contribute to other rural development projects including Project Vita initiatives. Lastly, the further work of implementing this community center project in practice is of the most importance to have an actual impact on the citizen's lives. Establishing a local engagement in the building process as well as involving local resources and labor to create a feeling of ownership are a few crucial factors identified from this community center project.

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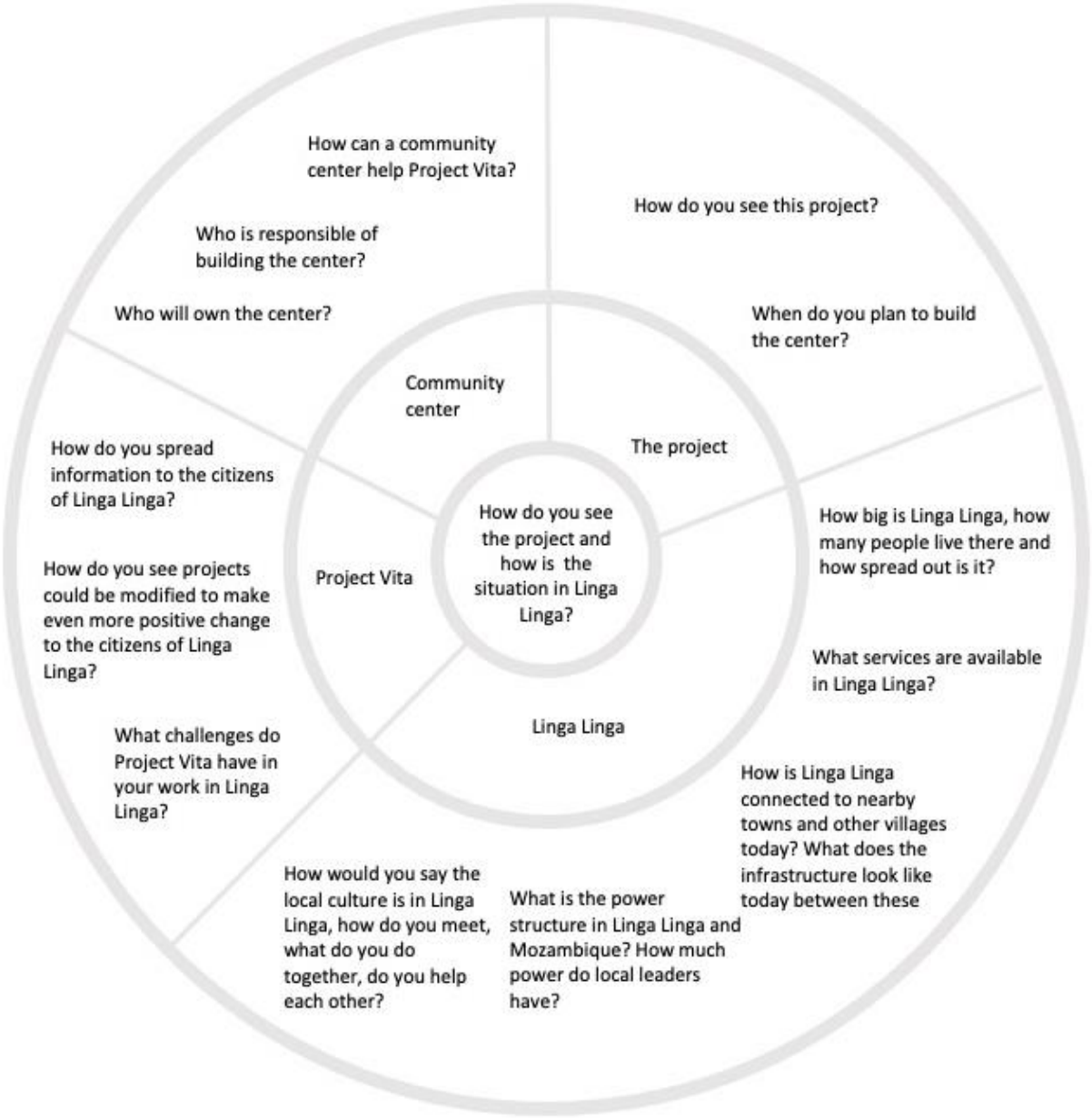
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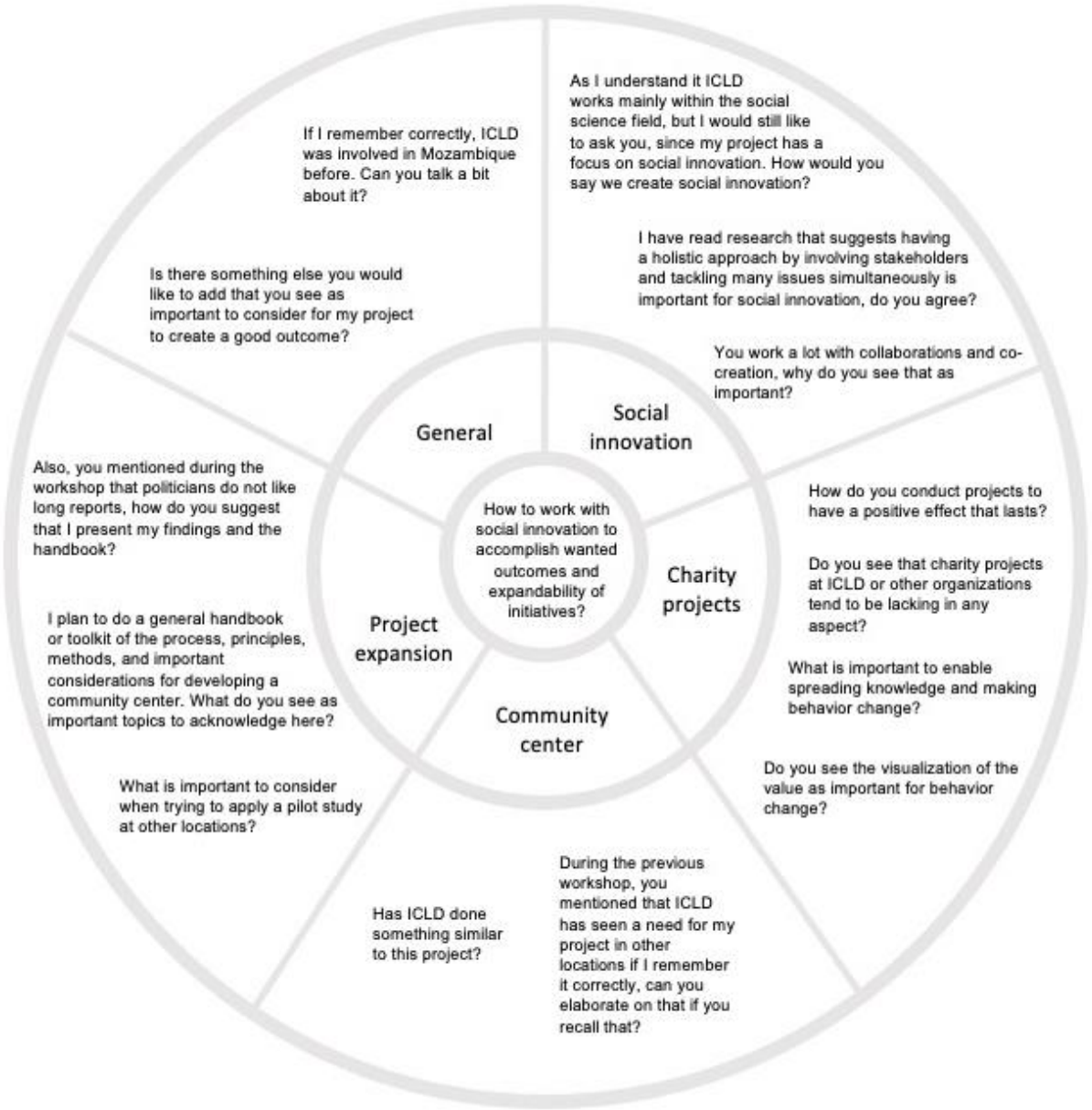
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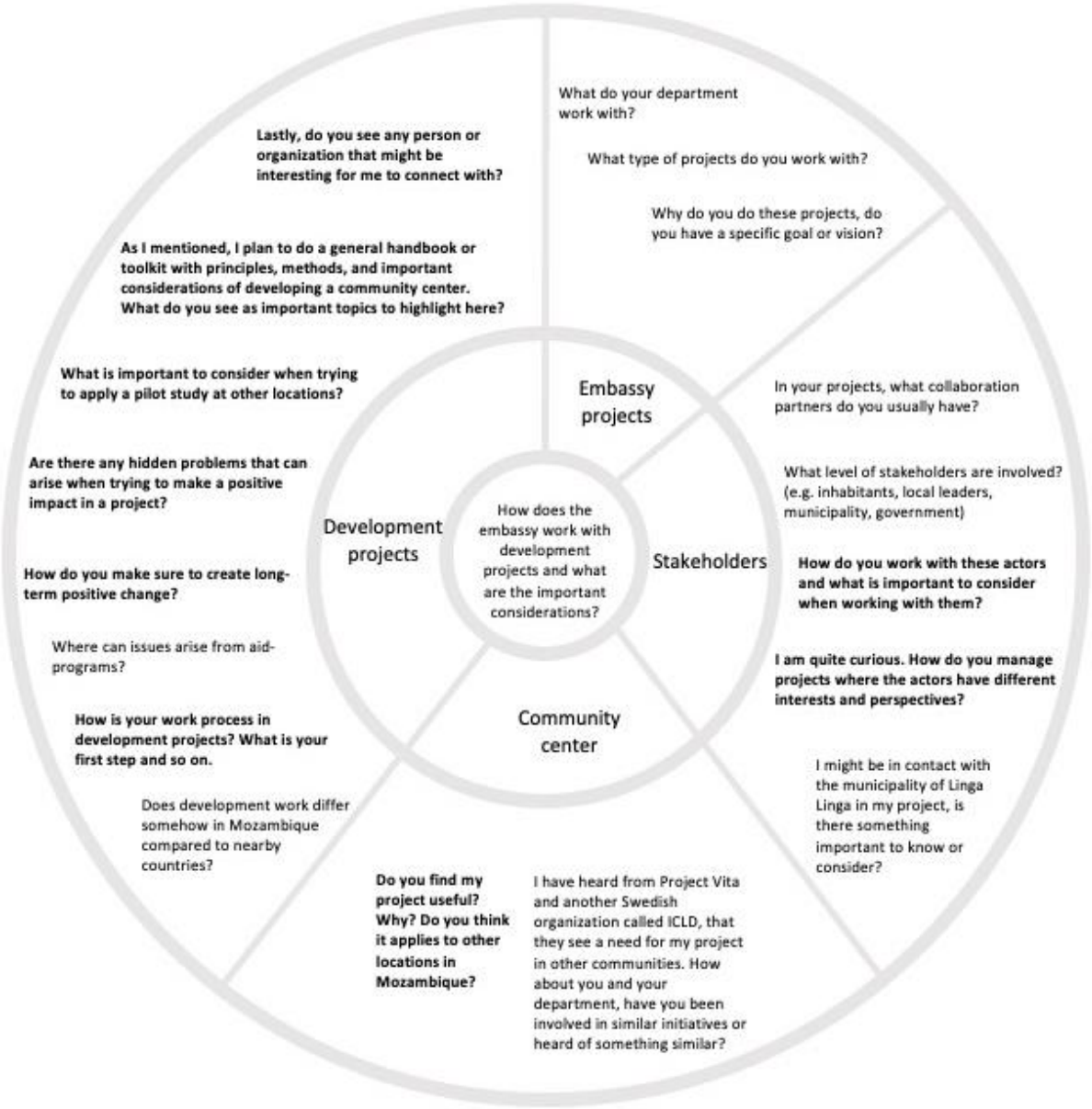
Appendix A - Interview guide Project Vita


How can a community center help Project Vita?
How do you see this project?
Who is responsible of building the center?
Who will own the center?
When do you plan to build the center?
Community center
The project
How do you spread information to the citizens of Linga Linga?
How do you see the project and how is the situation in Linga Linga?
How big is Linga Linga, how many people live there and how spread out is it?
Project Vita
How do you see projects could be modified to make even more positive change to the citizens of Linga Linga?
What services are available in Linga Linga?
Linga Linga
What challenges do Project Vita have in your work in Linga Linga?
How is Linga Linga connected to nearby towns and other villages today? What does the infrastructure look like today between these
How would you say the local culture is in Linga Linga, how do you meet, what do you do together, do you help each other?
What is the power structure in Linga Linga and Mozambique? How much power do local leaders have?

Appendix B - Interview guide ICLD



Appendix C - Interview guide SIDA



Appendix D - Interview guide The Kvinna till kvinna Foundation

A few open questions were formulated:

- Your colleague mentioned that you at Kvinna till kvinna, among other things, develop guidelines/tools that you contextualize to the needs. Is it the case that you adapt those tools to the needs of the new place you want to apply them?
- The idea of my project is also to develop a type of general toolbox with tools for how other communities can carry out similar projects. There can be large differences from place to place, even in the same country or region. How do you shape your guidelines/tools to be applicable to other places?
- How can these tools look like?
- What do you think is important in the creation of general guidelines/tools?

Appendix E - Interview guide previous thesis student

A few open questions were formulated:

- Do you have any advice on how to adapt interviews, questions, and workshops, and how does it differ from our projects in Sweden?
- How does it work to use an interpreter?
- You mentioned in your report that you experienced some problems with the card sorting workshop. Do you have any advice on how to conduct co-creation workshops?
- How much did you prepare before your field study started?
- How did your planned schedule agree with reality? Did some things take longer than others? How can I make my stay as efficient as possible?
- Any important cultural differences that might be good to know?

Appendix F - In-depth interview responses with residents in Linga Linga

Person 1 (Women)

- *How old are you?*
  41.

- *Where do you live?*
  Here.

- *Can you tell me about your family?*
  I have 4 kids and a husband

- *What do your family members do?*
  My husband is a fisherman and one of my children goes to school.

- *What do you do for a living?*
  Work at a hotel and do some farming here, these leaves.

- *What places do you go to during the day?*
  Here.

- *Do you go to other places outside of Linga Linga?*
  Maxixe and Morrumbene.

  (Why?) To buy food.

  (How often) In the end of the month.

  (How do you get there?) By Chapa.

- *How do you get information about what is happening in Linga Linga?*
  Meetings. The local leader tells me when there is a funeral for example.

  (…In the country Moçambique?) By the radio, I get information about countries.

- *Is there any information that you would like to access more/better?*
  -

- *Where do you express your ideas/opinions?*
  If I have a problem with, for example, my sister, I talk to an older man.

- *How do you meet the local leader?*
  I meet him at his house.

  (Do you talk directly to the local leader or does someone else speak for you?) I talk directly to him.

- *What do you do during a typical day?*
  Brush my teeth, clean the yard, feed the animals, work (clean and fix stuff), come home and cook, eat and sleep

- *What do you find difficult/not fun to do?*
  -

- *Is there something that you would like to learn?*
  Yes. Sewing.

  (Why?) So I am not dependent on one job. I want to help my kids and the citizents here.

- *Do you miss something in Linga Linga that you would like to access?*
  Yes! A market to buy vegetables and herbs. I want to sell my leaves because I have so much and also crabs that I find which I cannot sell today. I want to buy casava because the one in Linga Linga tastes bad. Also energy, I have a fridge but not energy. My sons could also come here from Maputo and work because they need electricity to work.

- *There is a plan to have a place in Linga Linga to offer different services. What would you like to have and do there?*
  Market and a shop.

- *Lastly, what do you think is fun to?*
  To work. To earn money. A market is also fun socially to talk to people. It also develops the country.

  (Would you be willing to pay to sell your things in the market?) Yes!

  (How much?) The government must say a price.

  (If you can choose e.g. 50 MZN) Yes of course!

Person 2 (Man)

- *How old are you?*
  20.

- *Where do you live?*
  Here.

- *Does anyone else live here?*
  Five people.

- *What do you do for a living?*
  Sewing.

- *What places do you go to during the day?*
  I am just here. Hanging out and playing with my friend.

- *Do you go to other places outside of Linga Linga?*
  Morrumbene.

  (Why?) To buy materials for the sewing.

  (How often) Five times per month.

  (How do you get there?) By Chapa.

- *Who do you meet during the day?*
  I meet friends and my family.

- *How do you get information about what is happening in Linga Linga?*
  Through the phone by messages or calls.

  (…In the country Moçambique?) On Facebook.

- *Is there any information that you would like to access more/better?*
  No, not now.

- *Where do you express your ideas/opinions?*
  To my friend.

- *What do you do during a typical day?*
  Bathing in the sea and then working all day.

- *What do you find difficult/not fun to do?*
  -

- *What do you think of this weather? What do you think about when it is raining? Why?*
  Rain? No!

- *Is there something that you would like to learn?*
  Music. All kinds.

- *Do you miss something in Linga Linga that you would like to access?*
  Energy and a school above 8th grade so kids (points at the younger boy in the room) do not have to travel to Morrumbene.

- *There is a plan to have a place in Linga Linga to offer different services. What would you like to have and do there?*
  A market so I do not need transport. To make life easier.

- *Lastly, what do you think is fun to?*
  Music. I like to play music and I am singing.

  (Would you like to do it more?) Yes.

  (Why don't you do it more often?) I want energy for a studio.

Person 3 (Women)

- *How old are you?*
  45.

- *Where do you live?*
  Close to here.

- *Does anyone else live there?*
  My mom and my son that is 24 years old.

- *What do your family members do?*
  My mom does bag weaving.

- *What do you do for a living?*
  I am in the kitchen, housekeeping and weaving bags on Sundays. I sell them in Inhambane. I use a “Guivar” (a seller) and I am the producer. I would like to keep

doing like that.

- *What places do you go to during the day?*
  At home fixing things, the church.

- *Do you go to other places outside of Linga Linga?*
  Morrumbene and Maxixe.

  (Why?) To buy food. Big shopping such as tomato, rice, sugar.

  (How often) Once per month. Sometimes not even once in a month.

  (How do you get there?) By Chapa.

- *Do you share things with people?*
  Rice, kitchen wear like pots.

- *How do you get information about what is happening in Linga Linga?*
  Through the phone or by talking to people. Also at funerals and community meetings. They are held at 8 o'clock, with the mayor. I do not go there because I am working.

- *Is there any information that you would like to access more/better?*
  More information from my family.

- *Where do you express your ideas/opinions?*
  Neighbors and family.

- *How do you meet the local leader?*
  I go to the leader's house.

  (Why do you talk with the leader?) To talk about things that must be going through the leader. For example, to fix my ID card, the leader has to verify it is me.

- *What do you find difficult/not fun to do?*
  I do not like problems. To not talk bad about people. Also, someone is stealing coconuts at my home.

- *Is there something that you would like to learn?*
  Yes. English. It is good to know. I can ask for help and ask for things.

- *Do you meet people in large groups in Linga Linga?*
  In Church.

- *Do you miss something in Linga Linga that you would like to access?*
  Much. Energy, a market, and a kiosk.

  (Why energy?) I have energy but others do not. They can use it for a fridge or freezer or to raise chickens with this lamp that you need.

  (Why market?) It would not be as much suffering. I would like to buy the same things as in Morrumbene. I would also like to sell clothes which I can get from UN donations.

- *Lastly, what do you think is fun to?*
  To work, to earn money.

  (Why?) So I do not have to steal *(laugh)*

Person 4 (Man)

- *How old are you?*
  19.

- *Where do you live?*
  Here.

- *Does anyone else live here?*
  I live by myself.

- *What do you do for a living?*
  General stuff. A janitor fixing stuff.

- *What places do you go to during the day?*
  Various kiosks, play a spinning wheel game with some money.

- *Do you go to other places outside of Linga Linga?*
  Morrumbene, Maxixe and Inhambane.

  (Why?) To follow my boss.

- *Who do you meet during the day?*
  Friends, grandma, mom, dad, my other grandma and my siblings.

- *What do you do together?*
  Talking with my friends, much about the future.

(What thoughts do you have about your future?) I want my own house. I have to work to get it. I have already bought some land.

- *Do you share things with people?*
  Only with people I trust. Lend some money to my uncle. My headphones to my friends. I only lend things.

- *How do you get information about what is happening in Linga Linga?*
  My friends share things when we meet by the road.

  (…In the country Moçambique?) By radio. A local radio channel.

- *Is there any information that you would like to access more/better?*
  No.

- *Where do you express your ideas/opinions?*
  Mom and grandma.

- *How do you meet the local leader?*
  Sometimes, just around, for example at the hospital.

  (Why do you talk with the leader?) Nothing important. I just say hi. I talk more to his assistant. Sometimes I give papers to him from my boss.

- *What do you find difficult/not fun to do?*

  I do not like fishing because I cannot swim. Also stealing.

- *Is there something that you would like to learn?*
  Yes. English.

  (Why?) I would like to talk with foreigners.

- *Do you meet people in large groups in Linga Linga?*
  At meetings, if for example, someone has stolen something. Some parties. For football. Projects, for example, when we hand out lamps to people.

- *Do you miss something in Linga Linga that you would like to access?*
  A market. Emergency hospital. Good water sources.

  (Why market?) It is tough to go to Morrumbene, for example, when it is raining on the chapa.

  (Why an emergency?) If people are pregnant they can be there first.

- *There is a plan to have a place in Linga Linga to offer different services. What would you like to have and do there?*
  A market, shop, small school.

  (Why market?) To buy most stuff.

  (Why a small school?) To give information, from computers. I would like to learn that. I have not tried a computer before. Computers do important things. I have seen my boss concentrate in front of the computer.

- *Lastly, what do you think is fun to?*
  To work.

  (Why?) To have my own stuff, my house, to not have to steal.

  I also like to watch football on tv and watch friends. There is one match per month here *(interpreter: matches between the villages here).*

Appendix G - In-depth interview response with leaders of Linga Linga

Leader assistant Linga Linga zone B

Leadership

- *How long have you been the leader of Linga Linga?*
  Since 2012

- *Can you tell me about your work?*
  I forward information from the local leader. Also, if it is a problem, for instance, someone is stealing or doing farming where it is not allowed, I forward it to the local leader. He is in charge. I hear complaints from the citizens. The local leader forwarded it to the district.

- *Where do you usually meet people in Linga Linga in your work?*
  It is a place close to me [where large meetings are held]. It was decided back in the 19th hundreds. However, now we have changed to close to Elmondo because it is better for cars. We sit in the sand under a tree.
  (Where do you initially meet people?) They come to my house. If there is a matter I can not solve I forward it to the local leader.

- *What do you talk about during the large meetings?*
  Election cards. It is mostly arranged by the district, from the various departments e.g. the police, infrastructure, and malaria.
  (How often do you have these meetings?) Around three per year that we arrange by ourselves. The district arranges most meetings. I would like to arrange a meeting per month. I have told the local leader this many times. I do not know why he has not decided this. The district also wants more meetings because they want to hear what is happening here every month since they have their monthly meeting in the district. Some matters are better to talk about during the meetings instead of the local leader doing it individually to citizens, for instance, if someone is stealing.

- *How many people usually attend?*
  Sometimes 10, sometimes 30. The mayor also complains that it is so few people at the meetings arranged by him. More people attended before. It is not well organized.

Communication & information

- *What information do they get from you?*
  Information about when meetings are held, and who has to do the cleaning for the meetings (district meetings).

(How can it be improved?) If there was money it would be easier. The local leader gets money but I do not. The local leader gives me assignments which I carry out.

- *How is information about large meetings spread to the local population?*
  Previously I used a microphone or I hit on a large bucket to gather people.

- *Is there some other information you see important to communicate?*
  To not bring people into the village with no good intentions, for instance, industrial crab fishers that fish the crabs.

- *How do you think these meetings could be improved?*
  It would be better if more people attended.

- *Linga Linga is exposed to varying weather and extreme weather. How do people get information about this?*
  There is an educated group of people which drives around with a megaphone. Some people can tell there is bad weather coming by feeling the temperature changes. They then call people about it. I experienced the cyclone in 2017, I did not get information about when it was going to come, only that it was coming at some point.

  (How did you feel at this time?) I was scared, I was wondering if we were going to survive this. I survived but not the houses.

  (What did you do?) I went to my neighbor who has a concrete house, I brought my wife there also.

Trading

- *When I have spoken to people, they say it is difficult to sell their own goods and buy goods in Linga Linga, which today is found in Morrumbene, what do you think about the problem?*
  This has been a problem for a long time. A market is needed. People has electricity for farming and fishing but not a place to sell this.

Transportation

- *When I have spoken to people, most people say that transport to Morrumbene is difficult and expensive, what do you think about the problem?*
  The chapas can not lower the price because they have to put money into the bad road. Therefore, they also search for many travelers.

Learning

- *How many children in Linga Linga do you think study after year 7?*
  All kids study to 8th grade here close by. All kids want to study after 8th grade but they cannot because it is too far. It requires money, 900 MZN per month.

- *When I have talked to people, everyone wants to learn different things, how do you think that can happen in Linga Linga?*
  People would like to learn things. People do bags, farming, boats, things are sold here.

Community center

- *Are you missing something in Linga Linga?*
  Somewhere to store fish. Chapas come at various times, if you miss one you have to wait for a long time.
  A market.
  Electricity. To connect the more wealthy people to the grid and have fish storage there.

- *If there was a place/a center in Linga Linga where many people gather for different services, how do you think it could help you in your work?*
  To have meetings there, would be easier because we do not have to clean up things when the district people are coming. We would be very happy.

- *What do you think are the challenges of such a place/center in the short and long term?*
  It would not be a problem. No partners are helping here, only Michel.

- *What are your thoughts on the future of Linga Linga, what do you want it to look like?*
  A place for meetings, today we lend chairs from neighbors for the big meetings.

Local leader of Linga Linga

Leadership

- *How long have you been the leader of Linga Linga?*
  10 years.

- *Can you tell me about your work?*
  I look over the citizens, forward messages from the district and the other way around.

- *How is your relationship with the local leaders of other villages?*
  Sometimes we go and meet each other and work, for instance, regarding a school since it affects all villages.

- *How many have access to electricity in Linga Linga?*
  Less than 50 people are connected to the grid. People buy solar panels by themselves.

- *Where do you usually meet people in Linga Linga in your work?*
  Under a tree close to here and other trees around.

- *What natural meeting places are in Linga Linga?*
  Carlitos (bar), Lino (bar), the two soccer fields.

- *What do you talk about during the large meetings?*
  Family conflicts. If it is not solved, the matter is taken forward. The conflicts can be about arguments and people being drunk.

- *Where do you have these first talks?*
  We meet at my home. We meet individually. The larger meetings is with the involved people and the leaders.

Communication & information

- *How is the process of forwarding messages from the citizens to the district?*
  I met with all the leaders from the other villages and the district representative once a month. Thereafter she forwards the matters to the district at her meeting with the mayor once a month. The answer from the district goes back the same route to the citizens.

- *How do people in Linga Linga get important information?*
  The different departments come here and inform the leaders. Then we forward it to the people. We go to the school, the churches, and along the street.

- *What important information do they get?*
  Education is important information. Health, diseases like malaria, where a health committee walks around to each home and informs.

- *Is there any other information you think would be good for people to get?*
  Yes, but I cannot say what it is.

- *Linga Linga is exposed to varying weather and extreme weather. How do people get information about this?*
  There is a committee in Morrumbene that sends a group to Linga Linga and informs. They use a megaphone so the people know around what day the cyclone may come. People hide in the school and hospital. People cannot prepare themselves, they can just wait.

- *How often do you have the community meetings?*
  2 times per month.

- *What do you talk about at those meetings?*
  We spread information from the district and events in Linga Linga.

- *How many people usually attend?*
  Few people.

(Why do you think few people are coming?) People are busy with their lives.

- *Do you want more people to attend the meetings?*
  Yes, I want many more people.

- *How do you think more people could attend the meetings?*
  Many people lives in different places, it is difficult to make them come to the same place. It could be helpful to offer drinks to attract more people. It is good for the people to attend these meetings because they develop themselves.

- *What do you miss in Linga Linga?*
  Better water, connected to water towers. Electricity. The road is bad, it needs asphalt. Market, Michel has already mentioned it. Transport, to improve the chapas. Asphalt makes it more accessible for better cars for example buses.

Trading

- *When I have spoken to people, they say it is difficult to sell their goods and buy goods in Linga Linga, which today is found in Morrumbene, what do you think about the problem?*
  It is a problem to transport people and goods, it would be good if there was a boat from Inhambane and to a hospital in another city.

Learning

- *How many children in Linga Linga do you think study after year 8?*
  The kids have to travel and it costs. Everyone needs studies, to improve life – more benefits.

- *What do you find important for children to learn?*
  They need what children in other cities have, light, computers.

Community center

- *If there was a place/a center in Linga Linga where many people gather for different services, how do you think it could help you in your work?*
  Education, trading, information (computer). Education involves school activities.

- *Do you see any relevant person being there?*
  Yes, a person can be there because then people know where you should meet. Also a person in control there.

- *Do you see this center could help you in your work?*
  Yes, it can help me. Meetings can be held here to spread information instead of walking around. All other meetings held around could be connected to one place, because the

meetings today are spread out around many places. The person in control of the center can inform me what is happening there.

- *Do you see any challenges with a place like this?*
  It is hard to say. I do not see any problems now since everyone needs this. Problems come once you start building, it comes later.

- *What are your thoughts on the future of Linga Linga?*
  A place that is developed so more investments for tourism come. It creates more jobs so people can improve their households.

Head of Locality Morrumbene district

Responsibilities

- *What do you do in your work?*
  Head of Locality.

- What are your responsibility areas?
  I am responsible for the villages around Morrumbene.

- *What are the district's responsibilities?*
  The district is broad. To improve and help the lives of those who live here. Focus areas are security which the justice department is also responsible for, education, agriculture, access to food, health, and communication with the population.

Communication

- *How do you communicate information to the population living in Linga Linga?*
  Through meetings with the people of Linga Linga. Sometimes small conferences that the population asks for. Also the finance department does this meeting.

- *What do you communicate?*
  About anything, for instance, how to structure society.

- *Do you think it could be improved in any way?*
  It can be improved by having places in Linga Linga to keep statistics. Also to add the services that society needs, health, education, security, and government agencies, decentralize the district's work. I would like to have an arm in Linga Linga to represent me, now the local leader does everything. I would like to have small parts of the institution around to get information more easily.

- *How often do you have community meetings with the population of Linga Linga?*
  1-3 times per month, the plans differ, it should be every month.

- *How can it be improved?*
  Very important to decentralize the institution, a place for one responsible for fishing, security, etc., so that people don't have to go to Morrumbene to talk to that person.

Community center

- *What do you think are the biggest difficulties with living in the countryside, for example in Linga Linga?*
  Road, transport, market, security no police station.

- *If there was a place/community center in Linga Linga where many people go to use different services and have access to different things. What do you wish was in this place to help the people of Linga Linga?*
  Generally everything. If energy, had a fishing market, also retirement home, and a police station because Linga Linga would have been safer, the market would make it easier for the population so they don't have to use the road.

- *If there was a market in Linga Linga, how do you see the groceries get delivered there?*
  It is a good question. To have an agent from here who is responsible, the Ministry of Finance which is responsible for goods can also get involved. Tourism is low in Linga Linga. Everything would be easier to get it delivered there if there were more tourists, it would become a tourist market. The population would receive goods more easily.

Transportation

- *Why isn't the road from Morrumbene to Linga Linga improved?*
  Money is missing. Castello has wanted to have a meeting with the investors (the hotels) next to the road about how the road can be resolved.

  (Interpreter asks if students can do a project about improving the road)
  It sounds good. Students can analyze how the road can be improved which can then be sent forward where we can apply for grants.

- *What do you think about if the villages got the responsibility of maintaining the road and receiving contributions to do it?*
  If the investors got together and talked, we could hire people from the villages to implement the construction, because today we hire an external company.

Learning

- *What do you see as important for children to learn things in Linga Linga? Adults?*
  They should learn how to do things, vocational school, carpentry, electrician, water, toilet. Today they cannot continue studying after school is over, but if they have learned a practice they can continue after school, how to build houses for example.

- *Many children in Linga Linga do not continue studying after year 8, what do you see needed to change this?*
  Build another school. A secondary school, to shorten the distance to school.

- *When I've talked to people, everyone wants to learn different things, how do you think that can happen in Linga Linga?*
  Build a place to learn, and also have teachers and machines if needed.

- *What do you want Linga Linga to be and look like in the future?*
  I want Linga Linga to become a point of reference in the Inhambane province, regarding tourism.

Appendix H – Co-creation workshop results

The results of the activities in the co-creation workshop are shown in Figure H1-H5.

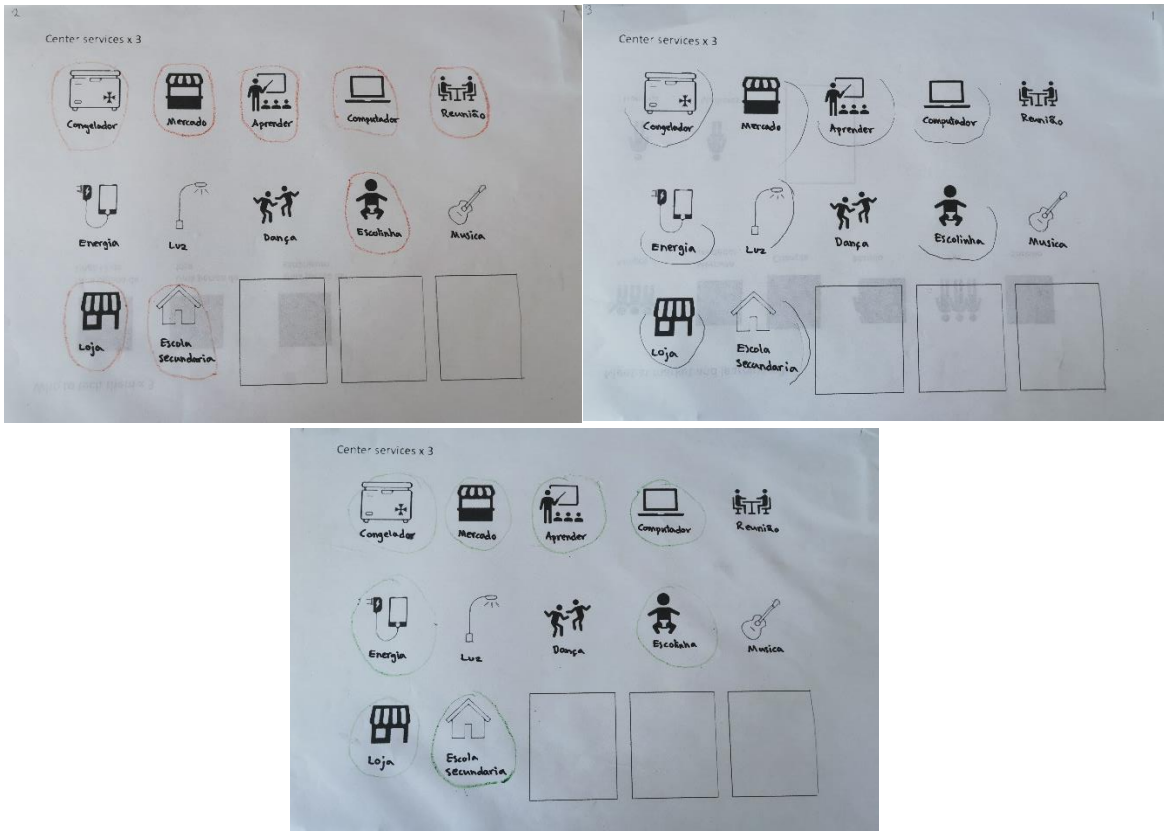


Figure H1. Activity 1, co-creation workshop.

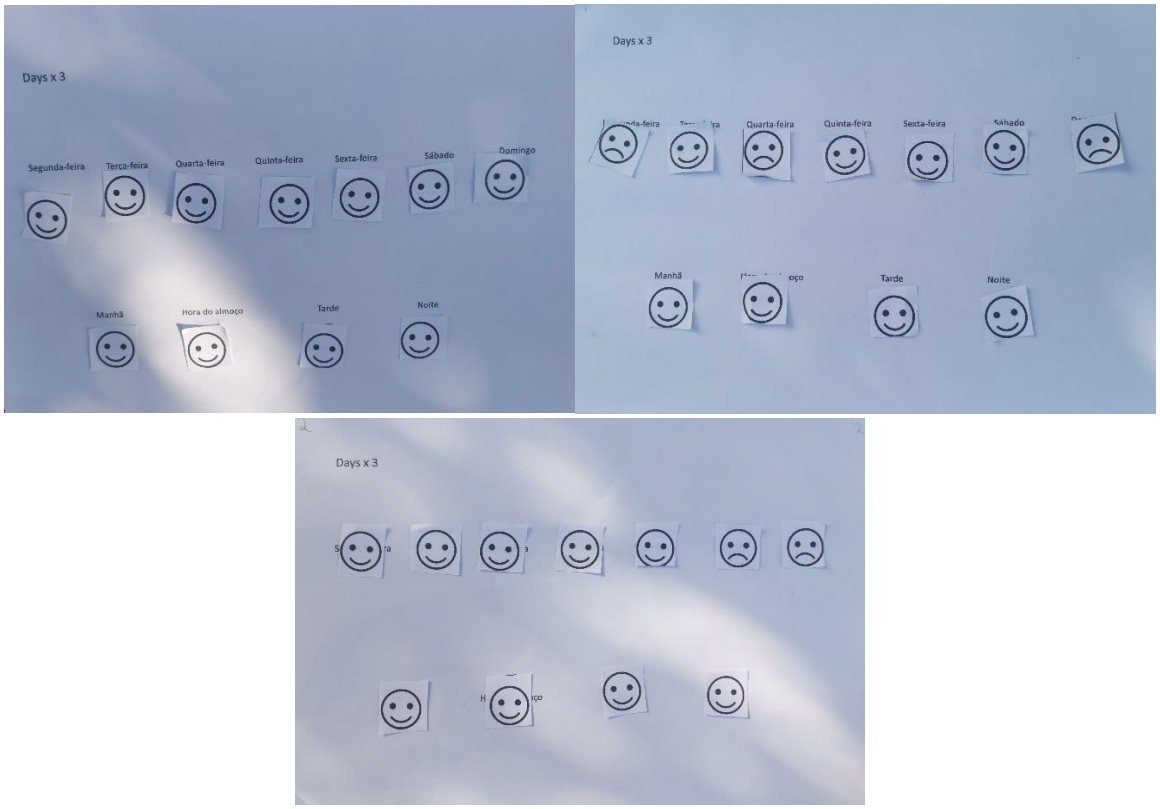


Figure H2. Activity 2 and 3, co-creation workshop.

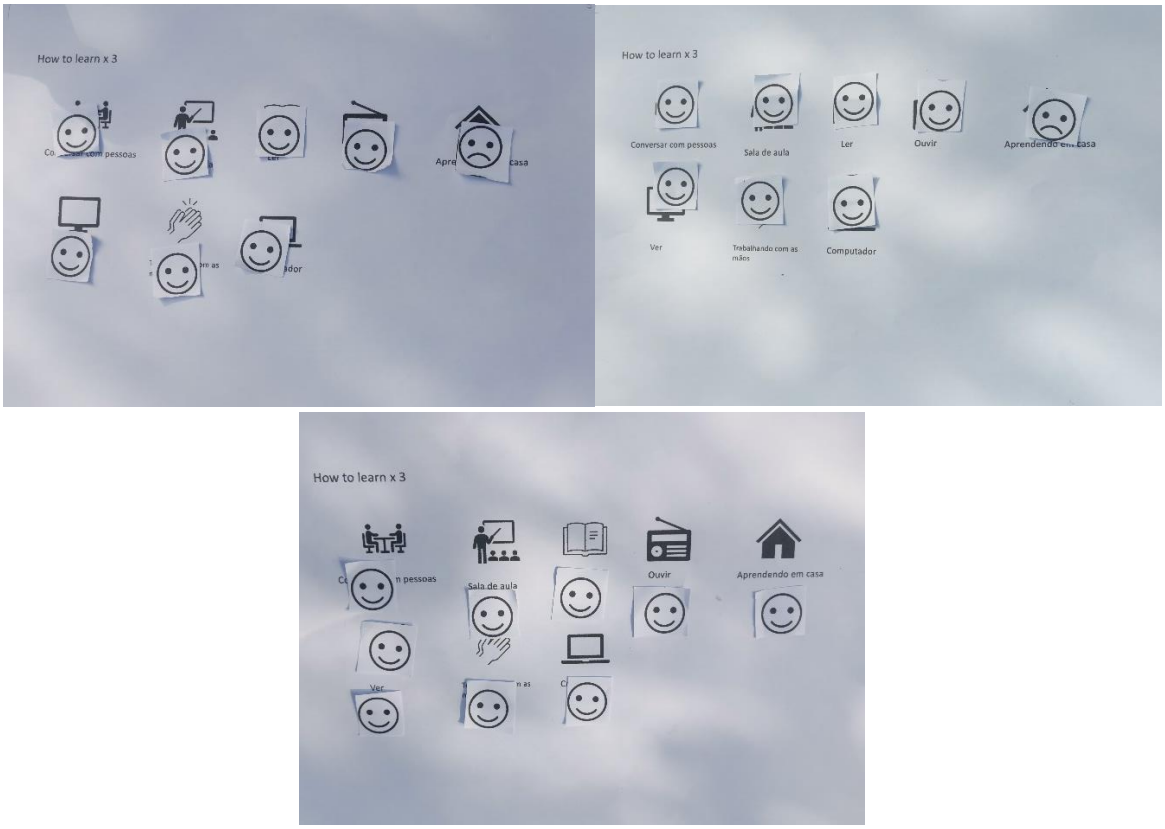


Figure H3. Activity 5, co-creation workshop.



Figure H4. Activity 6, co-creation workshop.

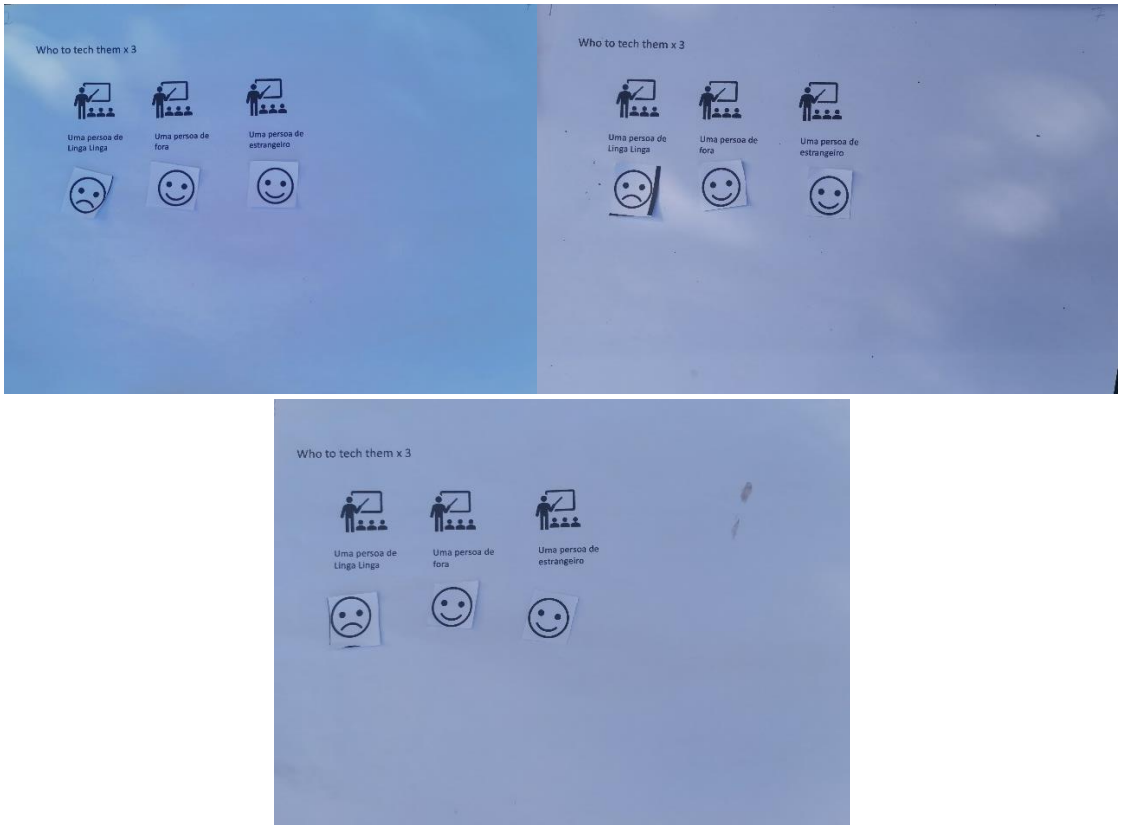


Figure H5. Activity 7, co-creation workshop.

TRITA – ITM-EX 2024:217
Stockholm, Sweden 2024

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